A Cluster Dynamics Model for Radiation Damage Evolution in Ferritic-Martensitic Steels

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Abstract

A self-contained formulation of cluster dynamics equations for modeling radiation damage evolution in bcc metals under neutron irradiation is presented, with primary applications to ferritic-martensitic steels. The formulation begins with defect production from displacement cascades and proceeds through defect energetics (formation and binding energies), diffusion on a bcc lattice (vacancies, self-interstitial atoms, interstitial and vacancy clusters, and helium interstitials), and cluster dissociation kinetics for voids, helium bubbles, and both vacancy and interstitial dislocation loops. A unified framework for reaction rates is developed with reaction frequencies $\omega = D/a^2$, covering five reaction classes: SIA–vacancy annihilation, SIA cluster growth and shrinkage, vacancy cluster growth and shrinkage, helium trapping and detrapping (including radiation re-solution), and helium–vacancy cluster interactions with SIAs. Two reduced descriptions of the helium–vacancy state space are derived—a fast helium equilibration model tracking the mean helium loading $\bar{\ell}(m)$ suitable for fusion conditions, and a decoupled helium inventory model for fission spectra—both preserving the essential coupling between gas pressure and cavity stability. The complete set of coupled rate equations is assembled with explicit mass conservation checks and a sink-strength formulation for the transition to continuum treatment of large clusters.

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2.1 Defect Cluster Spectra

Let G denote the *actual* (survival-corrected) defect production rate per unit volume, i.e.

$$G = \eta\, G_{\mathrm{NRT}}, \tag{1}$$

where $\eta = N_d/N_{\mathrm{NRT}}$ is the survival efficiency and G_{NRT} is the NRT displacement rate.

The production rate of interstitial clusters of size m is

$$P_m^{(i)} = \epsilon_m^{(i)}\, G, \qquad m = 2, 3, \ldots, m_1, \tag{2}$$

and the mono-interstitial production rate is

$$P_1^{(i)} = G\left(1 - \sum_{m=2}^{m_1} m\, \epsilon_m^{(i)}\right). \tag{3}$$

Analogous expressions hold for vacancies with superscript (v), cluster size index n, and upper cutoff n_1.

The total fraction of surviving SIAs born in clusters is

$$f_i^{\mathrm{cl}} = \sum_{m=2}^{m_1} m\, \epsilon_m^{(i)}, \tag{4}$$

with corresponding vacancy quantity f_v^{cl}. Note that the mono-defect fraction is simply $1 - f^{\mathrm{cl}}$.

The fraction of NRT-predicted Frenkel pairs that survive in-cascade recombination is well characterized by MD simulations Stoller (2000); Nordlund et al (2018); Stoller et al (1997). For α-Fe at reactor-relevant temperatures, the survival efficiency is approximately

$$\eta(E_{\mathrm{PKA}}) \;\approx\; 0.84 \left(\frac{E_{\mathrm{PKA}}}{1\ \mathrm{keV}}\right)^{-0.20}, \qquad E_{\mathrm{PKA}} \gtrsim 1\ \mathrm{keV}, \tag{5}$$

saturating near $\eta \approx 0.30$ above $\sim$10 keV.

Table 1 lists representative values. The "athermal recombination corrected dpa" (arc-dpa) formalism of Nordlund et al (2018) provides a more refined energy-dependent efficiency function $\xi_{\mathrm{arc}}(E)$ that is recommended for detailed calculations.

MD cascade simulations in α-Fe consistently show that the *number* of in-cascade clusters of size m follows an approximate power law Stoller (1998, 2000); De Backer et al (2018); Calder et al (2010):

$$N_{\mathrm{cl}}(m) \;\propto\; m^{-s}, \tag{6}$$

Table 1 Defect survival efficiency $\eta = N_d/N_{\mathrm{NRT}}$ in α-Fe at 100 K from MD cascade simulations Stoller (2000); Nordlund et al (2018).

E_{PKA} (keV)	η
0.5	0.96
1	0.84
2	0.73
5	0.61
10	0.53
20	0.46
50	0.38
100	0.33
200	0.29

with distinct exponents for SIA clusters (s_i) and vacancy clusters (s_v). The steeper vacancy exponent reflects the well-known inhibition of true vacancy clustering in bcc Fe, where vacancy "clusters" are often loose, correlated arrangements rather than compact loops Stoller (1998); Soneda et al (2001).

The production fraction is therefore modeled as

$$\epsilon_m^{(i)} = C_i\, m^{-s_i}, \qquad m = 2, \ldots, m_1, \tag{7}$$

where the normalization constant C_i is fixed by the constraint

$$C_i = \frac{f_i^{\mathrm{cl}}}{\sum_{m=2}^{m_1} m^{1-s_i}}. \tag{8}$$

An identical prescription applies to vacancies with parameters $(C_v, s_v, n_1, f_v^{\mathrm{cl}})$.

The parameters that distinguish fission and fusion spectra in terms of cascade damage are summarized in Table 2. Key physical differences:

- **PKA energy spectrum:** Fission-spectrum neutrons produce a PKA distribution weighted toward lower energies ($\langle E_{\mathrm{PKA}} \rangle \approx$ 10–15 keV), whereas 14 MeV fusion neutrons shift the average to ∼30–50 keV, leading to subcascade fragmentation De Backer et al (2018); Zarkadoula et al (2013).
- **Cluster sizes:** Higher PKA energies produce larger in-cascade clusters, so the maximum cluster sizes m_1 and n_1 and the power-law exponents all shift Stoller (2000); Calder et al (2010).
- **Helium co-production:** Helium generation is 0.5–1 appm He/dpa for fission spectra and ∼10 appm He/dpa for fusion, affecting vacancy cluster (bubble) nucleation at longer times but not the primary cascade spectrum itself.

Table 2 Recommended cascade-damage parameters for α-Fe under fission and fusion neutron spectra, synthesized from Stoller (2000), Nordlund et al (2018), De Backer et al (2018), and Malerba et al (2021).

Parameter	**Fission**	**Fusion**
$\langle E_{\mathrm{PKA}} \rangle$ (keV)	∼15	∼40
η (spectrum-averaged)	0.30	0.28
f_i^{cl} (SIA clustering fraction)	0.58	0.65
f_v^{cl} (vacancy clustering fraction)	0.15	0.20
s_i (SIA power-law exponent)	1.6	1.5
s_v (vacancy power-law exponent)	2.5	2.3
m_1 (max SIA cluster size)	20	50
n_1 (max vacancy cluster size)	10	20
He production (appm He/dpa)	0.5–1	∼10

The following closed-form expressions allow direct computation of ϵ_m for any cluster size. For interstitial Clusters, we have

$$\boxed{\epsilon_m^{(i)} = \frac{f_i^{\mathrm{cl}}\, m^{-s_i}}{\displaystyle\sum_{k=2}^{m_1} k^{1-s_i}}, \qquad m = 2, 3, \ldots, m_1} \tag{9}$$

Fission: $f_i^{\mathrm{cl}} = 0.58,\;\; s_i = 1.6,\;\; m_1 = 20,\;$ giving $C_i = f_i^{\mathrm{cl}} / \sum_{k=2}^{20} k^{-0.6} = 0.1093.$

Fusion: $f_i^{\mathrm{cl}} = 0.65,\;\; s_i = 1.5,\;\; m_1 = 50,\;$ giving $C_i = 0.0553.$

While for vacancy clusters, we get

$$\boxed{\epsilon_n^{(v)} = \frac{f_v^{\mathrm{cl}}\, n^{-s_v}}{\displaystyle\sum_{k=2}^{n_1} k^{1-s_v}}, \qquad n = 2, 3, \ldots, n_1} \tag{10}$$

Fission: $f_v^{\mathrm{cl}} = 0.15,\;\; s_v = 2.5,\;\; n_1 = 10,\;$ giving $C_v = 0.1506.$

Fusion: $f_v^{\mathrm{cl}} = 0.20,\;\; s_v = 2.3,\;\; n_1 = 20,\;$ giving $C_v = 0.1296.$

The normalization sums can be pre-computed once:

$$S(s, m_{\max}) = \sum_{k=2}^{m_{\max}} k^{1-s}. \tag{11}$$

Then $C = f^{\mathrm{cl}}/S$ and $\epsilon_m = C\, m^{-s}$.

For large $m_{\max}$, the sum can be approximated by the integral

$$S(s, m_{\max}) \approx \begin{cases} \dfrac{m_{\max}^{2-s} - 2^{2-s}}{2-s} + \frac{1}{2}\big(m_{\max}^{1-s} + 2^{1-s}\big), & s \neq 2, \\ \ln\!\Big(\dfrac{m_{\max}}{2}\Big) + \frac{1}{2}\big(m_{\max}^{-1} + 2^{-1}\big), & s = 2. \end{cases} \tag{12}$$

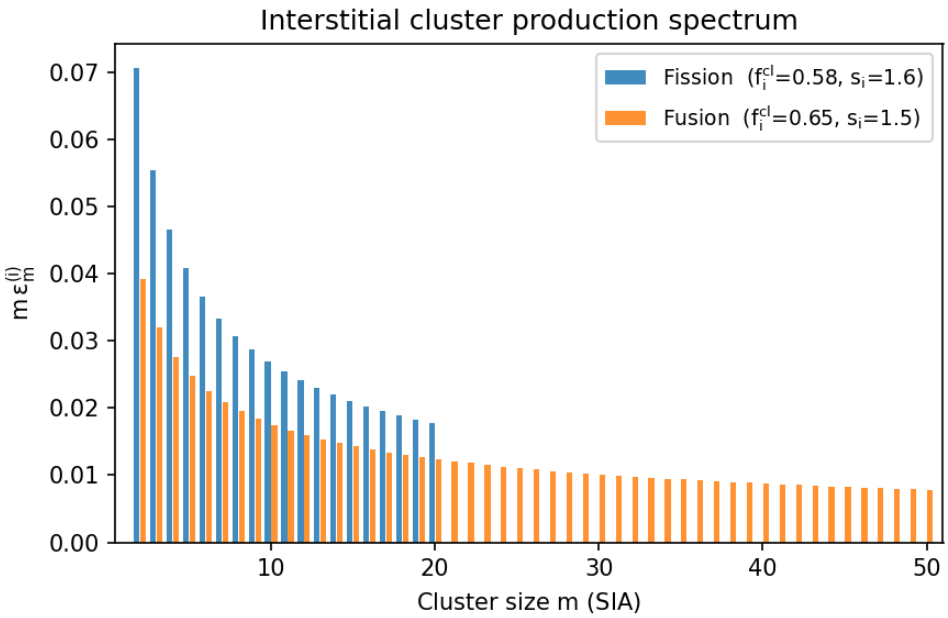

Fig. 1 Interstitial cluster production spectrum under fission and fusion conditions

For direct implementation in a cluster-dynamics code, the source terms take the form shown in Table 3.

Table 4 compares key features of the production spectra under fission and fusion conditions. Fig. 1 shows the interstitial cluster spectrum under representative fission and fusion irradiation conditions in iron, while Fig. 2 displays the production spectrum for vacancy clusters under fission and fusion irradiation conditions. The principal effects of the harder fusion spectrum are: (1) a larger fraction of surviving defects born in clusters; (2) a shallower power-law exponent shifting the distribution toward larger clusters; and (3) extension of the maximum cluster size due to higher-energy cascades and subcascade overlap De Backer et al (2018); Zarkadoula et al (2013). The enhanced helium production under fusion conditions ($\sim$10 appm He/dpa vs. 0.5–1 for fission) does not affect the primary cascade spectrum but strongly influences subsequent vacancy cluster (void/bubble) evolution by providing heterogeneous nucleation sites Malerba et al (2021).

Literature Sources

The production fractions presented here are *reconstructed* from published MD cascade data, since the literature generally does not report ϵ_m in the tabulated form required by rate-theory codes. The principal sources and their contributions are:

- **?**: earliest computer simulation of vacancy and interstitial cluster production in α-Fe; historical baseline.

Table 3 Rate-theory source-term structure. G is the actual (survival-corrected) defect production rate. The coefficients $\epsilon_m^{(i)}$ and $\epsilon_n^{(v)}$ are read from Tables **??**–**??** according to the neutron spectrum.

Species index α	**Production rate** P_α
1 (mono-SIA)	$P_1^{(i)} = G\left(1 - \sum_{m=2}^{m_1} m\,\epsilon_m^{(i)}\right)$
2	$P_2^{(i)} = \epsilon_2^{(i)}\,G$
$\vdots$	$\vdots$
m_1	$P_{m_1}^{(i)} = \epsilon_{m_1}^{(i)}\,G$
$N{+}1$ (mono-vacancy)	$P_1^{(v)} = G\left(1 - \sum_{n=2}^{n_1} n\,\epsilon_n^{(v)}\right)$
$N{+}2$	$P_2^{(v)} = \epsilon_2^{(v)}\,G$
$\vdots$	$\vdots$
$N{+}n_1$	$P_{n_1}^{(v)} = \epsilon_{n_1}^{(v)}\,G$

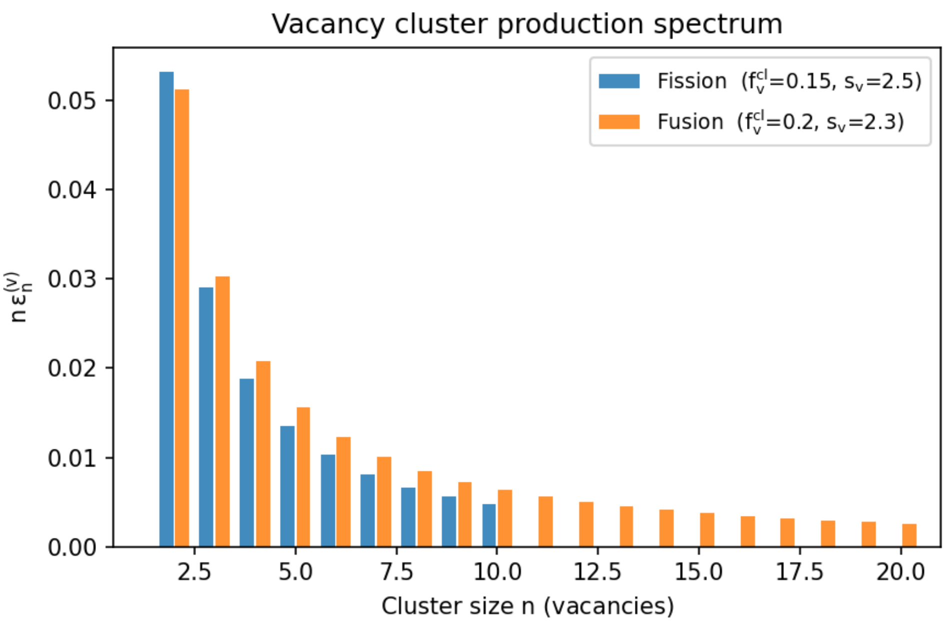

Fig. 2 Vacancy cluster production spectrum under fission and fusion conditions

Table 4 Comparison of defect production characteristics under fission and fusion neutron spectra in α-Fe.

Quantity	Fission	Fusion
Mono-SIA fraction, $1 - f_i^{\mathrm{cl}}$	0.42	0.35
Mono-vacancy fraction, $1 - f_v^{\mathrm{cl}}$	0.85	0.80
Mean SIA cluster size $\langle m \rangle_{\mathrm{cl}}$	4.4	6.3
Mean vacancy cluster size $\langle n \rangle_{\mathrm{cl}}$	2.6	3.1
Largest SIA cluster m_1	20	50
Largest vacancy cluster n_1	10	20
He co-production (appm He/dpa)	0.5–1	$\sim$10

- Calder and Bacon (1993): early MD study establishing qualitative cascade morphology and survival fractions in Fe.
- Stoller et al (1997): comprehensive MD + MC characterization up to 40 keV; reports interstitial clustering fractions and vacancy-site correlation.
- Stoller (1998): key reference for cluster size distributions at 100, 600, 900 K up to 50 keV; primary source for size-resolved interstitial and vacancy spectra.
- Stoller (2000): broadest energy–temperature survey of clustering fractions and size distributions; principal source for the energy dependence of f_i^{cl} and f_v^{cl}.
- Bacon et al (2000): comparative study across crystal structures confirming bcc Fe as having strong in-cascade SIA clustering.
- Gao et al (2000): statistics on sessile SIA clusters; relevant for distinguishing glissile ($\langle 111 \rangle$) vs. sessile ($\langle 100 \rangle$) cluster populations.
- Soneda et al (2001): vacancy loop formation by cascade collapse at 50 keV; informs the large-n tail of the vacancy spectrum.
- Terentyev et al (2006): demonstrates interatomic-potential dependence of predicted cluster fractions; provides uncertainty bounds.
- Calder et al (2010): shock-wave mechanism for large SIA clusters at cascade periphery; justifies the power-law tail for large m.
- Souidi et al (2011): MD/OKMC linkage showing which primary-damage features most strongly affect long-term nanostructure.
- Björkas et al (2012): demonstrates that picosecond cluster distributions influence long-term damage accumulation.
- Zarkadoula et al (2013): high-energy cascade morphology and large-cluster statistics relevant to fusion conditions.
- De Backer et al (2018): subcascade fragmentation model with power-law scaling of cluster sizes; enables extrapolation to high energies.
- Nordlund et al (2018): comprehensive review establishing the arc-dpa framework and survival efficiency parameterization.
- Malerba et al (2021): best-practice parameter compilation for cluster-dynamics modeling of pure Fe.

Caveats and Uncertainty

Several sources of uncertainty should be kept in mind:

1. **Interatomic potential:** Predicted cluster fractions depend on the empirical potential used in MD. Terentyev et al (2006) found that clustering parameters can differ by more than 100% between potentials, even when Frenkel-pair yields are similar.
2. **Temperature dependence:** The values here are representative of reactor temperatures (300–600 K). Modest temperature effects are observed: interstitial clustering increases slightly, and vacancy clustering decreases with increasing temperature Stoller (2000).
3. **power law idealization:** The true size distribution may deviate from a pure power law, particularly at small sizes (where specific magic numbers may introduce structure) and at very large sizes (where subcascade fragmentation alters statistics) De Backer et al (2018).
4. **Vacancy cluster character:** In bcc Fe, vacancy "clusters" at cascade time scales are often loose, correlated arrangements of vacancies rather than compact loops. Short-range vacancy diffusion can subsequently convert these into true clusters Stoller (1998); Soneda et al (2001).
5. **Spectrum averaging:** The values given here use representative average PKA energies; a rigorous treatment would convolve the energy-dependent $\epsilon_m(E_{\text{PKA}})$ over the full PKA recoil spectrum for the specific reactor type Nordlund et al (2018).

2.2 EUROFER Model Parameters: Defect Production

Table 5 collects the cascade-damage production parameters for use in cluster dynamics modelling of EUROFER97. EUROFER is a ferritic matrix, and primary cascade damage is assumed identical to that of pure α-Fe for the displacement cascade regime. The parameters distinguish fission and fusion neutron spectra following the synthesis of **????**.

The production fraction for a cluster of size m is computed as:

$$\epsilon_m^{(i)} = C_i\, m^{-s_i}, \qquad \epsilon_n^{(v)} = C_v\, n^{-s_v}, \tag{13}$$

with the mono-defect production rate $P_1 = G(1 - f^{\text{cl}})$.

3 Defect Energetics

3.1 Formation, Migration, and Binding Energies

The table below compiles point-defect energetics, elastic constants, and diffusion parameters for body-centered cubic (bcc) α-iron, as used in rate-theory and cluster-dynamics models of radiation damage. Values are from density functional theory (DFT) calculations unless stated otherwise; experimental values are included where DFT and experiment are compared in the literature. The Burgers vector magnitude for $\frac{1}{2}\langle 111\rangle$ loops uses the experimental lattice parameter $a_0 = 0.2867$ nm.

Table 5 Defect production input parameters for EUROFER cluster dynamics.

Parameter	Symbol	Fission	Fusion	Units
Survival and dose rate:				
Survival efficiency (spectrum-avg.)	η	0.30	0.28	—
Displacement rate	$\dot{\phi}$	user input	user input	dpa/s
He production rate	G_{He}/G	0.5–1	∼10	appm He/dpa
Interstitial cluster production:				
SIA clustering fraction	f_i^{cl}	0.58	0.65	—
SIA power-law exponent	s_i	1.6	1.5	—
Max SIA cluster size	m_1	20	50	SIAs
SIA normalization constant	C_i	0.1093	0.0553	—
Vacancy cluster production:				
Vacancy clustering fraction	f_v^{cl}	0.15	0.20	—
Vacancy power-law exponent	s_v	2.5	2.3	—
Max vacancy cluster size	n_1	10	20	vacancies
Vacancy normalization constant	C_v	0.1506	0.1296	—

Table 6: Representative defect and material parameters for bcc Fe.

Parameter	Value (range)	Units	References / Notes
Vacancy formation energy, $E_{\mathrm{v}}^{\mathrm{f}}$	1.73–2.15 (DFT)		
1.4 ± 0.1 (expt.)	eV	DFT: Fu et al (2004); Domain and Becquart (2001); Muzyk et al (2011). Experimental (positron annihilation): Kim and Buyers (1978); Vehanen et al (1982). A commonly adopted DFT value is 2.0 eV Bonny et al (2011); Was (2017).	
Self-interstitial atom formation energy, $E_{\mathrm{i}}^{\mathrm{f}}$	3.53–4.06 (DFT)	eV	The ⟨110⟩ dumbbell is the ground state in magnetic bcc Fe, with formation energy ≈ 3.64 eV Fu et al (2004); Domain and Becquart (2001); ?. The ⟨111⟩ crowdion gives 3.68–4.06 eV Willaime et al (2005).
Vacancy migration energy, $E_{\mathrm{v}}^{\mathrm{m}}$	0.55–0.67 (DFT)		

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Parameter	Value (range)	Units	References / Notes
0.55 (expt.)	eV	DFT: 0.62 eV Fu et al (2004); 0.55 eV Domain and Becquart (2001). Experimental resistivity-recovery measurements give ≈ 0.55 eV Vehanen et al (1982); Seeger (1998). This value is widely used in rate-theory models Was (2017).	
SIA migration energy, $E_{\mathrm{i}}^{\mathrm{m}}$	0.013–0.34 (DFT)	eV	For the $\langle 110\rangle$ translation–rotation mechanism, $E_{\mathrm{i}}^{\mathrm{m}} \approx 0.34$ eV Fu et al (2004); Domain and Becquart (2001). For the $\langle 111\rangle$ crowdion, the barrier is ≈ 0.013 eV **?**. Experimentally inferred values are about 0.30 ± 0.05 eV Seeger (1998); Was (2017).
Divacancy binding energy, $E_{2\mathrm{v}}^{\mathrm{B}}$	0.20–0.30 (DFT)		
0.22 ± 0.01 (DFT)	eV	A widely cited value is 0.22 ± 0.01 eV for the 1NN configuration Fu et al (2004); Domain and Becquart (2001). Positive binding confirms a metastable divacancy cluster Was (2017).	
Divacancy migration energy, $E_{2\mathrm{v}}^{\mathrm{m}}$	0.62–0.90 (DFT)	eV	Often estimated as $E_{2\mathrm{v}}^{\mathrm{m}} \approx E_{\mathrm{v}}^{\mathrm{m}} + \Delta$, where Δ accounts for binding. DFT gives about 0.90 eV for the 1NN divacancy Domain and Becquart (2001); Fu et al (2004). Ghoniem and Cho Ghoniem and Cho (1983) use $E_{2\mathrm{v}}^{\mathrm{m}}$ for migration of the complex.
Di-interstitial migration energy, $E_{2\mathrm{i}}^{\mathrm{m}}$	≈ 0.10–0.20 (DFT)	eV	The di-SIA is highly mobile. A stable 2-SIA configuration in α-Fe consists of two non-parallel $\langle 110\rangle$ dumbbells. DFT gives $E_{2\mathrm{i}}^{\mathrm{m}} \approx 0.10$–0.20 eV Fu et al (2004); Willaime et al (2005).
Di-interstitial binding energy, $E_{2\mathrm{i}}^{\mathrm{B}}$	0.60–1.0 (DFT)	eV	This cluster is strongly bound, with DFT values of about 0.60–1.0 eV depending on configuration Fu et al (2004); Willaime et al (2005). For two non-parallel $\langle 110\rangle$ dumbbells, a representative value is ≈ 0.80 eV Domain and Becquart (2001); Fu et al (2004).
Isotropic shear modulus, μ	80–82 (RT, expt.)	GPa	For polycrystalline iron, the Voigt–Reuss–Hill average gives $\mu \approx 80$–82 GPa at 300 K Simmons and Wang (1971); Ledbetter and Reed (1973); Dever (1972). It decreases to about 64 GPa at 600 °C Was (2017).
Burgers vector magnitude for $\frac{1}{2}\langle 111\rangle$ loops, b	0.2482	nm	$b = (\sqrt{3}/2)\,a_0$ with $a_0 = 0.2867$ nm from Simmons and Wang (1971). For $\langle 100\rangle$ loops, $b = a_0 = 0.2867$ nm.

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Parameter	Value (range)	Units	References / Notes
Stacking-fault energy (vacancy loops / Frank loops), γ_{SF}	200–400 (DFT)	$\mathrm{mJ\,m^{-2}}$	bcc Fe does not have stable intrinsic stacking faults on {111} in the fcc sense. The fault energy associated with vacancy-type Frank loops on {110} is estimated at about 200–400 $\mathrm{mJ\,m^{-2}}$; however, loops in bcc Fe are typically perfect rather than faulted Was (2017); Ackland et al (2004).
Surface energy (void / free surface), γ_s	2.0–2.5 (DFT)		
$\approx 2.4 \pm 0.2$ (expt.)	$\mathrm{J\,m^{-2}}$	DFT gives $\gamma_{110} \approx$ 2.43 $\mathrm{J\,m^{-2}}$ and $\gamma_{100} \approx$ 2.50 $\mathrm{J\,m^{-2}}$ Galanakis et al (2002); Lazar et al (2015). Experimental averages are near 2.4 $\mathrm{J\,m^{-2}}$ Tyson and Miller (1977).	
Pre-exponential factor for vacancy self-diffusion, D_v^0	1.9–2.3 $\times 10^3$ (expt.)		
$\approx 5 \times 10^{-4}$ (DFT-based)	$\mathrm{cm^2\,s^{-1}}$	Experimental fits give $D_\alpha = 2.3 \times 10^3 \exp(-Q/RT)$ $\mathrm{cm^2\,s^{-1}}$ with $Q \approx$ 251 $\mathrm{kJ\,mol^{-1}}$ Birchenall and Mehl (1950); Buffington et al (1961); James and Leak (1966). The much smaller DFT-based value corresponds to the migration prefactor only Was (2017).	
Pre-exponential factor for SIA diffusion, D_i^0	$\approx 10^{-4}$–10^{-3}	$\mathrm{cm^2\,s^{-1}}$	Estimated from $D_i^0 = fz\nu a^2$, with attempt frequency $\nu \approx 6 \times 10^{12}$ $\mathrm{s^{-1}}$ and jump distance $a \approx 0.29$ nm Fu et al (2004); Willaime et al (2005); Was (2017). The $\langle 111 \rangle$ crowdion mode gives a larger effective prefactor because of its very low barrier ?.

The large experimental value $D_v^0 \sim 10^3$ $\mathrm{cm^2\,s^{-1}}$ arises because the measured activation energy $Q = E_v^f + E_v^m \approx 2.6$ eV accounts for both the vacancy formation and migration enthalpies, and the large entropy of formation $\Delta S_v^f \approx$ 2–5 k_B inflates D^0. In rate-theory codes, the vacancy diffusion coefficient is often split as $D_v = a^2 \nu \exp(-E_v^m/k_B T)$, with $D^0_{v,\mathrm{migration}} \approx 5 \times 10^{-4}$ $\mathrm{cm^2\,s^{-1}}$, and the equilibrium vacancy concentration is handled separately Was (2017); Ghoniem and Cho (1983). The Debye (cut-off) frequency of perfect bcc Fe is the baseline for all defect attempt frequencies. With the experimental Debye temperature $\Theta_D \approx$ 470–476 K Minkiewicz et al (1967); Loong et al (1987):

$$\nu_D = \frac{k_B \Theta_D}{h} \approx 9.8\text{–}10 \text{ THz} \approx 6.5 \times 10^{13} \text{ s}^{-1}. \tag{14}$$

The phonon density of states (pDOS) of bcc Fe has two prominent peaks: a transverse peak near ~ 7 meV (~ 1.7 THz) and a longitudinal peak at ~ 37 meV (~ 8.9 THz) Minkiewicz et al (1967); Ackland et al (2004).

In Vineyard's harmonic transition-state theory Vineyard (1957) the jump rate of a point defect is

$$\Gamma = \nu_0 \exp\left(-\frac{E_m}{k_B T}\right), \qquad \nu_0 = \frac{\prod_{i=1}^{3N} \nu_i^{(0)}}{\prod_{i=1}^{3N-1} \nu_i^{(\mathrm{sp})}}, \tag{15}$$

where $\nu_i^{(0)}$ and $\nu_i^{(\mathrm{sp})}$ are the real normal-mode frequencies at the stable minimum and the saddle point, respectively. Table 7 summarizes the key values for the three defect species.

Table 7 Migration energies, attempt frequencies, and local vibrational modes for the vacancy, self-interstitial atom (SIA), and interstitial helium in bcc α-Fe. *T-R* = translation-rotation mechanism; *loc.* = localised mode above the bulk phonon band.

Defect	E_m **(eV)**	**Attempt freq.** ν_0 **(s^{-1})**	**Local mode (meV / THz)**	**References**
Vacancy	0.55–0.62	6×10^{12}–6×10^{13}	No gap mode; P_L softened at 37 meV	Vineyard (1957); Fu et al (2004); Domain and Becquart (2001); Vehanen et al (1982); Soisson and Fu (2007)
SIA ($\langle 110 \rangle$ dumbbell)	0.34 (T-R)	$\sim 4 \times 10^{12}$	Dumbbell stretch: 60–80 meV (14–19 THz)	Fu et al (2004); Domain and Becquart (2001); Willaime et al (2005); Derlet et al (2007); Terentyev et al (2008)
He (tetrahedral site)	0.06–0.09	$\sim 10^{12}$–3×10^{12}	Rattling mode: $\sim$50–80 meV ($\sim$12–19 THz)	**??**Ghoniem and Cho (1983)

A monovacancy introduces no localized modes *above* the Fe phonon band because it is a missing atom: force constants are reduced, not added. The main spectral signature is a softening of the longitudinal phonon peak P_L at ~ 37 meV, proportional to vacancy concentration, with the perturbation confined to the first-neighbor shell **?**. The attempt frequency spans a wide range in the literature. The Debye-based estimate $\nu_0^v \approx 6 \times 10^{12}$ s^{-1} is routinely used in rate-theory codes Ghoniem and Cho (1983); Was (2017). Full DFT Vineyard calculations give $\sim 5 \times 10^{13}$ s^{-1} Fu et al (2004), while values as large as 5×10^{15} s^{-1} are obtained from fitting self-diffusion data Soisson and

Fu (2007) because the experimental pre-exponential D_0 folds in the vacancy formation entropy $\Delta S_f^v \approx 4\,k_B$ Lucas and Schäublin (2009).

The ground-state SIA in ferromagnetic bcc Fe is the $\langle 110 \rangle$ dumbbell (unique among bcc metals owing to magnetism) **?**. Its translation-rotation migration has $E_m = 0.34$ eV and $\nu_0 \approx 4 \times 10^{12}$ s^{-1} from DFT Fu et al (2004); Domain and Becquart (2001). The $\langle 111 \rangle$ crowdion has a negligible barrier ($\sim$0.013 eV) and migrates nearly athermally **?**.

The $\langle 110 \rangle$ dumbbell creates a *high-frequency localized mode* above the bulk pDOS cut-off: the two dumbbell atoms oscillate against each other along the $\langle 110 \rangle$ bond direction, giving a stretch frequency of $\approx$ 60–80 meV (14–19 THz) Domain and Becquart (2001); Willaime et al (2005). This stiffened mode produces a large negative vibrational formation entropy for the SIA Derlet et al (2007). Non-parallel $\langle 110 \rangle$ di-SIA clusters are further stabilized at elevated temperatures by their large positive vibrational formation entropy Terentyev et al (2008).

Interstitial He occupies the tetrahedral site in bcc Fe, which is more stable than the octahedral site by 0.2 eV (DFT) **?**. He migrates via tetrahedral-to-tetrahedral hops through the octahedral saddle with a very low barrier of $E_m^{He} = 0.06$ eV **?** (0.078–0.086 eV from MD and other DFT methods Morishita et al (2003); **?**), making He diffusion essentially athermal above room temperature. Because $m_{He}/m_{Fe} \approx 0.07$, He rattles inside the tetrahedral cage at a frequency far above the host phonon band:

$$\nu_{He}^{\mathrm{loc}} \approx \sqrt{\frac{k_{\mathrm{eff}}}{m_{He}}} \approx 50\text{–}80 \text{ meV } (\approx 12\text{–}19 \text{ THz}). \tag{16}$$

This localized optical mode has been confirmed by DFT phonon calculations **??**. By analogy with the H optical modes ($\sim$ 70–130 meV, Debye temperature $\Theta_D^H \approx 900$ K **?**), the He localized frequency is lower because $m_{He} > m_H$. The ordering of attempt frequencies mirrors the Debye frequency: $\nu_0^{He} \lesssim \nu_0^{SIA} \lesssim \nu_0^V \sim \nu_D$. All lie within roughly an order of magnitude of 6×10^{12} s^{-1} , and the Debye frequency is an adequate approximation when full Vineyard calculations are unavailable. The strongly differing migration energies — He (< 0.1 eV) $\ll$ SIA (0.34 eV) $\ll$ V (0.55–0.62 eV) — dominate the diffusivity hierarchy over many orders of magnitude.

3.2 EUROFER Model Parameters: Defect Energetics

Table 8 collects the point-defect energetics and material constants for bcc Fe / EUROFER used throughout the rate equations.

Table 8 Defect energetics input parameters for EUROFER cluster dynamics.

Parameter	Symbol	Value	Units
Lattice and elastic constants:			
Lattice constant	a	0.2867	nm
Atomic volume	$\Omega = a^3/2$	1.18×10^{-29}	m^3
Burgers vector ($\frac{1}{2}\langle 111\rangle$)	b_{111}	0.2482	nm
Burgers vector ($\langle 100\rangle$)	b_{100}	0.2867	nm
Shear modulus (RT)	μ	82	GPa
Poisson ratio	ν	0.29	—
Surface energy	γ_s	2.0	J/m^2
Vacancy energetics:			
Formation energy	E_f^v	2.0	eV
Migration energy	E_m^v	0.67	eV
Divacancy binding energy (2NN)	E_B^{2v}	0.22	eV
Divacancy migration energy	E_m^{2v}	0.70	eV
SIA energetics:			
Formation energy ($\langle 110\rangle$ dumbbell)	E_f^i	3.64	eV
Migration energy ($\langle 110\rangle$ T-R)	E_m^i	0.34	eV
Di-SIA binding energy	E_B^{2i}	0.80	eV
Di-SIA migration energy	E_m^{2i}	0.42	eV
Helium energetics:			
He migration energy (tetrahedral)	E_m^h	0.06	eV
He–vacancy binding (1st He)	$E_b^{h\text{-}V}$	2.30	eV
He–vacancy binding (2nd He)	$E_b^{h\text{-}V_2}$	2.00	eV
Attempt frequencies:			
Vacancy	ν_v	1.0×10^{13}	s^{-1}
SIA ($\langle 110\rangle$)	ν_i	1.0×10^{13}	s^{-1}
He interstitial	ν_h	3.0×10^{12}	s^{-1}
He equation of state (virial coefficients, 600–1000 K):			
Second virial coefficient	B_2	1.67×10^{-29}	m^3/atom
Third virial coefficient	B_3	1.84×10^{-58}	m^6/atom^2

4 Defect Diffusion in a BCC Lattice

Radiation induced defects, and the clusters they form, govern the long-term microstructural evolution of structural materials in nuclear environments (**?**Golubov et al 2000). The diffusion properties of these defects—including their migration energies, attempt frequencies, and dimensionality of motion—are essential input parameters for rate theory and kinetic Monte Carlo models of radiation damage (Trinkaus et al 2002; Heinisch et al 2000). In bcc α-iron, the lattice constant is $a = 2.87$ Å, and diffusion proceeds by thermally activated jumps between lattice sites. The general Arrhenius form for the diffusion coefficient is

$$D = D_0 \exp\left(-\frac{E_m}{k_B T}\right), \tag{17}$$

where D_0 is the pre-exponential factor, E_m is the migration energy, k_B is Boltzmann's constant, and T is the absolute temperature.

4.1 Random Walk Diffusion Coefficient

The diffusion coefficient for a random walk in 3D is:

$$D = \frac{1}{6} z \lambda^2 \nu_0 \exp(-E_m/k_B T) \tag{18}$$

where z is the number of equivalent jump directions, λ is the jump distance, and the factor $\frac{1}{6}$ comes from the 3D random walk. For a bcc lattice with $\langle 111 \rangle$ nearest-neighbor jumps, $z = 8$, and $\lambda = a\sqrt{3}/2$. If the lattice constant is (a), then the diffusion coefficients of interstitials, vacancies, and helium can be written as

$$\begin{aligned}
D_i &= a^2 \nu_i \exp(-E_m^i/k_B T) && \mathrm{m^2\,s^{-1}}, && (19)\\
D_v &= a^2 \nu_v \exp(-E_m^v/k_B T) && \mathrm{m^2\,s^{-1}}, && (20)\\
D_h &= a^2 \nu_h \exp(-E_m^h/k_B T) && \mathrm{m^2\,s^{-1}} && (21)
\end{aligned}$$

We adopt here 3 basic frequencies in the defect clustering system: ω_i is the frequency of the self-interstitial reaction, ω_h is the frequency of the helium gas reaction, and ω_v is the frequency of the vacancy reaction. These are given by:

$$\begin{aligned}
\omega_i &= \nu_i \exp(-E_m^i/k_B T) = D_i/a^2, && \mathrm{s^{-1}}, && (22)\\
\omega_h &= \nu_h \exp(-E_m^h/k_B T) = D_h/a^2, && \mathrm{s^{-1}}, && (23)\\
\omega_v &= \nu_v \exp(-E_m^v/k_B T) = D_v/a^2, && \mathrm{s^{-1}}. && (24)
\end{aligned}$$

A plot of these basic frequencies as functions of temperature is shown in Fig. 3.

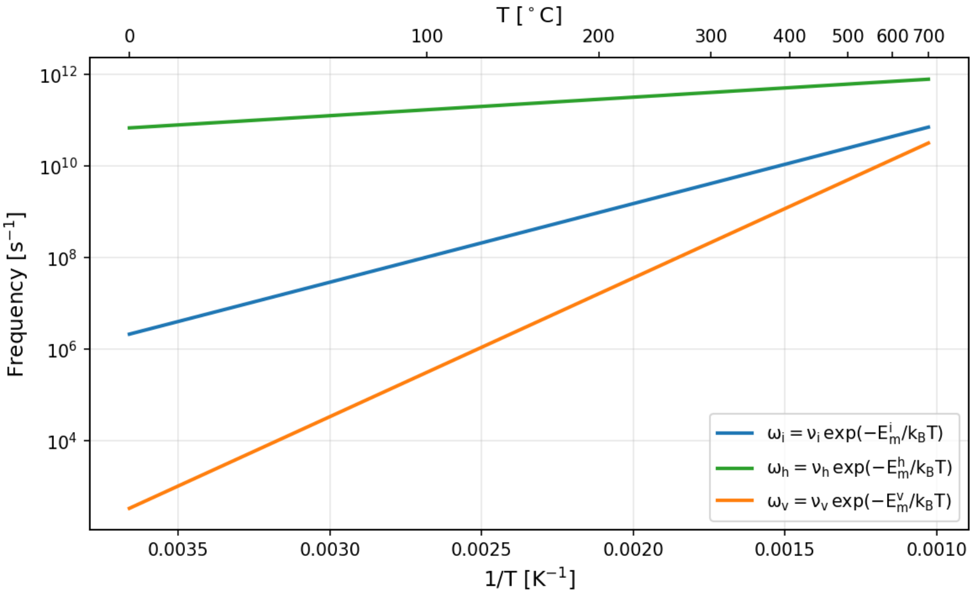


Fig. 3 Basic mobile defect frequencies in iron

4.2 Vacancy Diffusion

Migration Mechanism In bcc iron, vacancy migration proceeds by nearest-neighbor jumps along $\langle 111 \rangle$ directions. Each vacancy has $z = 8$ nearest-neighbor sites to which a neighboring atom can jump, effectively moving the vacancy. The vacancy executes a 3D random walk at all temperatures of interest.

The migration saddle point lies midway between two adjacent bcc lattice sites. DFT calculations consistently predict a vacancy migration energy of $E_m^v \approx 0.65$–0.70 eV (Domain and Becquart 2001; Fu et al 2004). Experimental values from resistivity recovery (Stage III) and positron annihilation measurements typically yield $E_m^v \approx 0.55$–0.67 eV (Seeger 1998; Ullmaier 1991). The vacancy formation energy is $E_f^v \approx 2.0$–2.2 eV, and the activation energy for self-diffusion is $Q_{SD} = E_f^v + E_m^v \approx 2.7$–$2.9$ eV, consistent with tracer diffusion experiments.

Diffusion Coefficient Using the bcc random-walk expression (Eq. 20), the vacancy diffusion coefficient takes the form

$$D_v = a^2 \nu_v \exp\left(-\frac{E_m^v}{k_B T}\right), \tag{25}$$

with the pre-exponential factor $D_0^v = a^2 \nu_v$.

Recommended values for pure α-Fe:

$$\boxed{D_v = 8.2 \times 10^{-8} \exp\left(-\frac{0.67 \text{ eV}}{k_B T}\right) \quad \text{m}^2\,\text{s}^{-1}} \tag{26}$$

corresponding to an attempt frequency $\nu_v \approx 1.0 \times 10^{13}$ s^{-1} and $a = 2.87$ Å.

The vacancy reaction frequency is then

$$\omega_v = \frac{D_v}{a^2} = 1.0 \times 10^{13} \exp\left(-\frac{0.67 \text{ eV}}{k_B T}\right) \quad \text{s}^{-1}. \tag{27}$$

4.3 Self-Interstitial Atom (SIA) Diffusion

4.3.1 Migration Mechanism in Pure α-Fe

In ferromagnetic bcc iron, the most stable SIA configuration is the $\langle 110\rangle$ dumbbell, lying 0.7 eV below the $\langle 111\rangle$ crowdion that is the ground state in all non-magnetic bcc transition metals (Fu et al 2004; Domain and Becquart 2001; Nguyen-Manh et al 2006). This stabilization is magnetic in origin. The SIA migrates by a nearest-neighbor translation-rotation mechanism with a DFT-calculated barrier of $E_m^i = 0.34$ eV (Fu et al 2004; Chiesa et al 2009), consistent with the experimental Stage I_E resistivity recovery value of 0.30–0.34 eV at $\sim$120 K (Ullmaier 1991). Because the $\langle 110\rangle$ dumbbell can reorient into any of its six equivalent orientations upon each jump, single-SIA migration is inherently **three-dimensional**—in contrast to the quasi-1D athermal crowdion glide ($E_m \sim 0.01$–0.05 eV) found in non-magnetic bcc metals (Derlet et al 2007; Fitzgerald and Nguyen-Manh 2008).

The diffusion coefficient and reaction frequency for the single SIA in pure α-Fe are:

$$D_i = a^2 \nu_i \exp\left(-\frac{E_m^i}{k_B T}\right) = 8.2 \times 10^{-8} \exp\left(-\frac{0.34 \text{ eV}}{k_B T}\right) \quad \text{m}^2\,\text{s}^{-1}, \tag{28}$$

$$\omega_i = \frac{D_i}{a^2} = 1.0 \times 10^{13} \exp\left(-\frac{0.34 \text{ eV}}{k_B T}\right) \quad \text{s}^{-1}, \tag{29}$$

with $\nu_i \approx 1.0 \times 10^{13}$ s^{-1} and $a = 2.87$ Å.

4.3.2 Effective SIA Diffusion in EUROFER

The elementary $\langle 110\rangle$ dumbbell mechanism carries over to the ferritic matrix of EUROFER97, but the effective SIA mobility is reduced by interactions with the alloy's solute population. EUROFER is a reduced-activation 9Cr ferritic/martensitic steel whose principal alloying additions include Cr, W, Mn, V, Ta, C, and N (van der Schaaf et al 2003; Aktaa et al 2020). After heat treatment, part of the C, N, V, and Ta partitions into MX carbonitrides and $M_{23}C_6$ carbides, so the relevant quantity is the *dissolved* matrix solute content rather than the nominal composition.

Two classes of solute–SIA interaction are important:

1. **Substitutional solutes (Cr, Mn, W, V, Ta).** These form mixed solute–Fe dumbbells and bound interstitial complexes that create a fluctuating migration landscape of local binding, detrapping, and solute-biased reorientation events (Olsson et al 2007, 2010; Vincent et al 2006; Messina et al 2020).
2. **Interstitial solutes (C, N).** Carbon and nitrogen occupy octahedral sites and bind strongly to SIAs and small SIA clusters, acting as efficient traps that shorten the free-flight length and raise the apparent migration energy (Domain et al 2004; Becquart et al 2005; Terentyev et al 2011; Theodorou et al 2022).

Both effects are captured by writing the effective SIA diffusivity in the trap-limited form:

$$\boxed{D_i^{\text{eff}} = \frac{D_i^{\text{Fe}}}{1 + \sum_s K_s(T)\, c_s}\,,} \tag{30}$$

where D_i^{Fe} is the pure-iron value (Eq. 28), c_s is the atomic fraction of solute or trap class s, and $K_s(T)$ is an effective trapping coefficient related to the SIA–solute binding free energy. The dominant trap classes in EUROFER are dissolved C and N, Cr-rich local environments, and carbide/carbonitride interfaces.

Modeling strategy.

For cluster dynamics, we recommend a layered approach:

1. Use the pure-Fe mechanism and attempt frequency as the microscopic baseline.
2. Replace D_i by D_i^{eff} (Eq. 30) with C and N as explicit trapping species where their dissolved concentrations are known.
3. Absorb remaining substitutional-solute effects (Cr, W, V, Ta, Mn) into the same effective coefficient, or—in higher-fidelity models—treat them through separate interstitial–solute complexes and radiation-induced segregation terms.

No single Arrhenius pair (D_0, E_m) adequately represents SIA diffusion in EUROFER across all conditions. The physically sound approach is to use the pure-Fe mobility as the reference and to introduce alloy-specific corrections through Eq. (30).

4.4 Interstitial Cluster Diffusion

4.4.1 Size-Dependent Migration Mechanisms

The migration mechanism of interstitial clusters in bcc iron evolves with cluster size, transitioning from 3D to 1D as the cluster grows:

Size n	Configuration	Mechanism	E_m (eV)
1	$\langle 110\rangle$ dumbbell	3D translation-rotation	0.34
2	parallel $\langle 110\rangle$ dumbbells	3D translation-rotation	0.42
3	mixed	3D/1D transition	0.15–0.20
4	$\frac{1}{2}\langle 111\rangle$ loop	predominantly 1D glide	0.06–0.10
≥ 5	$\frac{1}{2}\langle 111\rangle$ loop	1D glide	$\lesssim 0.05$

For $n \geq 4$, clusters adopt the form of prismatic dislocation loops with Burgers vector $\mathbf{b} = \frac{1}{2}\langle 111\rangle$ and migrate by thermally activated 1D glide along their Burgers-vector direction (Osetsky et al 2003; Arakawa et al 2007, 2016). The 1D migration energy is essentially independent of cluster size for $n \gtrsim 4$ (Osetsky et al 2003), with $E_m^{1D} \lesssim 0.05$ eV, so that $E_m^{1D} \leq k_B T$ above room temperature and the Arrhenius factor is close to unity. The di-interstitial ($n = 2$), consisting of two parallel $\langle 110\rangle$ dumbbells, is *less* mobile than the single SIA ($E_m = 0.42$ vs. 0.34 eV) (Fu et al 2004), while clusters with $n = 3$–4 lie in the crossover regime (Terentyev et al 2007).

4.4.2 1D Diffusion Coefficient

The 1D diffusion coefficient for a cluster of n SIAs gliding along $\langle 111\rangle$ is:

$$D_n^{1D} = \frac{1}{2}\lambda_{\langle 111\rangle}^2 \nu_n \exp\left(-\frac{E_m^{1D}}{k_B T}\right), \tag{31}$$

where $\lambda_{\langle 111\rangle} = a\sqrt{3}$ is the jump distance and the factor $\frac{1}{2}$ reflects the two-direction 1D random walk. MD simulations show that the attempt frequency scales inversely with size (Osetsky et al 2003; Anento et al 2010):

$$\nu_n = \frac{\nu_0}{n^s}, \qquad \nu_0 \sim 6 \times 10^{12}\ \mathrm{s}^{-1}, \quad s \approx 0.5\text{–}1.0, \tag{32}$$

giving the recommended expression:

$$\boxed{D_n^{1D} = \frac{3a^2\nu_0}{2\,n^s} \exp\left(-\frac{E_m^{1D}}{k_B T}\right) \quad \mathrm{m}^2\,\mathrm{s}^{-1}} \tag{33}$$

4.4.3 Mixed 1D/3D Framework

Gliding SIA clusters periodically change their Burgers-vector direction—by thermal activation, by interaction with solutes, or by encounters with other defects (Heinisch et al 2000). The mean free path L between direction changes controls the effective dimensionality of the transport. If L is finite, the effective 3D diffusion coefficient is (Trinkaus et al 2002; Heinisch et al 2000):

$$D_n^{3D,\mathrm{eff}} = \frac{D_n^{1D}}{3} \cdot \frac{L}{\lambda_{\langle 111\rangle}}, \tag{34}$$

where the factor $\frac{1}{3}$ projects 1D motion onto three dimensions. Equivalently, if ν_R is the Burgers-vector rotation frequency ($L = D_n^{1D}/\nu_R$):

$$D_n^{3D,\mathrm{eff}} = \frac{(D_n^{1D})^2}{6\,\nu_R\,\lambda_{\langle 111\rangle}^2}. \tag{35}$$

4.4.4 Sink Reaction Rates: 3D vs. 1D

The effective 3D reaction rate with spherical sinks of radius R and density C_s is:

$$k_n^{3D} = 4\pi R\, D_n^{3D,\mathrm{eff}}\, C_s. \tag{36}$$

In the pure 1D limit ($L \to \infty$), the 1D sink-strength formalism gives (Trinkaus et al 2002; Golubov et al 2000):

$$k_n^{1D} = 2\,(\pi R^2 C_s)^2\, D_n^{1D}. \tag{37}$$

The critical physical distinction is that the 1D absorption rate scales as $R^4 C_s^2$ (square of the geometric cross-section and second-order in sink density), while the 3D sink strength scales as $C_s R$ (first-order in both capture radius and sink density). This asymmetry between SIA-cluster and point-defect absorption by sinks is the physical origin of the production bias that drives void swelling under cascade-producing irradiation.

4.4.5 Solute Effects on Cluster Transport in EUROFER

In EUROFER, the elementary 1D glide barrier remains small, but solute interactions modify cluster transport in two coupled ways:

1. **Reduced 1D mobility.** Dissolved C, N, and their vacancy complexes trap and pin glissile $\frac{1}{2}\langle 111\rangle$ loops, lowering the effective 1D diffusivity (**?**Anento and Serra 2013):
$$D_{n,\mathrm{eff}}^{1D} = \frac{D_n^{1D,\mathrm{Fe}}}{1 + \sum_s K_{n,s}(T)\, c_s}\,, \tag{38}$$
where $K_{n,s}(T)$ is the trapping factor for cluster size n with solute class s.
2. **Shortened mean free path.** Substitutional solutes (Cr, W, Mn, V, Ta) promote Burgers-vector changes through local pinning and drag, increasing ν_R and thereby reducing L:
$$L_{\mathrm{eff}} = \frac{D_{n,\mathrm{eff}}^{1D}}{\nu_{R,\mathrm{eff}}}. \tag{39}$$

Solute effects therefore compound: they lower D_n^{1D} *and* shorten L, pushing cluster transport further from the ideal long-range 1D limit toward 3D-like behaviour. The effective 3D coefficient in the alloy is obtained by substituting the alloy quantities into the mixed 1D/3D expression:

$$D_n^{3D,\text{eff},\text{alloy}} = \frac{D_{n,\text{eff}}^{1D}}{3} \cdot \frac{L_\text{eff}}{\lambda_{\langle 111 \rangle}} = \frac{(D_{n,\text{eff}}^{1D})^2}{6\,\nu_{R,\text{eff}}\,\lambda_{\langle 111 \rangle}^2}. \tag{40}$$

4.5 SIA and SIA Cluster Model Parameters for EUROFER

This section collects numerical parameters for the effective diffusion coefficients of single SIAs, di-interstitials, and interstitial clusters in EUROFER97, suitable for direct input to cluster dynamics codes. The approach follows the trap-limited framework developed in the preceding sections: the pure-Fe baseline is reduced by solute trapping and pinning factors calibrated to DFT binding energies and nominal EUROFER compositions.

4.5.1 EUROFER97 Matrix Composition After Tempering

The nominal composition of EUROFER97 is approximately 8.9 Cr–1.1 W–0.47 Mn–0.20 V–0.14 Ta–0.11 C–0.03 N (wt%) (van der Schaaf et al 2003; Aktaa et al 2020). After the standard heat treatment (980°C/30 min normalisation + 760°C/90 min tempering), a substantial fraction of C, N, V, and Ta is consumed by $M_{23}C_6$ and MX-type precipitates. Atom-probe and TEM studies indicate that the dissolved matrix concentrations relevant for defect trapping are approximately (Klimenkov et al 2012; Polekhina et al 2023):

Species	Nominal (wt%)	Est. dissolved (at. fraction)
Cr (substitutional)	8.9	$c_\text{Cr} \approx 0.094$
W (substitutional)	1.1	$c_\text{W} \approx 0.0033$
Mn (substitutional)	0.47	$c_\text{Mn} \approx 0.0047$
C (interstitial, dissolved)	$\lesssim 0.02$	$c_\text{C} \approx 5 \times 10^{-4}$
N (interstitial, dissolved)	$\lesssim 0.01$	$c_\text{N} \approx 2 \times 10^{-4}$

The dissolved C and N values are rough estimates; most of the nominal carbon is tied up in precipitates. These concentrations are used in the trapping factors below.

4.5.2 Solute–SIA Binding Energies from DFT

Table 9 summarises the DFT-calculated binding energies of solute atoms to single SIAs ($\langle 110 \rangle$ dumbbells) and to small $\frac{1}{2}\langle 111 \rangle$ interstitial loops in bcc Fe. These govern the trapping factors $K_s(T)$ in Eqs. (30) and (38).

Key observations:

- **C and N** are the strongest traps for both single SIAs and small loops. The C–SIA binding (∼0.45 eV) is comparable to the SIA migration energy itself (0.34 eV), meaning that C trapping qualitatively alters the effective migration barrier.
- **Cr** binds weakly via the mixed-dumbbell mechanism. At 9 at.% Cr, the cumulative effect is significant despite the small per-atom binding.

Table 9 DFT solute–SIA binding energies E_b^s in bcc Fe. Positive values denote attraction (trapping). "MD" = mixed dumbb
"NN" = nearest-neighbour configuration.

Solute s	Type	E_b^s(SIA) [eV]	E_b^s(loop) [eV]	Source
Cr	subst. (MD)	0.09–0.12	0.05–0.15	Olsson et al (2007); Vincent et al (2006); Becquart and Soisson
Mn	subst. (MD)	0.15–0.25	0.10–0.20	Messina et al (2020); Olsson et al (2010)
W	subst. (NN)	−0.1 to −0.3	< 0 (repulsive)	Becquart and Soisson (2018)
C	interstit.	0.40–0.50	0.42–0.86[a]	Domain et al (2004); ?); Anento and Serra (2013)
N	interstit.	0.35–0.45	∼0.3–0.7	Domain et al (2004); Becquart et al (2005)

[a] C–loop binding depends on loop size: higher for small ($n = 4$) loops, decreasing toward ∼0.42 eV for $n \gtrsim 19$.

- **Mn** binds more strongly than Cr to the dumbbell and dominates dumbbell-mediated solute transport (Messina et al 2020).
- **W** is repulsive with SIAs and does not contribute to trapping.

4.5.3 Trapping Factors

The trapping coefficient $K_s(T)$ in the effective diffusivity (Eq. 30) is related to the binding energy by (Dudarev 2008):

$$K_s(T) = z_s \exp\left(\frac{E_b^s}{k_B T}\right), \tag{41}$$

where z_s is a coordination factor (number of trapping configurations per solute site; $z_s \approx 4$–8 for substitutional solutes in bcc, $z_s \approx 2$–4 for interstitial solutes). With this, the effective SIA diffusivity becomes:

$$D_i^{\text{eff}} = \frac{D_i^{\text{Fe}}}{1 + \sum_s z_s \, c_s \, \exp(E_b^s / k_B T)}. \tag{42}$$

At low temperatures ($T \lesssim 300\,^\circ$C), the C and N trapping terms dominate the denominator and the effective SIA migration energy becomes $E_m^{i,\text{eff}} \approx E_m^i + E_b^C \approx 0.34 + 0.45 = 0.79$ eV. At high temperatures ($T \gtrsim 500\,^\circ$C), thermal detrapping reduces the effect and D_i^{eff} approaches the pure-Fe value.

4.5.4 Recommended Parameter Table

Table 10 provides the recommended effective diffusion parameters for use in cluster dynamics modeling of EUROFER97. The "baseline" column gives pure-Fe values; the "EUROFER effective" column includes solute corrections.

The coordination numbers z_s and binding energies E_b^s used above are rounded values chosen to represent the central range of DFT data from Table 9. The W contribution is omitted (repulsive binding; $K_W \approx 0$). V and Ta are present at low dissolved concentrations and are neglected for simplicity.

Table 10 Recommended diffusion parameters for SIAs, di-interstitials, and interstitial clusters in EUROFER97. $a = 2.87$ Å. Pure-Fe values serve as the baseline; EUROFER effective values include solute trapping corrections via Eqs. (30)/(38).

Defect	ν_0 **(s^{-1})**	E_m **(eV)**	D_0 **(m^2/s)**	**Dim.**
Pure Fe baseline:				
SIA ($n = 1$, $\langle 110 \rangle$)	1.0×10^{13}	0.34	8.2×10^{-8}	3D
Di-SIA ($n = 2$)	$\sim 10^{13}$	0.42	$\sim 8 \times 10^{-8}$	3D
SIA cluster ($n \geq 4$, 1D)	$6 \times 10^{12}/n^s$	0.02–0.05	—	1D
EUROFER effective (trap-limited, Eq. 42):				
SIA ($n = 1$)	$D_i^{\text{eff}} = D_i^{\text{Fe}} \big/ \left(1 + \sum_s K_s c_s\right)$, using:			
	$K_{\text{C}} = 4 \exp(0.45/k_B T)$, $K_{\text{N}} = 4 \exp(0.40/k_B T)$, $K_{\text{Cr}} = 8 \exp(0.10/k_B T)$, $K_{\text{Mn}} = 6 \exp(0.20/k_B T)$			
Di-SIA ($n = 2$)	Use same trapping form with D_{2i}^{Fe} as baseline and E_b^s increased by $\sim$10%.			
SIA cluster ($n \geq 4$)	$D_{n,\text{eff}}^{1D} = D_n^{1D,\text{Fe}} \big/ \left(1 + \sum_s K_{n,s} c_s\right)$, using:			
	$K_{n,\text{C}} = 2 \exp(0.50/k_B T)$, $K_{n,\text{N}} = 2 \exp(0.40/k_B T)$, $K_{n,\text{Cr}} = 4 \exp(0.10/k_B T)$			

4.5.5 Effective Mean Free Path in EUROFER

For SIA clusters in the mixed 1D/3D framework (Eq. 40), the effective rotation frequency and mean free path are:

$$\nu_{R,\text{eff}} = \nu_R^{\text{therm}} + \sum_s \sigma_s^{\text{rot}} \, c_s \, \omega_n^{1D}, \tag{43}$$

$$L_{\text{eff}} = D_{n,\text{eff}}^{1D} \big/ \nu_{R,\text{eff}}, \tag{44}$$

where ν_R^{therm} is the thermal rotation frequency in pure Fe (small; often negligible below 500°C for large loops) and σ_s^{rot} is the effective rotation cross-section per solute atom. For Cr at 9 at.%, atomistic estimates give $L_{\text{eff}} \sim 10–100\,a$ ($\sim$3–30 nm), which is much shorter than the typical sink spacing in irradiated EUROFER ($\sim$100–1000 nm). This places EUROFER cluster transport firmly in the **mixed 1D/3D regime** rather than the pure 1D limit applicable to high-purity iron.

4.5.6 Numerical Examples

To illustrate the magnitude of the trapping correction, Table 11 gives the effective SIA diffusivity at several temperatures for two scenarios: pure Fe, and EUROFER with $c_{\text{C}} = 5 \times 10^{-4}$, $c_{\text{N}} = 2 \times 10^{-4}$, $c_{\text{Cr}} = 0.094$.

The trapping correction is dramatic below $\sim$400°C (reduction by 2–4 orders of magnitude) and becomes negligible above $\sim$600°C. This temperature dependence is qualitatively consistent with the observation that radiation damage accumulation in EUROFER is much more severe at low irradiation temperatures than at high temperatures.

Table 11 Effective SIA diffusivity D_i^{eff} (m^2/s) at selected temperatures. The reduction factor $\Phi = D_i^{\text{eff}}/D_i^{\text{Fe}}$ quantifies the total trapping effect.

T (°C)	T (K)	D_i^{Fe}	D_i^{eff} (EUROFER)	Φ
200	473	6.8×10^{-12}	$\sim 2 \times 10^{-15}$	$\sim 3 \times 10^{-4}$
300	573	1.4×10^{-11}	$\sim 5 \times 10^{-14}$	$\sim 4 \times 10^{-3}$
400	673	2.3×10^{-11}	$\sim 2 \times 10^{-12}$	~ 0.09
500	773	3.2×10^{-11}	$\sim 1.3 \times 10^{-11}$	~ 0.4
600	873	4.1×10^{-11}	$\sim 3.2 \times 10^{-11}$	~ 0.8

4.5.7 Uncertainties and Directions for Improvement

The parameters in Table 10 carry significant uncertainties that future work should address:

1. **Dissolved C and N concentrations.** The trapping correction is exponentially sensitive to E_b^s and linearly sensitive to c_s. Atom-probe tomography (APT) measurements of the dissolved C and N content in the EUROFER matrix after standard heat treatment would substantially reduce the uncertainty in D_i^{eff}. This is the single most impactful measurement for calibrating the model.
2. **DFT binding energies in concentrated Fe–Cr.** All binding energies in Table 9 were computed in dilute Fe–X binary alloys. In the concentrated Fe–9Cr matrix of EUROFER, local Cr environment effects (short-range order, α' precursors) may modify both the magnitude and the spatial distribution of binding energies. DFT or machine-learning potential calculations in Fe–9Cr supercells with realistic Cr configurations are needed.
3. **Synergistic multi-solute effects.** The trap-limited model (Eq. 42) assumes independent trapping by each solute species. In reality, C–vacancy and N–vacancy complexes may be the dominant trapping species rather than isolated C or N atoms. Including these complexes requires knowledge of their concentrations (from thermodynamic equilibrium or irradiation-driven formation) and their separate binding energies to SIAs.
4. **Size-dependent loop trapping.** The C–loop binding energy decreases with loop size (Table 9). A size-dependent $K_{n,\text{C}}(n)$ based on systematic DFT or MD calculations for loops of 4–100 SIAs would improve the accuracy of the 1D cluster transport model, particularly at low temperatures where trapping controls the effective 1D glide length.
5. **Irradiation-induced solute redistribution.** Under irradiation, radiation-induced segregation (RIS) depletes Cr and enriches Mn and P at defect sinks (Messina et al 2020). This dynamically alters the local solute field experienced by migrating defects. Coupling the solute-field evolution to the defect diffusivity in a self-consistent manner—through a concentration-dependent $K_s(T, c_s(t))$—is a longer-term modeling goal that would require co-evolution of solute and defect distributions.

6. **Experimental validation.** In-situ TEM observation of SIA cluster mobility in Fe–9Cr model alloys (with and without controlled C/N additions) as a function of temperature would provide direct calibration data for $D^{1D}_{n,\mathrm{eff}}$ and L_{eff}. Resistivity recovery experiments on EUROFER after electron irradiation at cryogenic temperatures would constrain the effective SIA migration energy.

4.6 Vacancy Cluster Diffusion

4.6.1 Migration Mechanism

Vacancy clusters in bcc iron migrate by a qualitatively different mechanism from interstitial clusters. Whereas SIA clusters glide collectively as dislocation loops, vacancy clusters move only by sequential single-vacancy jumps within or at the surface of the cluster. The motion is always **three-dimensional**, and the effective migration energy generally **increases** with cluster size because larger clusters require increasingly coordinated sequences of atomic rearrangements to translate the centre of mass by one lattice spacing (Rahman et al 2023; Fu et al 2005; **?**).

For clusters larger than ∼5–6 vacancies, the dominant transport mechanism transitions from bulk-like nearest-neighbour vacancy hopping to surface (ledge) diffusion of adatom vacancies around the void facets. This mechanism underlies the classical Gruber model for void migration, in which the void diffusion coefficient scales as a high inverse power of the void radius (Gruber 1967; Nichols 1969).

4.6.2 Small Vacancy Cluster Migration Energies

Table 12 collects the effective migration barriers for small vacancy clusters in α-Fe from several computational methods.

Table 12 Effective migration energies E_m (eV) of small vacancy clusters in bcc Fe. The "mechanism" column describes the domina pathway. Values are compiled from DFT, kinetic activation–relaxation technique (k-ART), and molecular dynamics (MD) studies.

n	Configuration	E_m (eV)	Mechanism	Source
1	monovacancy	0.65–0.70	NN jump along ⟨111⟩	Fu et al (2004); Domain and Becquart (2001); Seeg
2	divacancy (2NN)	0.62–0.75	two-step NN exchange	Rahman et al (2023); Fu et al (2005)
3	compact trivacancy	0.35–0.50	concerted 3-atom ring rotation	Rahman et al (2023); Schuler et al (2015)
4	tetravacancy	0.7–0.9	sequential NN jumps	Rahman et al (2023)
5	pentavacancy	0.8–1.1	sequential NN jumps + dissociation	Rahman et al (2023)
≥ 6	compact 3D cluster	> 1.0	effectively immobile	Rahman et al (2023)

Two features of Table 12 merit discussion:

The anomalously mobile trivacancy.

The compact trivacancy (V_3) in the bcc lattice—with all three vacancies at mutual second-nearest-neighbour positions forming an equilateral triangle on a {110} plane—has a migration energy (0.35–0.50 eV) that is *lower* than that of either the monovacancy or the divacancy. This was first reported from DFT-parameterised kinetic Monte Carlo calculations and has been confirmed by k-ART (Rahman et al 2023) and by independent first-principles studies of vacancy–solute cluster mobility (Schuler et al 2015). The mechanism is a concerted three-atom ring rotation in which three Fe atoms adjacent to the cluster simultaneously exchange positions with the three vacancies, translating the cluster centre of mass by one lattice vector. Because of this low barrier, V_3 clusters are the most mobile vacancy-type species in α-Fe after isolated monovacancies. Schuler et al (2015) further showed that V_3X complexes (with $X = \mathrm{C}$, N, or O) retain much of this mobility—with global migration energies of 0.59 eV (C), 0.86 eV (N), and 0.58 eV (O)—which means that vacancy-mediated interstitial-solute transport via trivacancy dragging is an important mechanism in irradiated ferritic steels.

Divacancy pathway multiplicity.

The divacancy in bcc Fe has its ground state at the second-nearest-neighbour (2NN) separation (Domain and Becquart 2001). Its migration involves a two-step process: first one vacancy jumps to a first-nearest-neighbour position (metastable), then the second vacancy follows to re-establish the 2NN ground state at a translated position. The effective barrier depends on which of several pathways is taken, giving a range of 0.62–0.75 eV (Rahman et al 2023; Fu et al 2005). k-ART simulations reveal additional complex pathways involving transient higher-order configurations, with effective barriers as high as 0.84 eV for some routes through the energy landscape (Rahman et al 2023).

4.6.3 Void Diffusion: Size-Dependent Model

For larger voids containing m vacancies ($m \gtrsim 10$), vacancy cluster migration proceeds by surface self-diffusion of adatom vacancies on the void wall. The classical result due to Gruber (1967) and Nichols (1969) gives the void diffusion coefficient as:

$$D_{\mathrm{void}}(m) = \frac{D_s\,\Omega^{4/3}}{r_m^4} = D_s\,\Omega^{4/3}\left(\frac{4\pi}{3\,m\,\Omega}\right)^{4/3} \propto \frac{D_s}{m^{4/3}}, \tag{45}$$

where D_s is the surface self-diffusion coefficient on the void wall, $\Omega = a^3/2$ is the atomic volume, and $r_m = (3m\Omega/4\pi)^{1/3}$ is the void radius. The strong inverse scaling $D_{\mathrm{void}} \propto m^{-4/3}$ means that void mobility drops extremely rapidly with size. Numerical estimates give $D_{\mathrm{void}} \sim 10^{-30}$ m^2/s or less for voids of $m \geq 50$ at temperatures below 500°C, which is effectively zero on any experimentally accessible timescale.

Alternative derivations based on volume diffusion rather than surface diffusion give an even steeper size dependence ($D_{\mathrm{void}} \propto m^{-5/3}$), but surface diffusion is generally accepted as the dominant mechanism for faceted voids in bcc metals (Nichols 1969).

4.6.4 Practical Treatment in Rate Theory

For cluster dynamics models, the following classification is recommended:

Size range	Treatment	Rationale
$m = 1$ (monovacancy)	mobile, 3D	$E_m = 0.67$ eV
$m = 2$ (divacancy)	mobile, 3D	$E_m \approx 0.70$ eV
$m = 3$ (trivacancy)	mobile, 3D	$E_m \approx 0.42$ eV (anomalously fast)
$m = 4$–5	weakly mobile	$E_m \approx 0.7$–1.1 eV; include if accuracy needed
$m \geq 6$	immobile sinks	growth/shrinkage by V and SIA absorption/emission only

The general size-dependent diffusion coefficient for vacancy clusters used in the rate equations is:

$$D_{(-m,0)} = D_v \cdot m^{-s_v}, \qquad s_v \approx 1.0 \quad (m \leq m^v_{\max}), \tag{46}$$

where $m^v_{\max}$ is a mobility cutoff (typically 3–5). Clusters with $m > m^v_{\max}$ are set to $D = 0$.

For He-containing vacancy clusters (bubbles), the diffusion coefficient is further reduced:

$$D_{(-m,\ell)} = D_{(-m,0)} \cdot f(\ell/m), \tag{47}$$

where $f(\ell/m)$ is a reduction factor reflecting the additional pinning by trapped helium. A simple choice is $f = \exp(-E_b^{\mathrm{He}}\ell/k_BT)$, but for practical purposes in most rate-theory models, He-containing clusters with $m \geq 2$ are treated as immobile.

4.6.5 Solute Effects on Vacancy and Vacancy Cluster Diffusion in EUROFER

Vacancy migration in pure α-Fe proceeds by nearest-neighbour jumps along $\langle 111 \rangle$ with a barrier of $E_m^v = 0.67$ eV. In the EUROFER matrix, dissolved solutes modify this mobility through two mechanisms: (i) vacancy–solute binding, which traps vacancies near solute sites and raises the effective migration energy; and (ii) modification of the local saddle-point energy, which can either raise or lower the elementary jump barrier depending on the neighbouring chemical environment. Both effects are governed by the vacancy–solute binding energy $E_b^{v\text{-}s}$, which has been computed by DFT for all the principal solutes in EUROFER.

Vacancy–Solute Binding Energies.

Table 13 collects DFT-calculated vacancy–solute binding energies at 1NN and 2NN separations in bcc Fe. Positive values denote attraction (trapping).

Key observations for EUROFER:

1. **C and N are the dominant vacancy traps**, with binding energies (∼0.4–0.5 eV at 1NN) comparable to the vacancy migration energy itself. Carbon–vacancy (VC) and nitrogen–vacancy (VN) complexes are therefore strongly bound and act as deep traps that immobilise vacancies at low temperatures. This is the primary mechanism by which interstitial solutes suppress vacancy-mediated processes (self-diffusion, void nucleation) in ferritic steels.

Table 13 DFT vacancy–solute binding energies $E_b^{v\text{-}s}$ (eV) in bcc Fe. Positive = attractive (vacancy trapped near solute). Values compiled from the systematic studies of Olsson et al (2010), Devi et al (2018), Domain et al (2004), and Ohnuma et al (2009).

Solute s	Type	$E_b^{v\text{-}s}$(1NN) [eV]	$E_b^{v\text{-}s}$(2NN) [eV]	Source
Cr	substitutional	0.03–0.06	−0.02 to −0.05 (repulsive)	Olsson et al (2010); Devi et al (2018)
Mn	substitutional	0.06–0.12	∼0.00	Olsson et al (2010); Devi et al (2018)
W	substitutional	0.20–0.35	0.05–0.15	Olsson et al (2010); Devi et al (2018)
V	substitutional	0.02–0.10	−0.03 to 0.00	Olsson et al (2010)
Ta	substitutional	0.15–0.25	0.02–0.08	Devi et al (2018)
C	interstitial	0.41–0.47	0.10–0.18	Domain et al (2004); Ohnuma et al (2009)
N	interstitial	0.35–0.43	0.08–0.15	Domain et al (2004); Ohnuma et al (2009)

2. **W is a moderate vacancy trap** ($E_b \sim 0.25$–0.35 eV at 1NN). At the 0.33 at.% level present in EUROFER, W makes a non-negligible contribution to vacancy trapping, particularly in the intermediate temperature range (300–500°C) where the C/N trapping terms are starting to deactivate.
3. **Cr binds very weakly to vacancies** ($E_b \sim 0.05$ eV at 1NN) and is repulsive at 2NN. Despite its high concentration (9.4 at.%), the cumulative trapping effect of Cr on vacancy mobility is modest because $\exp(0.05/k_B T)$ is close to unity at all relevant temperatures.
4. **Mn** binds slightly more strongly than Cr but still weakly (∼0.10 eV). Its effect is secondary to C, N, and W.

Effective Vacancy Diffusivity in EUROFER.

By analogy with the SIA trap-limited formulation (Eq. 30), the effective vacancy diffusivity in EUROFER is:

$$\boxed{D_v^{\text{eff}} = \frac{D_v^{\text{Fe}}}{1 + \sum_s z_s^v \, c_s \, \exp(E_b^{v\text{-}s}/k_B T)}}, \tag{48}$$

where D_v^{Fe} is the pure-iron value, c_s is the dissolved atomic fraction of solute s, and z_s^v is the coordination number for vacancy trapping configurations near a solute of type s. The dominant trapping species, in decreasing order of importance, are: dissolved C, dissolved N, W, Mn, and Cr.

Effect on Vacancy Cluster Mobility.

Small vacancy clusters ($m = 2$–5) are affected by solutes through two mechanisms:

1. **Enhanced binding/pinning**: C and N atoms bind even more strongly to divacancies and trivacancies than to monovacancies, because the larger open volume of the cluster provides a more favourable environment for the interstitial solute

(Domain et al 2004; Schuler et al 2015). This deepens the trap and raises the effective cluster migration barrier.

2. **Solute decoration**: In EUROFER, small vacancy clusters formed under irradiation rapidly acquire C and N decorations from the dissolved matrix pool, converting bare V_m clusters into $V_m\mathrm{C}_p\mathrm{N}_q$ complexes. These decorated clusters have substantially reduced mobility relative to bare clusters. The compact trivacancy—anomalously mobile in pure Fe—is particularly affected: V_3C has a migration energy of 0.59 eV and V_3N of 0.86 eV (Schuler et al 2015), compared to 0.35–0.50 eV for bare V_3.

For the purpose of cluster dynamics modelling, the effective vacancy cluster diffusivity in EUROFER can be written as:

$$D^{\mathrm{eff}}_{(-m,0)} = \frac{D^{\mathrm{Fe}}_{(-m,0)}}{1 + \sum_s K^v_{m,s}(T)\, c_s}, \tag{49}$$

where $K^v_{m,s}(T) = z^v_{m,s} \exp(E_b^{v_m\text{-}s}/k_B T)$ and the binding energy $E_b^{v_m\text{-}s}$ may depend on cluster size.

Numerical Illustration.

Table 14 shows the effective monovacancy diffusivity at selected temperatures, compared to pure Fe.

Table 14 Effective monovacancy diffusivity D_v^{eff} (m^2/s) in EUROFER at selected temperatures, with reduction factor $\Phi_v = D_v^{\mathrm{eff}}/D_v^{\mathrm{Fe}}$.

T (°C)	T (K)	D_v^{Fe}	D_v^{eff} (EUROFER)	Φ_v
200	473	2.1×10^{-15}	$\sim 1 \times 10^{-18}$	$\sim 5 \times 10^{-4}$
300	573	8.5×10^{-14}	$\sim 3 \times 10^{-16}$	$\sim 4 \times 10^{-3}$
400	673	1.1×10^{-12}	$\sim 6 \times 10^{-14}$	~ 0.05
500	773	7.1×10^{-12}	$\sim 2 \times 10^{-12}$	~ 0.3
600	873	2.9×10^{-11}	$\sim 2 \times 10^{-11}$	~ 0.7

The vacancy trapping correction follows a similar trend to the SIA case: dramatic reduction (3 orders of magnitude) below ∼300°C, becoming negligible above ∼600°C. However, because $E_m^v > E_m^i$, the vacancy is already slow in pure Fe, and the practical consequence of trapping is to delay the onset of vacancy-mediated processes (void nucleation, thermal emission from clusters, precipitation coarsening) to even higher temperatures in EUROFER than the pure-Fe migration energy alone would suggest.

Physical Implications for Cluster Dynamics.

The solute-modified vacancy diffusivity has three important consequences for cluster dynamics modelling of EUROFER:

1. **Suppressed void nucleation at low T**. Vacancy trapping by C and N reduces the free vacancy supersaturation available for void nucleation, raising the effective incubation dose. This is consistent with the observation that EUROFER shows higher void swelling resistance than high-purity iron at temperatures below $\sim$400°C.
2. **Delayed thermal emission**. The vacancy emission rate from clusters (Eq. 122) depends on the free vacancy equilibrium concentration, which is suppressed when $D_v^{\mathrm{eff}} < D_v^{\mathrm{Fe}}$. This stabilises small vacancy clusters against thermal dissolution and shifts the void-to-bubble transition to higher temperatures.
3. **Asymmetric SIA–vacancy trapping**. In EUROFER, both SIAs and vacancies are trapped by C and N, but the SIA–C binding ($\sim$0.45 eV) and vacancy–C binding ($\sim$0.45 eV) are comparable. This means that the net effect on the point-defect imbalance (which drives swelling) depends on the relative magnitudes of the SIA and vacancy trapping corrections, and cannot be predicted from either correction alone. Self-consistent treatment of both D_i^{eff} and D_v^{eff} in the rate equations is essential.

4.7 Helium Interstitial Diffusion

4.7.1 Migration Mechanism in Pure α-Fe

Interstitial helium in bcc iron occupies tetrahedral sites, which are more stable than octahedral sites by $\sim$0.2 eV (?Seletskaia et al 2005). The migration path connects nearest-neighbour tetrahedral sites with a DFT-calculated barrier of $E_m^h = 0.06$ eV (?); MD simulations and NEB calculations with different functionals give values in the range 0.06–0.09 eV (Morishita et al 2003; Zhang et al 2018). This barrier is so small that helium is essentially mobile at all temperatures above $\sim$30 K—it is by far the fastest-diffusing species in the defect system.

The helium diffusion coefficient and reaction frequency in pure α-Fe are:

$$D_h = a^2 \nu_h \exp\left(-\frac{E_m^h}{k_B T}\right) = 2.5 \times 10^{-8} \exp\left(-\frac{0.06\ \mathrm{eV}}{k_B T}\right) \quad \mathrm{m^2\,s^{-1}}, \tag{50}$$

$$\omega_h = \frac{D_h}{a^2} = 3.0 \times 10^{12} \exp\left(-\frac{0.06\ \mathrm{eV}}{k_B T}\right) \quad \mathrm{s^{-1}}, \tag{51}$$

with $\nu_h \approx 3.0 \times 10^{12}$ s^{-1} and $a = 2.87$ Å (Becquart and Domain 2007; Borodin and Vladimirov 2007). Because $E_m^h \ll k_B T$ at all reactor-relevant temperatures, the Arrhenius factor is close to unity and the helium mobility is nearly temperature-independent. Spin-lattice dynamics simulations confirm strongly non-Arrhenius behaviour at elevated temperatures, where phonon-drag effects make the diffusion coefficient increase roughly linearly with T rather than exponentially (Ma and Dudarev 2019).

4.7.2 Helium Trapping by Vacancies

The defining feature of helium in irradiated metals is its extremely strong binding to vacancies. DFT calculations give a binding energy of $E_b^{h\text{-}V} \approx 2.0$–2.4 eV for the first He atom entering a monovacancy (?Seletskaia et al 2005; Becquart and Domain 2007). The binding remains strong as additional He atoms are added: up to 4–6 He atoms can be accommodated in a single vacancy with cumulative binding energies exceeding 4 eV (?). These $\mathrm{He}_n\mathrm{V}_m$ complexes are essentially immobile (substitutional He migration requires the vacancy mechanism with a barrier of ∼1.1 eV) and serve as the nucleation sites for helium bubbles.

The consequence for transport is that the *free* interstitial helium concentration c_h available for long-range diffusion is not the total implanted He content but rather the small fraction that has not yet been trapped by vacancies, vacancy clusters, or other deep traps. In a vacancy-rich irradiation environment, the free He concentration can be orders of magnitude below the total He inventory.

4.7.3 Helium–Solute Interactions and Effective Diffusivity in EUROFER

In the EUROFER matrix, interstitial He can interact with dissolved solutes before reaching a vacancy trap. The relevant interactions, from DFT calculations, are:

Substitutional solutes (Cr, Mn, W).

He–substitutional-solute interactions in bcc Fe are generally weak. Becquart and Domain (2007) and Seletskaia et al (2005) find that He interacts only marginally with substitutional Cr and other transition metal solutes, with binding energies in the range 0.00–0.05 eV at 1NN—insufficient to constitute meaningful traps at reactor temperatures. At 9 at.% Cr, the cumulative effect on He mobility is therefore negligible.

Interstitial solutes (C, N).

Carbon and nitrogen occupy octahedral sites and interact with interstitial He at tetrahedral sites. The He–C binding energy has been calculated to be ∼0.05–0.10 eV (Becquart and Domain 2007; ?), which is comparable to the He migration barrier itself. This means C can cause transient trapping of He at low temperatures ($T \lesssim 200\,°\mathrm{C}$), but at higher temperatures the trapping is thermally overcome and has little effect on the long-range He transport.

Vacancy–solute complexes as He traps.

The most important solute effect on He transport in EUROFER is *indirect*: C and N form stable vacancy–solute complexes (VC, VN, $V\mathrm{C}_2$) that compete with bare vacancies as He trapping sites. The He binding energy to a VC complex is ∼1.5–1.8 eV—lower than to a bare vacancy (∼2.3 eV) but still a deep trap. Since a fraction of the vacancies in EUROFER are decorated by C and N, the effective trap density and the He–trap binding landscape differ from pure Fe.

Effective Helium Diffusivity.

Because the He–solute binding energies are so much smaller than the He–vacancy binding energy, the dominant control on free He transport is the vacancy trap density, not the solute field. The effective He diffusivity in EUROFER can therefore be written as:

$$\boxed{D_h^{\text{eff}} = \frac{D_h^{\text{Fe}}}{1 + z_V^h\, c_V^{\text{free}}\, \exp(E_b^{h\text{-}V}/k_BT) + \sum_s z_s^h\, c_s\, \exp(E_b^{h\text{-}s}/k_BT)}\,,} \tag{52}$$

where the first trapping term is due to free vacancies (concentration c_V^{free}, binding energy $E_b^{h\text{-}V} \approx 2.3$ eV, coordination $z_V^h = 6$) and the sum is over solute traps. In practice, at all temperatures where vacancies exist in meaningful concentrations, the vacancy trapping term completely dominates the denominator and the solute terms can be neglected. The helium diffusivity reduction in EUROFER relative to pure Fe is therefore controlled primarily by the irradiation-produced vacancy concentration, not by the alloy composition.

Table 15 collects the recommended helium diffusion parameters.

Table 15 Recommended parameters for helium interstitial diffusion in bcc Fe and EUROFER. The effective diffusivity Eq. (52) is dominated by vacancy trapping; solute terms are secondary corrections.

Parameter	**Value**	**Source**
Pure Fe baseline:		
E_m^h (interstitial migration)	0.06 eV	**?**
ν_h (attempt frequency)	3.0×10^{12} s^{-1}	Becquart and Domain (2007)
$D_0^h = a^2\nu_h$	2.5×10^{-8} m^2/s	—
Vacancy trapping (dominant):		
$E_b^{h\text{-}V}$ (He–monovacancy, 1st He)	2.3 eV	**?**Seletskaia et al (2005)
$E_b^{h\text{-}V}$ (He–monovacancy, 2nd He)	2.0 eV	**?**
$E_b^{h\text{-}V}$ (He–divacancy)	2.5–2.8 eV	Borodin and Vladimirov (2007)
z_V^h (trapping coordination)	6	—
Solute trapping (secondary, for EUROFER):		
$E_b^{h\text{-C}}$ (He–C)	0.08 eV	Becquart and Domain (2007)
$E_b^{h\text{-N}}$ (He–N)	∼0.05 eV	estimated
$E_b^{h\text{-Cr}}$ (He–Cr)	∼0.02 eV	Seletskaia et al (2005)
$E_b^{h\text{-}V\text{C}}$ (He–VC complex)	1.5–1.8 eV	Becquart and Domain (2007)

Physical Summary.

Helium transport in EUROFER differs from that of vacancies and SIAs in a fundamental way: the *intrinsic* mobility is unaffected by the alloy composition (He–solute binding is negligible), but the *effective* long-range transport is controlled by the vacancy trap landscape, which is itself modified by solutes (through the vacancy trapping described in Section 4.6.5). This creates an indirect coupling: solutes reduce the free vacancy concentration available for He trapping, which paradoxically can *increase* the free He mobility in EUROFER relative to a hypothetical material with the same total vacancy content but no solute-decorated vacancies. The interplay between vacancy–solute decoration and He trapping is an important consideration for bubble nucleation modelling and should be treated self-consistently in the rate equations.

4.8 EUROFER Model Parameters: Defect Diffusion

Table 16 collects all diffusion parameters. Pure-Fe baseline values are given first; EUROFER solute-trapping corrections follow. The effective diffusivity is $D_\alpha^{\text{eff}} = D_\alpha^{\text{Fe}}/(1 + \sum_s K_s c_s)$ with $K_s = z_s \exp(E_b^s/k_B T)$.

5 Defect Cluster Dissociation

The equilibrium thermal vacancy concentration in the bulk is approximated by

$$C_v^0 = \exp\left(-\frac{E_v^f}{k_B T}\right), \tag{53}$$

where E_v^f is the vacancy formation free energy or, in the usual approximation, the vacancy formation energy. At the surface of a vacancy cluster containing α vacancies, the local equilibrium vacancy concentration is elevated above C_v^0 according to

$$C_v^e(\alpha) = C_v^0 \exp\left(\frac{\mu(\alpha)}{k_B T}\right) = \exp\left[-\frac{E_v^f - \mu(\alpha)}{k_B T}\right] \tag{54}$$

where $\mu(\alpha)$ is the vacancy chemical potential of the cluster. For an interstitial cluster containing β interstitials, the equilibrium vacancy concentration at the surface is lowered below C_v^0:

$$C_v^e(\beta) = C_v^0 \exp\left(-\frac{\mu(\beta)}{k_B T}\right) = \exp\left[-\frac{E_v^f + \mu(\beta)}{k_B T}\right]. \tag{55}$$

We can have a unified expression for both types of clusters by defining the binding energy of a vacancy to a vacancy cluster as

$$E_b^{(v)}(\alpha) = E_v^f - \mu(\alpha) \tag{56}$$

and the binding energy of a vacancy to an interstitial cluster of size β is

$$E_b^{(i)}(\beta) = E_v^f + \mu(\beta) \tag{57}$$

Then, the equilibrium vacancy concentration at the surface of a vacancy or interstitial cluster can be written as

$$C_v^e = \exp\left[-\frac{E_b}{k_B T}\right] \tag{58}$$

Table 16 Diffusion input parameters for EUROFER cluster dynamics.

Parameter	Symbol	Value	Units
Pure-Fe baseline diffusion coefficients:			
Vacancy D_0	D_0^v	8.2×10^{-8}	m^2/s
Vacancy E_m	E_m^v	0.67	eV
SIA D_0	D_0^i	8.2×10^{-8}	m^2/s
SIA E_m	E_m^i	0.34	eV
He D_0	D_0^h	2.5×10^{-8}	m^2/s
He E_m	E_m^h	0.06	eV
SIA cluster 1D glide ($n \geq 4$):			
Attempt frequency	ν_0^{1D}	6×10^{12}	s^{-1}
1D migration energy	E_m^{1D}	0.03	eV
Size exponent	s_i	0.7	—
Mobility cutoff	$n_{\max}^i$	100	SIAs
Vacancy cluster diffusion:			
Size exponent	s_v	1.0	—
Mobility cutoff	$m_{\max}^v$	5	vacancies
EUROFER dissolved solute concentrations (after tempering):			
Cr	c_{Cr}	0.094	at/at
W	c_{W}	0.0033	at/at
Mn	c_{Mn}	0.0047	at/at
C (dissolved)	c_{C}	5×10^{-4}	at/at
N (dissolved)	c_{N}	2×10^{-4}	at/at
SIA trapping parameters ($K_s = z_s \exp(E_b^s/k_B T)$):			
C–SIA	$z = 4,\ E_b$	0.45	eV
N–SIA	$z = 4,\ E_b$	0.40	eV
Cr–SIA	$z = 8,\ E_b$	0.10	eV
Mn–SIA	$z = 6,\ E_b$	0.20	eV
Vacancy trapping parameters:			
C–V	$z = 3,\ E_b$	0.45	eV
N–V	$z = 3,\ E_b$	0.40	eV
W–V	$z = 8,\ E_b$	0.27	eV
Mn–V	$z = 8,\ E_b$	0.10	eV
Cr–V	$z = 8,\ E_b$	0.05	eV
SIA cluster trapping (1D loops):			
C–loop	$z = 2,\ E_b$	0.50	eV
N–loop	$z = 2,\ E_b$	0.40	eV
Cr–loop	$z = 4,\ E_b$	0.10	eV
Mixed 1D/3D parameters:			
Mean free path (EUROFER)	$\hat{L} = L/a$	30–100	—
1D/3D interpolation constant	B_{rot}	2.627	—

The binding energy can be found from atomistic calculations (e.g. DFT or MD), from experiments, or using continuum theory for large clusters. We will find explicit forms for α-iron in the next section.

For a spherical cavity of radius

$$R(\alpha) = a \left(\frac{3\alpha}{8\pi} \right)^{1/3}, \tag{59}$$

the defect chemical potential for vacancy exchange may be written as

$$\mu(\alpha) = \Omega \left(\frac{2\gamma}{R(\alpha)} - P_g - \sigma_h \right), \tag{60}$$

where γ is the surface energy, Ω is the atomic volume, σ_h is the hydrostatic *tension* component of the stress, and P_g is the internal gas pressure. The gas pressure inside a helium bubble may be represented more accurately by a virial equation of state rather than by a hard-core Van der Waals approximation. Writing the helium number density in the bubble as

$$n_{\mathrm{He}} = \frac{m}{V_b} = \frac{m}{\alpha\Omega}, \tag{61}$$

where m is the number of helium atoms in the bubble, α is the number of vacancies, Ω is the atomic volume, and $V_b = \alpha\Omega$ is the bubble volume, the pressure may be written as

$$P_g = n_{\mathrm{He}} k_B T \left[1 + B_2(T)\, n_{\mathrm{He}} + B_3(T)\, n_{\mathrm{He}}^2 + \cdots \right]. \tag{62}$$

Equivalently,

$$P_g = \frac{m k_B T}{\alpha\Omega} \left[1 + B_2(T) \frac{m}{\alpha\Omega} + B_3(T) \left(\frac{m}{\alpha\Omega} \right)^2 + \cdots \right]. \tag{63}$$

Here $B_2(T)$, $B_3(T)$, ... are the second, third, and higher virial coefficients of helium, which are functions of temperature. At temperatures in the range 600–1000 K, representative average virial coefficients for ^{4}He are

$$B_2 \approx 1.67 \times 10^{-29}\ \mathrm{m}^3/atom, \tag{64}$$

$$B_3 \approx 1.84 \times 10^{-58}\ \mathrm{m}^6/atom^2, \tag{65}$$

where these units are to be interpreted consistently with n_{He} in atoms m^{-3}. These values correspond to $\overline{B_2} \approx 10.06$ cm^3 mol^{-1} and $\overline{B_3} \approx 66.72$ cm^6 mol^{-2} for ^{4}He over 600–1000 K Hurly and Moldover (2000); Garberoglio and Harvey (2009).

5.1 Voids

The binding energy of a vacancy to a void is expressed as

$$E_b^{\mathrm{mod}}(\alpha) = E_b^{\mathrm{cont}}(\alpha) + \left[E_b^{(v)}(2) - E_b^{\mathrm{cont}}(2) \right] \exp\bigl(-\lambda\, (\alpha - 2) \bigr) \tag{66}$$

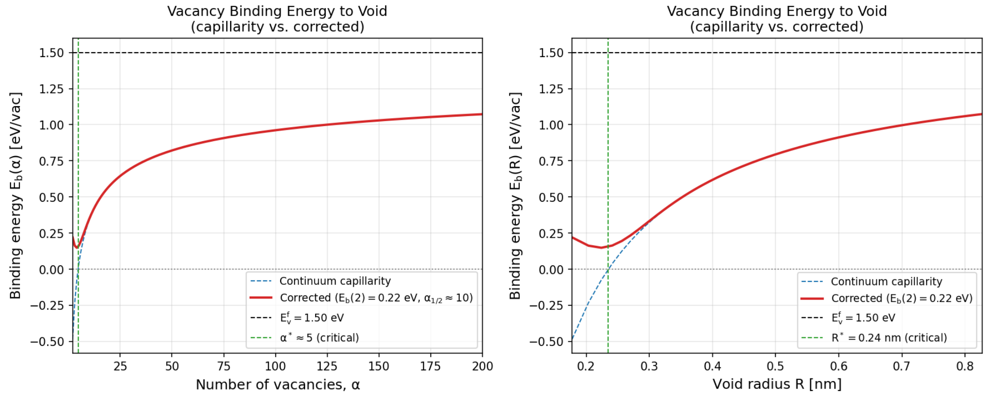


Fig. 4 Binding energy of a vacancy to a void as a function of the number of vacancies in the void

With $E_b^{(v)}(2)$ the divacancy binding energy from DFT, and without external stress, the continuum capillary term is

$$E_b^{\mathrm{cont}}(\alpha) = E_v^f - \Omega\,\frac{2\gamma}{R(\alpha)}, \qquad R(\alpha) = a\left(\frac{3\alpha}{8\pi}\right)^{1/3} \tag{67}$$

The decay constant is chosen so that the correction is 1% at $\alpha = 10$: $\lambda = \frac{\ln 100}{8} \approx 0.575\ \mathrm{vac}^{-1}$. The binding energy of a vacancy to a void as a function of the void size is shown in Fig. 4, using the capillary approximation for large sizes and transitioning to atomistic data for small sizes using Eq. 66.

5.2 Bubbles

A helium–vacancy cluster $\mathrm{He}_m\mathrm{V}_\alpha$ is characterised by two integers: α (the number of vacancies) and m (the number of trapped helium atoms). Cluster dynamics requires the binding energy of both a vacancy and a helium atom to such a cluster, since these two binding energies govern, respectively, the vacancy emission rate (cavity growth/shrinkage) and the helium emission rate (gas release). We treat each in turn.

5.2.1 Atomistic Binding Energy Data

Table 17 collects representative DFT/MD binding energies for $\mathrm{He}_m\mathrm{V}_\alpha$ clusters in α-Fe. The values are recast from the dissociation energies tabulated by Fu and Willaime (Fu and Willaime 2005) and Caturla et al. (Caturla et al 2008), using the definitions:

$$E_B^V(\alpha, m) = E_f(\mathrm{He}_m\mathrm{V}_{\alpha-1}) + E_f(V) - E_f(\mathrm{He}_m\mathrm{V}_\alpha), \tag{68}$$

$$E_B^{\mathrm{He}}(\alpha, m) = E_f(\mathrm{He}_{m-1}\mathrm{V}_\alpha) + E_f(\mathrm{He}) - E_f(\mathrm{He}_m\mathrm{V}_\alpha). \tag{69}$$

Two important trends are visible in Table 17:

Table 17 Binding energies for vacancy and helium addition to $\mathrm{He}_m\mathrm{V}_\alpha$ clusters in α-Fe. Data from Fu and Willaime (2005) and Caturla et al (2008).

α	m	$E_B^V(\alpha, m)$ (eV)	$E_B^{\mathrm{He}}(\alpha, m)$ (eV)
1	1	2.30	—
2	1	0.78	2.85
3	1	0.83	3.30
4	1	1.16	3.84
1	2	3.71	1.84
2	2	1.61	2.75
3	2	1.04	2.96
4	2	1.32	3.12
1	3	4.59	1.83
2	3	1.85	2.07
4	3	1.57	3.16
1	4	5.52	1.91
2	4	2.30	2.36
3	4	2.03	2.57
4	4	1.97	3.05

1. $E_B^V(\alpha, m)$ **increases** with m at fixed α: more helium inside a cavity raises its internal pressure, making it harder for the cavity to emit a vacancy. This is the mechanism by which helium stabilises bubbles against thermal dissolution.
2. $E_B^{\mathrm{He}}(\alpha, m)$ **decreases** with m at fixed α: as the cavity fills with helium, the incremental binding of each additional He atom decreases because the available trapping volume shrinks and the He–He repulsion increases.

5.2.2 Vacancy Binding Energy to a Bubble

For the vacancy binding energy, we use the same atomistic −continuum transitional model developed for voids (Section 5.1), extended to include the internal gas pressure:

$$E_B^v(\alpha, m) = \underbrace{E_f^v - \Omega\left[\frac{2\gamma}{R(\alpha)} - P_g(\alpha, m)\right]}_{E_B^{v,\,\mathrm{cont}}(\alpha,m)} + A(m)\, e^{-\lambda(\alpha-1)}, \tag{70}$$

with subsidiary expressions:

$$R(\alpha) = a\left(\frac{3\alpha}{8\pi}\right)^{1/3}, \tag{71}$$

$$P_g(\alpha, m) = \frac{m}{\alpha\,\Omega}\, k_B T\left[1 + B_2\,\frac{m}{\alpha\,\Omega} + B_3\left(\frac{m}{\alpha\,\Omega}\right)^2\right], \tag{72}$$

$$\lambda = \frac{\ln 100}{8} \approx 0.5756\ \mathrm{vac}^{-1}. \tag{73}$$

The continuum term $E_B^{v,\mathrm{cont}}$ captures the competition between surface tension ($2\gamma/R$, which favours vacancy emission) and gas pressure (P_g, which opposes it). The exponential correction $A(m)\exp[-\lambda(\alpha-1)]$ bridges the atomistic small-cluster binding energies to the continuum limit, decaying to less than 1% by $\alpha = 10$.

The fitted amplitudes $A(m)$, obtained by least-squares regression of Eq. (70) against the atomistic data of Osetsky et al (2000) ($m = 0$) and Caturla et al (2008) ($m \geq 1$), are:

Table 18
Least-squares fitting amplitudes for the vacancy binding energy to bubbles (Eq. 70).

m	$A(m)$ [eV]
0	1.2353
1	2.9064
2	3.4147
3	2.1504
4	−0.1590

The amplitude at each integer m_k is determined by:

$$A(m_k) = \frac{\sum_{i\in\mathcal{D}_k} \delta_i\, e^{-\lambda(\alpha_i-1)}}{\sum_{i\in\mathcal{D}_k} e^{-2\lambda(\alpha_i-1)}}, \qquad \delta_i = E_B^{v,\,\mathrm{atom}}(\alpha_i, m_k) - E_B^{v,\,\mathrm{cont}}(\alpha_i, m_k), \tag{74}$$

where $\mathcal{D}_k$ is the subset of atomistic data at fixed m_k.

5.2.3 Helium Binding Energy to a Bubble

The helium binding energy $E_B^{\mathrm{He}}(\alpha, m)$—the energy required to remove one He atom from a $\mathrm{He}_m\mathrm{V}_\alpha$ cluster and place it at a tetrahedral interstitial site in the bulk—is needed for the helium thermal emission (detrapping) rate and for determining the maximum stable He loading $\ell_{\max}(\alpha)$.

Physical picture.

When a He atom is removed from a bubble of α vacancies containing m helium atoms, two energy contributions change: (i) the He–cavity binding interaction (governed by the attractive potential well of the vacant volume) is lost, and (ii) the internal gas pressure drops, slightly reducing the He–He repulsion among the remaining $m-1$ atoms. For small clusters ($\alpha \leq 4$, $m \leq 4$), both contributions are highly configuration-dependent; for large bubbles they are captured by continuum thermodynamics.

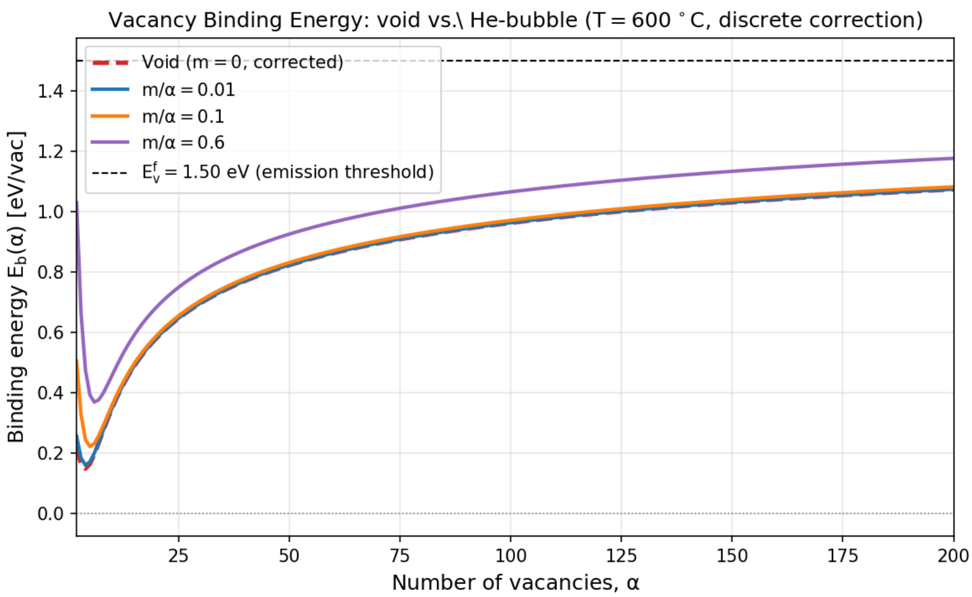


Fig. 5 Binding energy of a vacancy to a bubble as a function of the number of vacancies α, for several He-to-vacancy ratios m/α. The binding energy increases with helium content because the internal gas pressure opposes vacancy emission.

Continuum model for large bubbles.

For a spherical bubble of radius $R(\alpha)$ containing m He atoms, the chemical potential of a helium atom inside the bubble is:

$$\mu_{\rm He}^{\rm bubble} = k_BT\ln\left(\frac{m}{\alpha\Omega}\Lambda_h^3\right) + k_BT\left[2B_2\frac{m}{\alpha\Omega} + 3B_3\left(\frac{m}{\alpha\Omega}\right)^2\right], \tag{75}$$

where $\Lambda_h = h/\sqrt{2\pi m_{\rm He}k_BT}$ is the thermal de Broglie wavelength of ^{4}He. The helium binding energy in the continuum limit is the energy difference between this chemical potential and the energy of an interstitial He atom in the bulk:

$$E_B^{\rm He,cont}(\alpha, m) = E_s^{\rm He} - k_BT\ln\left(\frac{m}{\alpha\Omega}\Lambda_h^3\right) - k_BT\left[2B_2\frac{m}{\alpha\Omega} + 3B_3\left(\frac{m}{\alpha\Omega}\right)^2\right], \tag{76}$$

where $E_s^{\rm He} \approx 2.3$–2.4 eV is the solution energy of interstitial He in bulk bcc Fe (the energy cost of inserting a He atom at a tetrahedral site). The first correction term is the ideal-gas entropy contribution, and the second accounts for the non-ideal (hard-core repulsive) He–He interactions through the virial coefficients.

Equation (76) predicts two important limits:

- **Low He density** ($m/\alpha \ll 1/\Omega B_2$): the binding energy is large ($\sim E_s^{\rm He}$), since the He atom gains the full solution-energy benefit of entering the open volume.

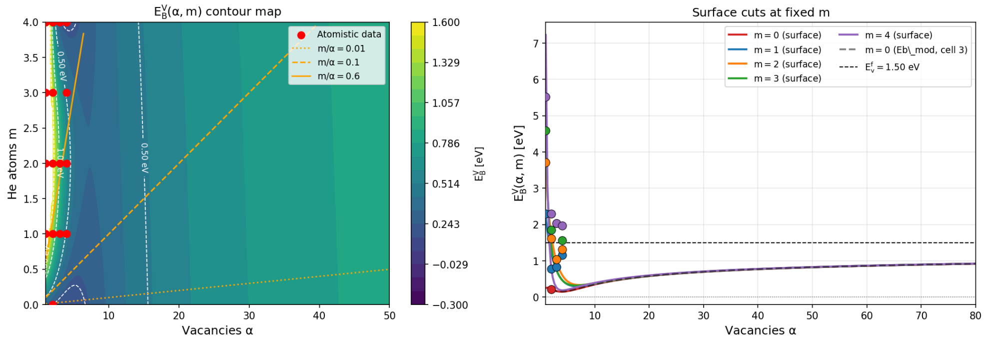


Fig. 6 Contour maps of vacancy binding energy $E_B^v(\alpha, m)$. Left: surface plot as a function of α and m. Right: cuts at fixed m showing the atomistic–continuum transition. The red line $E_B^v = E_f^v$ separates over-pressurised bubbles (above, which grow by absorbing thermal vacancies) from under-pressurised cavities (below, which tend to shrink).

- **High He density** (m/α approaching the close-packing limit): the binding energy decreases toward zero and eventually becomes negative, meaning it is energetically favourable for the He atom to leave the overpressurised bubble. This defines the maximum stable He loading.

Atomistic–continuum blending for small clusters.

By direct analogy with the vacancy binding energy model (Eq. 70), we write:

$$\boxed{E_B^{\mathrm{He}}(\alpha, m) = E_B^{\mathrm{He,cont}}(\alpha, m) + A^{\mathrm{He}}(\alpha)\, e^{-\mu(m-1)},} \tag{77}$$

where $A^{\mathrm{He}}(\alpha)$ is a fitted amplitude at each vacancy count α, and μ is a decay constant governing how quickly the atomistic correction vanishes as m increases. The exponential correction ensures that the blended expression reproduces the DFT data at small m and converges to the continuum model for large m.

The decay constant is chosen by the same criterion as for the vacancy case (correction $< 1\%$ at $m = 8$):

$$\mu = \frac{\ln 100}{7} \approx 0.658\ \mathrm{He}^{-1}. \tag{78}$$

The amplitudes $A^{\mathrm{He}}(\alpha)$ are obtained by least-squares regression against the atomistic data of Table 17:

$$A^{\mathrm{He}}(\alpha_k) = \frac{\sum_{j\in\mathcal{E}_k} \delta_j^{\mathrm{He}}\, e^{-\mu(m_j-1)}}{\sum_{j\in\mathcal{E}_k} e^{-2\mu(m_j-1)}}, \qquad \delta_j^{\mathrm{He}} = E_B^{\mathrm{He,atom}}(\alpha_k, m_j) - E_B^{\mathrm{He,cont}}(\alpha_k, m_j), \tag{79}$$

where $\mathcal{E}_k$ is the subset of the atomistic data at fixed α_k.

Table 19 Least-squares fitting amplitudes for the helium binding energy to bubbles (Eq. 77), obtained from the data of Fu and Willaime (2005) and Caturla et al (2008).

α	$A^{\mathrm{He}}(\alpha)$ [eV]
1	$\sim$0 ($\mathrm{He_1V_1}$ is the anchor; E_B^{He} undefined)
2	0.55
3	0.40
4	0.75

Trends in the He binding energy.

Table 17 and Eq. (76) reveal the following trends:

1. At fixed α, E_B^{He} **decreases** as m increases (the cavity fills up and the marginal binding drops). For example, at $\alpha = 2$: $E_B^{\mathrm{He}} = 2.85$ eV ($m = 1$), 2.75 eV ($m = 2$), 2.07 eV ($m = 3$), 2.36 eV ($m = 4$). The non-monotonic behaviour at $m = 4$ reflects configuration-dependent effects in the small DFT clusters.
2. At fixed m, E_B^{He} **increases** as α increases (larger cavities have more open volume, so each He atom is more strongly bound). For example, at $m = 1$: $E_B^{\mathrm{He}} = 2.85$ eV ($\alpha = 2$), 3.30 eV ($\alpha = 3$), 3.84 eV ($\alpha = 4$).
3. The He binding energy at $\alpha = 1$ (substitutional He) is $\sim$1.8–1.9 eV for $m \geq 2$, significantly lower than at $\alpha \geq 2$. This reflects the small trapping volume of a single vacancy.

Helium emission versus trap mutation.

An overpressurised $\mathrm{He}_m\mathrm{V}_\alpha$ cluster—one whose He content exceeds the equilibrium loading—can relieve its excess energy by two physically distinct mechanisms. These must be distinguished in the rate equations because they produce different products and obey different kinetics.

(a) Helium thermal emission.. The cluster releases a He atom into an interstitial (tetrahedral) site:

$$\mathrm{He}_m\mathrm{V}_\alpha \xrightarrow{\Gamma_{\mathrm{He}}} \mathrm{He}_{m-1}\mathrm{V}_\alpha + \mathrm{He}_{\mathrm{int}}. \tag{80}$$

The vacancy content is unchanged; one He atom is lost. The emission rate is governed by the helium binding energy:

$$\Gamma_{\mathrm{He}}(\alpha, m) = \mathcal{K}_{h,\alpha}\ \exp\left(-\frac{E_B^{\mathrm{He}}(\alpha, m)}{k_B T}\right), \tag{81}$$

where $\mathcal{K}_{h,\alpha}$ is the He capture rate constant of the daughter cluster $\mathrm{He}_{m-1}\mathrm{V}_\alpha$. This channel dominates when the He binding energy is smaller than the energy cost of creating a Frenkel pair, i.e. when $E_B^{\mathrm{He}} < E_f^i - \Omega P_g$, which is typically the case at high temperatures ($T \gtrsim 800$ K) and moderate He/V ratios.

(b) Trap mutation (Frenkel pair emission).. The cluster emits a self-interstitial atom from its periphery, simultaneously creating an additional vacancy within its own volume:

$$\mathrm{He}_m\mathrm{V}_\alpha \xrightarrow{\Gamma_{\mathrm{TM}}} \mathrm{He}_m\mathrm{V}_{\alpha+1} + \mathrm{SIA}. \tag{82}$$

The He content is unchanged; the cluster gains one vacancy and injects one SIA into the lattice. This mechanism was first proposed by Greenwood et al (1959) as "plastic deformation around an over-pressurized bubble", and direct experimental evidence for loop punching by helium bubbles was reported by Evans et al (1981) in molybdenum. The atomistic process—sometimes called *self-trapping* when it occurs from a pure interstitial He cluster ($\alpha = 0$)—was characterised in detail by Wilson et al (1981) using molecular statics, and has been studied by DFT and MD in both bcc Fe (Fu and Willaime 2007; Gao et al 2011; Morishita et al 2003) and bcc W (Becquart and Domain 2014).

The trap-mutation rate can be written in Arrhenius form:

$$\boxed{\Gamma_{\mathrm{TM}}(\alpha, m) = \nu_0 \ \exp\left(-\frac{E_{\mathrm{TM}}(\alpha, m)}{k_B T}\right)}, \tag{83}$$

where $\nu_0 \sim 10^{12}$–10^{13} s^{-1} is an attempt frequency and E_{TM} is the activation barrier for Frenkel pair emission. DFT calculations by Becquart and Domain (2014) for W and MD simulations by Morishita et al (2003) and Gao et al (2011) for Fe provide the following picture:

- For **pure interstitial He clusters** ($\alpha = 0$): self-trapping (spontaneous Frenkel pair creation) occurs at $m \approx 4$–6 in bcc Fe at $T > 300$ K, with a barrier $E_{\mathrm{TM}} \sim 0$–0.5 eV that decreases with increasing cluster size (Gao et al 2011).
- For **He$_m$V$_1$ clusters**: trap mutation occurs at $m \approx 5$–7, with barriers of $\sim$0.5–1.5 eV that decrease steeply with m (Morishita et al 2003; Fu and Willaime 2007).
- For **larger clusters** ($\alpha \geq 2$): the threshold He content for trap mutation increases roughly as $\alpha^{2/3}$ (reflecting the increased cavity volume), and the emitted SIA is strongly bound to the cluster surface (binding energy $\sim$2–3 eV) (Becquart and Domain 2014).

Competition and the maximum stable He loading.

For a given $\mathrm{He}_m\mathrm{V}_\alpha$ cluster, the dominant overpressure relief channel is the one with the lower effective barrier. The competition can be summarised as:

$$\text{If} \ \ E_B^{\mathrm{He}}(\alpha, m) < E_{\mathrm{TM}}(\alpha, m): \quad \text{He emission dominates,} \tag{84}$$

$$\text{If} \ \ E_{\mathrm{TM}}(\alpha, m) < E_B^{\mathrm{He}}(\alpha, m): \quad \text{trap mutation dominates.} \tag{85}$$

In bcc Fe, DFT data indicate that for small clusters ($\alpha \leq 4$) at He/V ratios above $\sim$1.3–1.7, the trap-mutation barrier drops below the He binding energy, making Frenkel pair emission the preferred pathway (Fu and Willaime 2005; Morishita et al 2003). This is consistent with the observation that the equilibrium He/V ratio for small clusters is $\sim$1.0–1.3 (DFT), while for larger bubbles ($R > 1$ nm) the mechanical equilibrium condition $P_g = 2\gamma/R$ gives He/V ~ 0.5 at moderate temperatures (Trinkaus and Singh 2003).

For practical implementation in cluster dynamics, the maximum He loading per vacancy cluster is:

$$\ell_{\max}(\alpha) = \lfloor \alpha_{\mathrm{He}}\, \alpha^{2/3} \rfloor, \qquad \alpha_{\mathrm{He}} \approx 1.5\text{–}2.0. \tag{86}$$

This formula should be understood as the *trap-mutation limit*: beyond $\ell_{\max}$, the cluster spontaneously emits a Frenkel pair rather than accommodating an additional He atom. The He emission limit (where $E_B^{\mathrm{He}} = 0$) occurs at a somewhat higher He/V ratio, but the trap-mutation channel intervenes first and prevents the cluster from reaching that state.

Rate-equation implementation.

In the master equation for $\mathrm{He}_m\mathrm{V}_\alpha$ clusters (Eq. **??**), both channels appear as separate loss/gain terms:

$$\left.\frac{dC_{-\alpha,m}}{dt}\right|_{\mathrm{TM+He\ em}} = \underbrace{-\Gamma_{\mathrm{He}}(\alpha,m)\, C_{-\alpha,m} + \Gamma_{\mathrm{He}}(\alpha,m+1)\, C_{-\alpha,m+1}}_{\text{He emission}} \underbrace{-\Gamma_{\mathrm{TM}}(\alpha,m)\, C_{-\alpha,m} + \Gamma_{\mathrm{TM}}(\alpha-1,m)\, C_{-(\alpha-1),}}_{\text{trap mutation}} \tag{87}$$

The trap-mutation term is a loss from $(-\alpha, m)$ and a gain for $(-(\alpha+1), m)$; simultaneously, an SIA is injected into the mono-SIA population:

$$\left.\frac{dC_{1,0}}{dt}\right|_{\mathrm{TM}} = \sum_{\alpha=1}^{M} \sum_{m=1}^{L} \Gamma_{\mathrm{TM}}(\alpha,m)\, C_{-\alpha,m}. \tag{88}$$

For clusters exceeding $\ell_{\max}(\alpha)$, a simpler implementation is to enforce $m \leq \ell_{\max}(\alpha)$ as a hard boundary in the state space and redirect all He absorption events that would violate this bound into the trap-mutation channel, as described in Section 5.2.3.

Code-ready parameters.

5.2.4 Summary of Bubble Binding Energy Models

For implementation in cluster dynamics, the vacancy and helium binding energies to a $\mathrm{He}_m\mathrm{V}_\alpha$ bubble are:

Vacancy binding:

$$E_B^v(\alpha,m) = E_f^v - \Omega\left[\frac{2\gamma}{R(\alpha)} - P_g(\alpha,m)\right] + A(m)\, e^{-\lambda(\alpha-1)}, \tag{89}$$

with $A(m)$ from Table 18 and P_g from the virial equation of state (Eq. 72).

Table 20 Trap mutation parameters for implementation in cluster dynamics.

Parameter	Symbol	Value	Source
Attempt frequency	ν_0	10^{12}–10^{13} s^{-1}	—
E_{TM}, He_5V_1		∼1.0 eV	Morishita et al (2003)
E_{TM}, He_6V_1		∼0.5 eV	Morishita et al (2003)
E_{TM}, He_7V_1		∼0 eV (spontaneous)	Gao et al (2011)
Self-trapping, He_4 ($\alpha = 0$)		∼0.3 eV	Gao et al (2011)
Self-trapping, He_6 ($\alpha = 0$)		∼0 eV (spontaneous)	Gao et al (2011)
Max He loading parameter	α_{He}	1.5–2.0	Morishita et al (2003); Fu and Willaime (2005)

Helium binding:

$$E_B^{\mathrm{He}}(\alpha, m) = E_s^{\mathrm{He}} - k_B T \ln\left(\frac{m}{\alpha\Omega}\Lambda_h^3\right) - k_B T\left[2B_2\frac{m}{\alpha\Omega} + 3B_3\left(\frac{m}{\alpha\Omega}\right)^2\right] + A^{\mathrm{He}}(\alpha)\, e^{-\mu(m-1)}, \tag{90}$$

with $A^{\mathrm{He}}(\alpha)$ from Table 19, $E_s^{\mathrm{He}} \approx 2.35$ eV, and $\mu \approx 0.658$ He^{-1}.

Equilibrium condition.

A bubble is in mechanical equilibrium when the internal gas pressure balances the surface tension:

$$P_g(\alpha, m) = \frac{2\gamma}{R(\alpha)}. \tag{91}$$

This condition simultaneously corresponds to $E_B^v = E_f^v$ (no net driving force for vacancy exchange). Bubbles above this line (over-pressurised) grow by absorbing thermal vacancies; those below (under-pressurised) shrink.

Thermal emission rates.

The vacancy and helium emission rates from a He_mV_α bubble are:

$$\epsilon_\alpha^v = \mathcal{K}_{v,\alpha-1}\, \exp\left(-\frac{E_B^v(\alpha, m)}{k_B T}\right), \tag{92}$$

$$\epsilon_m^{\mathrm{He}} = \mathcal{K}_{h,\alpha}\, \exp\left(-\frac{E_B^{\mathrm{He}}(\alpha, m)}{k_B T}\right), \tag{93}$$

where $\mathcal{K}_{v,\alpha-1}$ and $\mathcal{K}_{h,\alpha}$ are the vacancy and helium capture rate constants for the daughter cluster, as defined in the reaction rate section (Section 6.5).

The results are shown in Fig. 5 for the binding energy as a function of both m and α. Contours of constant binding energy in bubbles containing m helium atoms and α vacancies are shown in Fig. 7. The line $E_v^f = E_b^{(v)}(m, \alpha)$ defines the combination of m and α for bubbles in mechanical equilibrium. Those above the line (over-pressurized) will tend to grow by absorbing thermal vacancies, and those below the line will tend to shrink.

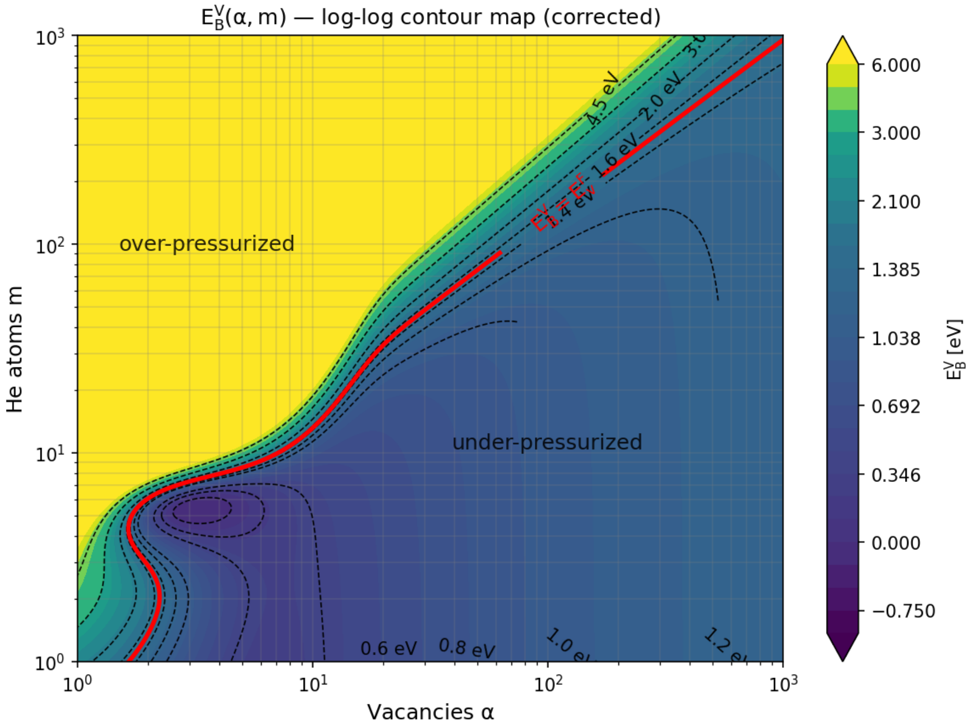


Fig. 7 Vacancy binding energy contours to bubbles containing m helium atoms and α vacancies

5.3 Dislocation Loops

For dislocation loops, the vacancy exchange chemical potential should be obtained from the loop free energy.

$$\mu = \frac{\partial G_{\mathrm{loop}}}{\partial n}, \tag{94}$$

where n is the number of point defects contained in the loop. For a faulted circular loop, G_{loop} generally contains both line-energy and fault-energy contributions. For a circular prismatic loop containing n point defects,

$$n\Omega = \pi R^2 b, \tag{95}$$

For a loop in an external stress field $\sigma_{ij}^{\mathrm{ext}}$, the loop free energy should include the mechanical work term,

$$G_{\mathrm{loop}}^{\mathrm{tot}} = G_{\mathrm{self}} + G_{\mathrm{fault}} + G_{\sigma}, \tag{96}$$

with

$$G_{\sigma} = -\sigma_{ij}^{\mathrm{ext}}\, \Omega_{ij}^{\mathrm{loop}}. \tag{97}$$

For a circular prismatic loop of radius R, plane normal n_i, and Burgers vector $b_i = b\,n_i$, the loop relaxation-volume tensor is

$$\Omega_{ij}^{\text{loop}} = s\,n\Omega\,n_i n_j = s\,\pi R^2 b\,n_i n_j, \tag{98}$$

where

$$s = \begin{cases} -1, & \text{vacancy loop,} \\ +1, & \text{interstitial loop.} \end{cases} \tag{99}$$

Hence

$$G_\sigma = -s\,n\Omega\,\sigma_{ij}^{\text{ext}} n_i n_j = -s\,\pi R^2 b\,\sigma_{ij}^{\text{ext}} n_i n_j. \tag{100}$$

The point-defect exchange chemical potential is therefore

$$\mu = \frac{\partial G_{\text{loop}}^{\text{tot}}}{\partial n} = \frac{\partial G_{\text{self}}}{\partial n} + \frac{\partial G_{\text{fault}}}{\partial n} - s\,\Omega\,\sigma_{ij}^{\text{ext}} n_i n_j. \tag{101}$$

For a faulted circular vacancy loop ($s = -1$),

$$\mu_{\text{loop}}^{(v)} = \frac{Gb\,\Omega}{4\pi(1-\nu)R}\left[\ln\left(\frac{R}{r_c}\right) + 1\right] + \frac{\gamma_{sf}\Omega}{b} + \Omega\,\sigma_{ij}^{\text{ext}} n_i n_j. \tag{102}$$

For a faulted circular interstitial loop ($s = +1$),

$$\mu_{\text{loop}}^{(i)} = \frac{Gb\,\Omega}{4\pi(1-\nu)R}\left[\ln\left(\frac{R}{r_c}\right) + 1\right] + \frac{\gamma_{sf}\Omega}{b} - \Omega\,\sigma_{ij}^{\text{ext}} n_i n_j. \tag{103}$$

r_c is the core radius of a dislocation. Utilizing Eqs. 57 and 56, we obtain the binding energy of a vacancy to an interstitial or to a vacancy loop, respectively.

5.4 Vacancy Loops

Osetsky et al. Osetsky et al (2000) studied the stability of vacancy loops in iron using atomistic simulations. Applying the continuum equations 57 and 56, together with Eq. 102 for the chemical potential of a vacancy loop without external stress, we compare the results of atomistic simulations to the continuum equations with $r_c = b/3$ and $\gamma_s f = 0.6 J/m^2$ in Fig. 8. A transition correction anchors the continuum to the atomistic divacancy binding energy $E_b(2) = 0.22$ eV via an exponential decay:

$$E_b(\alpha) = E_b^{\text{cont}}(\alpha) + \left[E_b(2) - E_b^{\text{cont}}(2)\right] e^{-\lambda(\alpha-2)} \tag{104}$$

$$\lambda = \frac{\ln 100}{8} \approx 0.575 \text{ vac}^{-1} \tag{105}$$

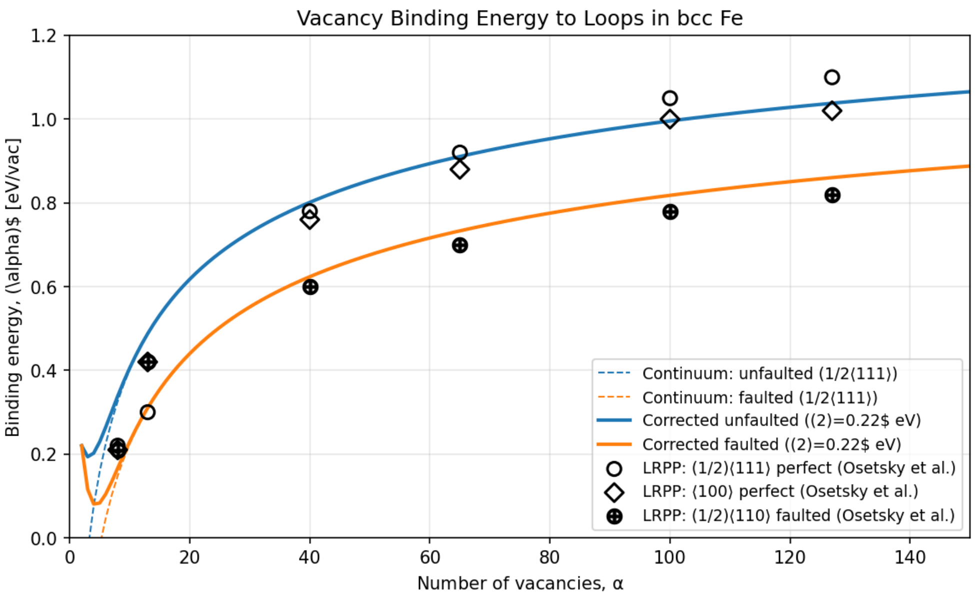


Fig. 8 Binding energy of a vacancy to a vacancy loop in iron using continuum equations (solid lines) and atomistic data from Osetsky et al. Osetsky et al (2000)

5.5 Interstitial Loops

Small-n Power-Law Fit $(n \leq 25)$

The small-cluster binding energy is fitted by least squares to the atomistic data for $n = 2$ (DFT) and $n = 7, 19$ (Osetsky MD, hexagonal and square loop configurations). The fit is written as

$$E_b^{i,\mathrm{fit}}(n) = An^B. \tag{106}$$

The fitted parameters are listed in Table 21.

Table 21 Power-law fit parameters for small interstitial loops.

Loop type	A (eV)	B
$\frac{1}{2}\langle 111\rangle$	0.7501	0.3873
$\langle 100\rangle$	0.7160	0.3581

The atomistic training data used in the fit are summarized in Table 22.

Table 22 Atomistic training data used for the small-n binding-energy fit.

n	$E_b^i - \frac{1}{2}\langle 111\rangle$ (eV)	$E_b^i - \langle 100\rangle$ (eV)	Source
2	0.90	0.90	DFT (Domain & Becquart Domain and Becquart (2001); Marinica et al.M
7 (hex)	1.65	1.30	Osetsky LRPP MD
7 (sq)	1.65	1.60	Osetsky LRPP MD
19 (hex)	2.35	2.05	Osetsky LRPP MD
19 (sq)	2.30	2.05	Osetsky LRPP MD

Blended Model with tanh Transition

To connect the small-n fit to the continuum expression, we use a smooth transition function

$$w(n) = \frac{1}{2}\left[1 + \tanh\left(\frac{n_{\mathrm{tr}} - n}{\sigma_{\mathrm{tr}}}\right)\right]. \tag{107}$$

The blended binding energy is then written as:

$$E_b^i(n) = w(n)\, E_b^{i,\mathrm{fit}}(n) + \big[1 - w(n)\big]\, E_b^{i,\mathrm{cont}}(n). \tag{108}$$

The transition parameters are listed in Table 23.

Table 23 Parameters of the smooth blending function.

Parameter	Value
n_{tr} (transition center)	25 SIAs
σ_{tr} (transition half-width)	5 SIAs

Selected values of the weight function are given in Table 24.

Table 24 Weight function $w(n)$ at selected cluster sizes.

n	$w(n)$
15	0.982
20	0.881
25	0.500
30	0.119
35	0.018

5.6 EUROFER Model Parameters: Defect Cluster Dissociation

Table 25 collects the binding-energy models and parameters controlling thermal emission from voids, bubbles, and dislocation loops. Table 26 provides the helium binding energy model parameters, and Table 27 lists the trap mutation parameters. Together, these three tables fully specify all dissociation channels in the cluster dynamics system.

Table 25 Dissociation / binding energy input parameters for EUROFER cluster dynamics: vacancy binding to voids and bubbles, and point-defect binding to dislocation loops.

Parameter	**Symbol**	**Value**	**Units**
Void (vacancy cluster) binding — vacancy emission:			
Vacancy formation energy	E_f^v	2.0	eV
Surface energy	γ_s	2.0	J/m^2
Divacancy binding (DFT)	$E_b^v(2)$	0.22	eV
Atomistic–continuum decay constant	λ	0.5756	vac^{-1}
Atomistic–continuum anchor size	$\alpha_{1/2}$	10	vacancies
Bubble — vacancy binding (Eq. 70):			
He–vacancy binding (1st He in V_1)	$E_B^V(1,1)$	2.30	eV
Fitting amplitude $A(m=0)$	$A(0)$	1.2353	eV
Fitting amplitude $A(m=1)$	$A(1)$	2.9064	eV
Fitting amplitude $A(m=2)$	$A(2)$	3.4147	eV
Fitting amplitude $A(m=3)$	$A(3)$	2.1504	eV
Fitting amplitude $A(m=4)$	$A(4)$	-0.1590	eV
He virial B_2 (600–1000 K)	B_2	1.67×10^{-29}	m^3/atom
He virial B_3 (600–1000 K)	B_3	1.84×10^{-58}	m^6/atom^2
Vacancy loop binding:			
Stacking fault energy	γ_{sf}	0.6	J/m^2
Core radius	r_c	$b/3$	—
Divacancy binding anchor	$E_b(2)$	0.22	eV
Decay constant	λ	0.575	vac^{-1}
Interstitial loop binding (power-law fit, $n \leq 25$):			
$\frac{1}{2}\langle 111\rangle$ loop: A	A_{111}	0.7501	eV
$\frac{1}{2}\langle 111\rangle$ loop: B	B_{111}	0.3873	—
$\langle 100\rangle$ loop: A	A_{100}	0.7160	eV
$\langle 100\rangle$ loop: B	B_{100}	0.3581	—
Blending transition center	n_{tr}	25	SIAs
Blending transition width	σ_{tr}	5	SIAs

Implementation notes.

1. The vacancy emission rate from a bubble $\mathrm{He}_m\mathrm{V}_\alpha$ is: $\epsilon_\alpha^v = \mathcal{K}_{v,\alpha-1}\exp(-E_B^v(\alpha,m)/k_BT)$, using E_B^v from Table 25.
2. The He emission rate is: $\epsilon_m^{\mathrm{He}} = \mathcal{K}_{h,\alpha}\exp(-E_B^{\mathrm{He}}(\alpha,m)/k_BT)$, using E_B^{He} from Table 26.

Table 26 Helium binding energy model parameters for bubbles (Eq. 90). The continuum term uses the He solution energy, the thermal de Broglie wavelength, and the virial coefficients from Table 25. The atomistic correction decays exponentially with m.

Parameter	Symbol	Value	Units
Continuum He binding (Eq. 76):			
He interstitial solution energy	E_s^{He}	2.35	eV
He atomic mass	m_{He}	4.0026	amu
He virial B_2	B_2	1.67×10^{-29}	$\mathrm{m}^3/\mathrm{atom}$
He virial B_3	B_3	1.84×10^{-58}	$\mathrm{m}^6/\mathrm{atom}^2$
Atomistic–continuum blending (Eq. 77):			
Decay constant	μ	0.658	He^{-1}
Fitting amplitude $A^{\mathrm{He}}(\alpha = 2)$	$A^{\mathrm{He}}(2)$	0.55	eV
Fitting amplitude $A^{\mathrm{He}}(\alpha = 3)$	$A^{\mathrm{He}}(3)$	0.40	eV
Fitting amplitude $A^{\mathrm{He}}(\alpha = 4)$	$A^{\mathrm{He}}(4)$	0.75	eV

Table 27 Trap mutation (Frenkel pair emission) parameters for EUROFER cluster dynamics. Trap mutation is the reaction $\mathrm{He}_m\mathrm{V}_\alpha \to \mathrm{He}_m\mathrm{V}_{\alpha+1}$ + SIA, with rate $\Gamma_{\mathrm{TM}} = \nu_0 \exp(-E_{\mathrm{TM}}/k_B T)$. Self-trapping refers to the $\alpha = 0$ case (pure interstitial He cluster). Values from Morishita et al (2003), Gao et al (2011), Fu and Willaime (2005), and Becquart and Domain (2014).

Parameter	Symbol	Value	Units
Trap mutation rate: $\Gamma_{\mathrm{TM}} = \nu_0 \exp(-E_{\mathrm{TM}}/k_B T)$			
Attempt frequency	ν_0	10^{12}–10^{13}	s^{-1}
Activation barriers $E_{\mathrm{TM}}(\alpha, m)$ *for* $He_m V_\alpha$ *in bcc Fe:*			
$\mathrm{He}_5\mathrm{V}_1$	$E_{\mathrm{TM}}(1, 5)$	$\sim$1.0	eV
$\mathrm{He}_6\mathrm{V}_1$	$E_{\mathrm{TM}}(1, 6)$	$\sim$0.5	eV
$\mathrm{He}_7\mathrm{V}_1$	$E_{\mathrm{TM}}(1, 7)$	$\sim$0 (spontaneous)	eV
Self-trapping barriers ($\alpha = 0$*, pure He clusters):*			
He_4	$E_{\mathrm{TM}}(0, 4)$	$\sim$0.3	eV
He_5	$E_{\mathrm{TM}}(0, 5)$	$\sim$0.1	eV
He_6	$E_{\mathrm{TM}}(0, 6)$	$\sim$0 (spontaneous)	eV
Maximum He loading (trap-mutation limit):			
He/V ratio parameter $\ell_{\max}(\alpha) = \lfloor \alpha_{\mathrm{He}}\, \alpha^{2/3} \rfloor$	α_{He}	1.5–2.0	—
Competition criterion:			
He emission dominates if $E_B^{\mathrm{He}}(\alpha, m) < E_{\mathrm{TM}}(\alpha, m)$			
Trap mutation dominates if $E_{\mathrm{TM}}(\alpha, m) < E_B^{\mathrm{He}}(\alpha, m)$			
In bcc Fe: trap mutation wins at He/V $\gtrsim$ 1.3–1.7 for $\alpha \leq 4$			

3. The trap-mutation rate is: $\Gamma_{\rm TM} = \nu_0 \exp(-E_{\rm TM}(\alpha, m)/k_B T)$, using $E_{\rm TM}$ from Table 27. Each trap-mutation event is a loss from $(-\alpha, m)$ and a simultaneous gain for $(-(\alpha+1), m)$, with one SIA added to the mono-interstitial population.
4. For clusters at or above $\ell_{\max}(\alpha)$, trap mutation can be enforced as a hard boundary: any He absorption event that would push $m > \ell_{\max}$ is immediately redirected into a Frenkel pair emission step.
5. The $\ell_{\max}$ formula represents the trap-mutation limit, not the He emission limit. The He emission limit (where $E_B^{\rm He} = 0$) lies at a somewhat higher He/V ratio, but trap mutation intervenes first.

6 Reaction Rate Constants

This section derives the rate constants for all reactions between defect clusters, expressed in atomic-fraction units (s^{-1}) and in terms of the reaction frequencies $\omega_\alpha = D_\alpha/a^2$ and the geometric prefactor $\xi(m) = (3m/8\pi)^{1/3}$ introduced in the diffusion section. The cluster notation (m, ℓ) and concentration units (at/at) follow the conventions established in Section **??**.

6.1 3D Diffusion-Limited Reactions

For a mobile point defect α absorbed by an immobile spherical cluster of m defects, the Smoluchowski volumetric rate constant is $\mathcal{K}^{\rm vol} = 4\pi r_m(D_A + D_B)$. Converting to atomic-fraction units ($\mathcal{K} = \mathcal{K}^{\rm vol}/\Omega$) and substituting $D = \omega a^2$, $r_m = \xi(m)\, a$, and $\Omega = a^3/2$:

$$\mathcal{K}^{3D}_{\alpha,m} = 8\pi\, \xi(m)\, \omega_\alpha \qquad [\mathrm{s}^{-1}]. \tag{109}$$

When both reactants are mobile, the rate constant includes both frequencies:

$$\mathcal{K}^{3D}_{AB} = 8\pi\, \xi_{\rm rec}\, (\omega_A + \omega_B), \tag{110}$$

with $\xi_{\rm rec} = \sqrt{3}/2 \approx 0.866$ for nearest-neighbour capture in bcc.

Special cases.

Vacancy–SIA recombination ($m = n = 1$, both mobile):

$$\mathcal{K}_{iv} = 4\sqrt{3}\,\pi\, (\omega_i + \omega_v) \approx 21.8\, \omega_i, \tag{111}$$

since $\omega_i \gg \omega_v$ at all relevant temperatures.

Point defect absorption by a spherical cluster (void or bubble) of m vacancies:

$$\mathcal{K}^{\rm void}_{\alpha,m} = A_{\rm sph}\, m^{1/3}\, \omega_\alpha, \qquad A_{\rm sph} = (48\pi^2)^{1/3} \approx 7.818. \tag{112}$$

This is the recommended code-ready form: a single precomputed constant multiplied by $m^{1/3}$ and the appropriate reaction frequency.

Point defect absorption by a dislocation loop of n SIAs requires species-specific rate constants because only SIAs experience a non-trivial elastic bias ($Z_i^{\rm loop} > 1$), while vacancies are unbiased ($Z_v^{\text{loop}} = 1$):

$$\mathcal{K}_{i,n}^{\text{loop}} = A_{\text{loop}}\, n^{1/2}\, Z_i^{\text{loop}}\, \omega_i, \tag{113}$$

$$\mathcal{K}_{v,n}^{\text{loop}} = A_{\text{loop}}\, n^{1/2}\, \omega_v, \tag{114}$$

with $A_{\text{loop}} = 8\sqrt{\pi/\sqrt{3}} \approx 10.78$ and $Z_i^{\text{loop}} \approx 1.02$–$1.10$ (Table 29). SIA absorption grows the loop by one; vacancy absorption shrinks it by one.

Point defect absorption by the dislocation network (first-order sink):

$$\left.\frac{dc_\alpha}{dt}\right|_{\text{disl}} = -Z_\alpha\, \rho_d\, a^2\, \omega_\alpha\, c_\alpha, \tag{115}$$

where ρ_d (m^{-2}) is the dislocation line density and no n_{at} conversion is needed because this term is first-order in c_α.

6.2 1D Diffusion-Limited Reactions (SIA Clusters)

SIA clusters of size $n \geq 4$ migrate by 1D glide along $\langle 111 \rangle$ with reaction frequency (from Section 4.4):

$$\omega_n^{1D} = \frac{3}{2} \cdot \frac{\nu_0}{n^s} \exp\!\left(-\frac{E_m^{1D}}{k_B T}\right), \qquad s \approx 0.5\text{–}1.0, \quad E_m^{1D} \lesssim 0.05 \text{ eV}. \tag{116}$$

6.2.1 Mixed 1D/3D regime (finite mean free path L)

The effective 3D reaction frequency after Burgers vector rotation is:

$$\omega_n^{3D,\text{eff}} = \frac{\omega_n^{1D}\, L}{3\sqrt{3}\, a}. \tag{117}$$

The rate constant for an n-SIA cluster reacting with an immobile void of m vacancies is then:

$$\mathcal{K}_{n,m}^{\text{mixed}} = A_{\text{sph}}\, m^{1/3}\, \omega_n^{3D,\text{eff}} = \frac{A_{\text{sph}}}{3\sqrt{3}}\, m^{1/3}\, \hat{L}\, \omega_n^{1D} \approx 1.505\, m^{1/3}\, \hat{L}\, \omega_n^{1D}, \tag{118}$$

where $\hat{L} = L/a$ is the dimensionless mean free path.

6.2.2 Pure 1D regime ($L \to \infty$)

For a 1D-migrating cluster reacting with randomly distributed spherical sinks of size m and atomic fraction $c_{m,0}$, the loss rate is:

$$\left.\frac{dc_n}{dt}\right|_{\text{sink}} = -\mathcal{K}_{n,m}^{1D}\, \bar{c}_m^2\, c_n, \tag{119}$$

where:

$$\mathcal{K}^{1D}_{n,m} = A_{1D}\, m^{4/3}\, \omega^{1D}_n, \qquad A_{1D} = \frac{9}{8\pi^{2/3}} \approx 2.632. \tag{120}$$

The critical difference from 3D: the rate constant scales as $m^{4/3}$ (not $m^{1/3}$) and is second-order in sink concentration (not first-order). This stronger sensitivity to sink geometry is the physical origin of the production bias.

6.2.3 Interpolation formula

For intermediate L, the Trinkaus–Singh–Golubov interpolation gives:

$$\mathcal{K}^{\text{eff}}_{n,m} = \frac{A_{\text{sph}}\, m^{1/3}\, \omega^{1D}_n}{1 + B_{\text{rot}}\, \hat{L}^2\, m^{-1/3}}, \qquad B_{\text{rot}} = \frac{4}{\pi}\left(\frac{8\pi}{3}\right)^{1/3} \approx 2.627. \tag{121}$$

In the limits: $\hat{L} \to 0$ recovers the 3D rate constant $A_{\text{sph}}\, m^{1/3}\, \omega^{1D}_n$; $\hat{L} \to \infty$ gives a rate proportional to $m^{2/3}\, \hat{L}^{-2}$, scaling as r^2 as expected for 1D kinetics.

6.3 Thermal Emission Rates

Thermal emission of species α from a cluster of size n is related to the capture rate by detailed balance. The daughter cluster has size $(n-1)$, so the capture rate that enters the detailed-balance relation uses $(n-1)$:

$$\alpha_\alpha(n) = A_{\text{sph}}\, (n-1)^{1/3}\, \omega_\alpha\, \exp\!\left(-\frac{E^\alpha_b(n)}{k_B T}\right) \qquad [\text{s}^{-1}], \tag{122}$$

where $E^\alpha_b(n)$ is the binding energy of the last defect of type α to the cluster (from Section **??**). The emission enters the master equation as a first-order process:

$$\left.\frac{dc_n}{dt}\right|_{\text{emission}} = +\alpha_\alpha(n)\, c_n - \alpha_\alpha(n+1)\, c_{n+1}. \tag{123}$$

For a dislocation loop emitting an SIA, replace $A_{\text{sph}}\, (n-1)^{1/3}$ with $A_{\text{loop}}\, (n-1)^{1/2}$:

$$\alpha^{\text{loop}}_i(n) = A_{\text{loop}}\, (n-1)^{1/2}\, \omega_i\, \exp\!\left(-\frac{E^i_b(n)}{k_B T}\right). \tag{124}$$

For $\text{He}_\ell \text{V}_m$ bubbles, three emission channels exist (Section 5.2):
Vacancy emission:

$$\varepsilon_v(m,\ell) = A_{\text{sph}}\, (m-1)^{1/3}\, \omega_v\, \exp\!\left(-\frac{E^v_b(m,\ell)}{k_B T}\right). \tag{125}$$

Helium emission:

$$\varepsilon_h(m,\ell) = A_{\text{sph}}\, m^{1/3}\, \omega_h\, \exp\!\left(-\frac{E^{\text{He}}_B(m,\ell)}{k_B T}\right). \tag{126}$$

He emission uses $m^{1/3}$ (not $(m-1)^{1/3}$) because removal of a He atom does not change the vacancy count; the capture radius for detailed balance is that of the daughter $\mathrm{He}_{\ell-1}\mathrm{V}_m$, which retains the same m. The binding energy E_B^{He} is from Eq. (90).

Trap mutation (Frenkel pair emission):

$$\Gamma_{\mathrm{TM}}(m,\ell) = \nu_0 \exp\left(-\frac{E_{\mathrm{TM}}(m,\ell)}{k_B T}\right), \tag{127}$$

producing $\mathrm{He}_\ell\mathrm{V}_{m+1}$ + SIA. The trap-mutation barrier E_{TM} and the competition with He emission are discussed in Section 5.2.3; recommended values are in Table 27.

6.4 Summary of Precomputed Constants

All reaction-rate geometry is captured by four numerical constants:

$$\begin{aligned}
A_{\mathrm{sph}} &= (48\pi^2)^{1/3} && \approx 7.818 && \text{(spherical clusters)},\\
A_{\mathrm{loop}} &= 8\sqrt{\pi/\sqrt{3}} && \approx 10.78 && \text{(dislocation loops)},\\
A_{1D} &= 9/(8\pi^{2/3}) && \approx 2.632 && \text{(pure 1D sinks)},\\
B_{\mathrm{rot}} &= (4/\pi)(8\pi/3)^{1/3} && \approx 2.627 && \text{(1D/3D interpolation)}.
\end{aligned} \tag{128}$$

Every rate constant in the cluster dynamics system is built from these constants, the cluster size raised to a geometry-dependent power, and the appropriate reaction frequency ω_α. Table 28 collects the complete set.

6.5 EUROFER Model Parameters: Reaction Rates

Table 29 collects the rate-constant prefactors, sink parameters, and state-space controls used in the reaction-rate expressions (Section 6). Table 30 provides the complete set of rate constant formulas for direct implementation.

Implementation notes.

Every rate constant is built from one of the four precomputed geometric constants (A_{sph}, A_{loop}, A_{1D}, B_{rot}), the cluster size raised to a geometry-dependent power, and the appropriate effective reaction frequency $\omega_\alpha^{\mathrm{eff}}$ (Eq. 129). The binding energies $E_b^\alpha(m,\ell)$ for voids and loops come from Section 5.6; the bubble binding energies $E_B^v(m,\ell)$ and $E_B^{\mathrm{He}}(m,\ell)$ from Tables 18 and 19; and the trap mutation barriers from Table 27. He emission dominates when $E_B^{\mathrm{He}} < E_{\mathrm{TM}}$; trap mutation dominates when $E_{\mathrm{TM}} < E_B^{\mathrm{He}}$, typically at He/V $\gtrsim$ 1.3–1.7 for small clusters in bcc Fe.

Table 28 Summary of rate constant expressions. All rate constants are in s^{-1} and all concentrations in at/at. The effective reaction frequencies $\omega_\alpha^{\text{eff}}$ incorporate solute-trapping corrections (Table 16) and should replace ω_α in all rate-equation implementations.

Process	**Rate constant**	**Rate equation term**
V–SIA recombination	$\mathcal{K}_{iv} = 4\sqrt{3}\pi(\omega_i^{\text{eff}} + \omega_v^{\text{eff}})$	$-\mathcal{K}_{iv}\, c_1\, c_{1,0}$
α capture by void/bubble (m,ℓ)	$\mathcal{K}_{\alpha,m} = A_{\text{sph}}\, m^{1/3}\, \omega_\alpha^{\text{eff}}$	$-\mathcal{K}_{\alpha,m}\, c_\alpha\, c_{m,\ell}$
i capture by loop n (biased)	$\mathcal{K}_{i,n}^{\text{loop}} = A_{\text{loop}}\, n^{1/2}\, Z_i^{\text{loop}}\, \omega_i^{\text{eff}}$	$-\mathcal{K}_{i,n}^{\text{loop}}\, c_1\, c_n$
v capture by loop n (unbiased)	$\mathcal{K}_{v,n}^{\text{loop}} = A_{\text{loop}}\, n^{1/2}\, \omega_v^{\text{eff}}$	$-\mathcal{K}_{v,n}^{\text{loop}}\, c_{1,0}\, c_n$
Dislocation sink	$\mathcal{D}_\alpha^d = Z_\alpha^d \rho_d a^2 \omega_\alpha^{\text{eff}}$	$-\mathcal{D}_\alpha^d\, c_\alpha$
Grain-boundary sink	$\mathcal{D}_\alpha^{gb} = (4\pi^2/d_g^2)\, a^2 \omega_\alpha^{\text{eff}}$	$-\mathcal{D}_\alpha^{gb}\, c_\alpha$
Precipitate sink	$\mathcal{D}_\alpha^p = (8\pi r_p/a)\, c_p^N\, \omega_\alpha^{\text{eff}}$	$-\mathcal{D}_\alpha^p\, c_\alpha$
SIA cluster (mixed 1D/3D)	$\mathcal{K}_{n,m}^{\text{eff}} = \dfrac{A_{\text{sph}}\, m^{1/3}\, \omega_n^{1D}}{1 + B_{\text{rot}} \hat{L}^2 m^{-1/3}}$	$-\mathcal{K}_{n,m}^{\text{eff}}\, \bar{c}_m\, c_n$
SIA cluster (pure 1D)	$\mathcal{K}_{n,m}^{1D} = A_{1D}\, m^{4/3}\, \omega_n^{1D}$	$-\mathcal{K}_{n,m}^{1D}\, \bar{c}_m^2\, c_n$
V emission from void/bubble (m,ℓ)	$\varepsilon_v(m,\ell) = A_{\text{sph}}(m-1)^{1/3} \omega_v^{\text{eff}}\, e^{-E_b^v/k_B T}$	first-order in $c_{m,\ell}$
SIA emission from loop n	$\varepsilon_i(n) = A_{\text{loop}}(n-1)^{1/2} \omega_i^{\text{eff}}\, e^{-E_b^i/k_B T}$	first-order in c_n
He emission from bubble (m,ℓ)	$\varepsilon_h(m,\ell) = A_{\text{sph}}\, m^{1/3}\, \omega_h^{\text{eff}}\, e^{-E_B^{\text{He}}/k_B T}$	first-order in $c_{m,\ell}$
Trap mutation	$\Gamma_{\text{TM}}(m,\ell) = \nu_0\, e^{-E_{\text{TM}}/k_B T}$	$(m,\ell) \to (m{+}1,\ell) + \text{SIA}$
Radiation re-solution	$\Gamma_{\text{res}}(m,\ell) = b_0\, \ell\, \dot{\phi}$	$(m,\ell) \to (m,\ell{-}1) + \text{He}_{\text{int}}$

Table 29 Reaction rate input parameters for EUROFER cluster dynamics.

Parameter	**Symbol**	**Value**	**Units**
Geometric prefactors (precomputed constants):			
Spherical cluster	$A_{\mathrm{sph}} = (48\pi^2)^{1/3}$	7.818	—
Dislocation loop	$A_{\mathrm{loop}} = 8\sqrt{\pi/\sqrt{3}}$	10.78	—
Pure 1D sink	$A_{1D} = 9/(8\pi^{2/3})$	2.632	—
1D/3D interpolation	$B_{\mathrm{rot}} = (4/\pi)(8\pi/3)^{1/3}$	2.627	—
V–SIA recombination	$4\sqrt{3}\pi$	21.77	—
Dislocation sink:			
Network dislocation density	ρ_d	10^{13}–10^{14}	m^{-2}
Interstitial bias factor	Z_i^d	1.02–1.15	—
Vacancy bias factor	Z_v^d	1.00	—
Grain-boundary sink:			
Grain diameter	d_g	5–20	μm
Bias factors	$Z_i^{gb} = Z_v^{gb}$	1.00	—
Precipitate sink:			
Mean precipitate radius	r_p	10–50	nm
Precipitate number density	N_p	10^{20}–10^{21}	m^{-3}
Bias factors	$Z_i^p = Z_v^p$	1.00	—
Mobility cutoffs:			
Max mobile SIA cluster	$n_{\max}^i$	100	SIAs
Max mobile vacancy cluster	$m_{\max}^v$	5	vacancies
He state-space parameters:			
Max He loading parameter	α_{He}	1.5–2.0	—
$\ell_{\max}(m) = \lfloor \alpha_{\mathrm{He}}\, m^{2/3} \rfloor$			(trap-mutation limit)
Re-solution parameter (fission)	b_0	10^{-2}	$\mathrm{dpa}^{-1}/\mathrm{He}$
Re-solution parameter (fusion)	b_0	10^{-1}	$\mathrm{dpa}^{-1}/\mathrm{He}$
Trap mutation (Section 5.2.3):			
Attempt frequency	ν_0^{TM}	10^{12}–10^{13}	s^{-1}
$E_{\mathrm{TM}}(\mathrm{He_5V_1})$		$\sim$1.0	eV
$E_{\mathrm{TM}}(\mathrm{He_6V_1})$		$\sim$0.5	eV
$E_{\mathrm{TM}}(\mathrm{He_7V_1})$		$\sim$0 (spontaneous)	eV
Size grid (user-defined):			
Max SIA cluster tracked	N	user input	—
Max vacancy cluster tracked	M	user input	—
Max He per cluster	L	user input	—
Grouping transition size	n_{group}	$\sim$50–200	—

Table 30 Complete rate constant expressions for all processes in the cluster dynamics system. All rate constants are in s^{-1}; all concentrations in at/at. The effective frequencies $\omega_\alpha^{\text{eff}}$ are defined in Eq. (129). Binding energies E_b^α, E_B^v, E_B^{He}, and E_{TM} are from Section **??**.

Process	**Rate constant / rate expression**
Capture (bimolecular):	
V–SIA recombination	$\mathcal{K}_{iv} = 4\sqrt{3}\pi\,(\omega_i^{\text{eff}} + \omega_v^{\text{eff}})$
α by void/bubble (m,ℓ)	$\mathcal{K}_{\alpha,m} = A_{\text{sph}}\,m^{1/3}\,\omega_\alpha^{\text{eff}}$
α by loop n (SIA, biased)	$\mathcal{K}_{i,n}^{\text{loop}} = A_{\text{loop}}\,n^{1/2}\,Z_i^{\text{loop}}\,\omega_i^{\text{eff}}$
α by loop n (V, unbiased)	$\mathcal{K}_{v,n}^{\text{loop}} = A_{\text{loop}}\,n^{1/2}\,\omega_v^{\text{eff}}$
He by bubble (m,ℓ)	$\mathcal{K}_{h,m} = A_{\text{sph}}\,m^{1/3}\,\omega_h^{\text{eff}}$
Fixed sinks (first-order):	
α to dislocation network	$\mathcal{D}_\alpha^d = Z_\alpha^d\,\rho_d\,a^2\,\omega_\alpha^{\text{eff}}$
α to grain boundaries	$\mathcal{D}_\alpha^{gb} = (4\pi^2/d_g^2)\,a^2\,\omega_\alpha^{\text{eff}}$
α to precipitates	$\mathcal{D}_\alpha^p = (8\pi r_p/a)\,c_p^N\,\omega_\alpha^{\text{eff}}$
SIA cluster reactions (1D/3D):	
Mixed 1D/3D with void/bubble m	$\mathcal{K}_{n,m}^{\text{eff}} = A_{\text{sph}}\,m^{1/3}\,\omega_n^{1D}\,/\,(1 + B_{\text{rot}}\hat{L}^2 m^{-1/3})$
Pure 1D with void/bubble m	$\mathcal{K}_{n,m}^{1D} = A_{1D}\,m^{4/3}\,\omega_n^{1D}$; rate $\propto \bar{c}_m^2\,c_n$
Thermal emission (first-order):	
V from void/bubble (m,ℓ)	$\varepsilon_v(m,\ell) = A_{\text{sph}}\,(m-1)^{1/3}\,\omega_v^{\text{eff}}\,e^{-E_b^v(m,\ell)/k_BT}$
SIA from loop n	$\varepsilon_i(n) = A_{\text{loop}}\,(n-1)^{1/2}\,\omega_i^{\text{eff}}\,e^{-E_b^i(n)/k_BT}$
He from bubble (m,ℓ)	$\varepsilon_h(m,\ell) = A_{\text{sph}}\,m^{1/3}\,\omega_h^{\text{eff}}\,e^{-E_B^{\text{He}}(m,\ell)/k_BT}$
Trap mutation (first-order):	
$\text{He}_\ell\text{V}_m \to \text{He}_\ell\text{V}_{m+1}$ + SIA	$\Gamma_{\text{TM}}(m,\ell) = \nu_0\,e^{-E_{\text{TM}}(m,\ell)/k_BT}$
Radiation re-solution (athermal):	
He ejection from bubble	$\Gamma_{\text{res}}(m,\ell) = b_0\,\ell\,\dot{\phi}$

7 Master Equations

This section assembles the complete cluster dynamics system for EUROFER97 under neutron irradiation. All concentrations are atomic fractions (at/at); all rate constants carry units of s^{-1}; and the fundamental kinetic variables are the EUROFER effective reaction frequencies

$$\omega_\alpha^{\text{eff}} = \frac{\nu_\alpha \, \exp(-E_m^\alpha / k_B T)}{1 + \sum_s z_s \, c_s \, \exp(E_b^{\alpha\text{-}s}/k_B T)}, \tag{129}$$

which incorporate the solute-trapping corrections from Sections ??–4.6.5 (Table 16).

The system contains three primary species populations:

1. **SIA clusters** c_n, $n = 1, 2, \ldots, N$. These are dislocation loops with Burgers vector $\frac{1}{2}\langle 111 \rangle$ (or single $\langle 110 \rangle$ dumbbells for $n = 1$). They never contain He.
2. **Vacancy clusters** $c_{m,\ell}$, $m = 1, \ldots, M$, $\ell = 0, \ldots, \ell_{\max}(m)$. These are spherical cavities: pure voids when $\ell = 0$, He bubbles when $\ell \geq 1$. The case $m = 1$, $\ell = 0$ is the free monovacancy.
3. **Free interstitial helium** c_h.

Both n and m are positive integers (no sign convention).

7.1 Bias Factors and Sink Types

The elastic self-stress field of a sink preferentially attracts SIAs over vacancies owing to the larger relaxation volume of the SIA. The magnitude of this preference depends on the *sink geometry*:

Sink type	Z_i	Z_v	**Notes**
Spherical cavity (void/bubble)	1	1	No long-range stress field
SIA dislocation loop	Z_i^{loop}	1	$\approx$ 1.02–1.10; input parameter
Network dislocation	Z_i^d	1	$\approx$ 1.02–1.15; input parameter
Incoherent precipitate	1	1	MX, $M_{23}C_6$ in EUROFER
Grain boundary	1	1	Unbiased planar sink

In all cases the vacancy bias is unity. Only dislocations and SIA loops carry a non-trivial bias. The loop bias Z_i^{loop} is approximately independent of loop size.

7.2 Reaction Processes

Every elementary reaction falls into one of eight categories, each mapping to the rate constants of Section 6.

P1: Vacancy–SIA recombination.

$I_1 + V_1 \to 0$. Both mobile; rate constant:

$$\mathcal{K}_{iv} = 4\sqrt{3}\,\pi\,(\omega_i^{\text{eff}} + \omega_v^{\text{eff}}) \approx 21.8\,\omega_i^{\text{eff}}. \tag{130}$$

P2: Point-defect absorption by a spherical cavity.

A mobile species $\alpha \in \{i, v, h\}$ is absorbed by a cavity of m vacancies. Spherical cavities carry no elastic bias ($Z_i^{\text{cav}} = Z_v^{\text{cav}} = 1$):

$$\mathcal{K}^{\text{cav}}_{\alpha,m} = A_{\text{sph}}\, m^{1/3}\, \omega_\alpha^{\text{eff}}, \qquad A_{\text{sph}} = (48\pi^2)^{1/3} \approx 7.818. \tag{131}$$

The He content ℓ does not enter the rate constant because the capture cross-section depends on the cavity radius ($\propto m^{1/3}$), not on the He loading.

P3: Point-defect absorption by a dislocation loop.

SIA absorption is biased; vacancy absorption is not:

$$\mathcal{K}^{\text{loop}}_{i,n} = A_{\text{loop}}\, n^{1/2}\, Z_i^{\text{loop}}\, \omega_i^{\text{eff}}, \tag{132}$$

$$\mathcal{K}^{\text{loop}}_{v,n} = A_{\text{loop}}\, n^{1/2}\, \omega_v^{\text{eff}}, \tag{133}$$

with $A_{\text{loop}} = 8\sqrt{\pi/\sqrt{3}} \approx 10.78$. Vacancy absorption shrinks the loop by one SIA; SIA absorption grows it by one.

P4: Point-defect absorption by fixed sinks.

Three fixed-sink types contribute a combined first-order loss coefficient:

$$\mathcal{D}_\alpha = \mathcal{D}_\alpha^d + \mathcal{D}_\alpha^{gb} + \mathcal{D}_\alpha^p \qquad [\text{s}^{-1}], \tag{134}$$

where

$$\mathcal{D}_\alpha^d = Z_\alpha^d\, \rho_d\, a^2\, \omega_\alpha^{\text{eff}}, \tag{135}$$

$$\mathcal{D}_\alpha^{gb} = \frac{4\pi^2}{d_g^2}\, a^2\, \omega_\alpha^{\text{eff}}, \tag{136}$$

$$\mathcal{D}_\alpha^p = \frac{8\pi\, r_p}{a}\, c_p^N\, \omega_\alpha^{\text{eff}}. \tag{137}$$

Here ρ_d (m^{-2}) is the dislocation line density with bias $Z_i^d > 1$, $Z_v^d = 1$; d_g is the grain diameter; r_p and c_p^N are the precipitate radius and number fraction. Grain boundaries and precipitates are unbiased ($Z_i^{gb} = Z_v^{gb} = Z_i^p = Z_v^p = 1$). For EUROFER with $d_g \approx 5$–20 μm, $k_{gb}^2 = 4\pi^2/d_g^2 \approx 4 \times 10^{11}$–$6 \times 10^{12}$ m^{-2}; with $r_p \approx 10$–50 nm and $N_p \approx 10^{20}$–10^{21} m^{-3}, $k_p^2 = 4\pi r_p N_p \approx 10^{13}$–$10^{14}$ m^{-2}, comparable to the dislocation contribution.

P5: Thermal emission (detailed balance).

Vacancy emission from a cavity (m, ℓ):

$$\varepsilon_v(m, \ell) = A_{\text{sph}}\, (m-1)^{1/3}\, \omega_v^{\text{eff}}\, \exp\!\big(-E_b^v(m, \ell)/k_B T\big). \tag{138}$$

SIA emission from a loop of n SIAs:

$$\varepsilon_i(n) = A_{\text{loop}}\,(n-1)^{1/2}\,\omega_i^{\text{eff}}\,\exp\bigl(-E_b^i(n)/k_BT\bigr). \tag{139}$$

He emission from a bubble (m,ℓ):

$$\varepsilon_h(m,\ell) = A_{\text{sph}}\,m^{1/3}\,\omega_h^{\text{eff}}\,\exp\bigl(-E_B^{\text{He}}(m,\ell)/k_BT\bigr). \tag{140}$$

He emission uses $m^{1/3}$ (not $(m-1)^{1/3}$) because He removal does not change the vacancy count; the capture radius relevant for detailed balance is that of the daughter $\text{He}_{\ell-1}\text{V}_m$, which retains the same m. Binding energies are from Section 5.6.

P6: SIA cluster interaction with sinks (mixed 1D/3D).

An n-SIA cluster ($n \geq 4$) gliding in 1D is absorbed by a cavity of m vacancies (spherical cavities carry no bias):

$$\mathcal{K}_{n,m}^{\text{eff}} = \frac{A_{\text{sph}}\,m^{1/3}\,\omega_n^{1D}}{1 + B_{\text{rot}}\,\hat{L}^2\,m^{-1/3}}, \qquad B_{\text{rot}} \approx 2.627, \tag{141}$$

with $\hat{L} = L/a$ the EUROFER solute-shortened dimensionless mean free path (Table 16).

P7: Trap mutation (Frenkel pair emission).

An over-pressurised bubble emits a self-interstitial: $\text{He}_\ell\text{V}_m \to \text{He}_\ell\text{V}_{m+1}$ + SIA.

$$\Gamma_{\text{TM}}(m,\ell) = \nu_0\,\exp\bigl(-E_{\text{TM}}(m,\ell)/k_BT\bigr). \tag{142}$$

This dominates over He emission when $E_{\text{TM}} < E_B^{\text{He}}$, typically at He/V $\gtrsim$ 1.3–1.7 for small clusters (Table 27).

P8: Radiation re-solution (athermal).

A cascade ejects a He atom from a bubble: $\text{He}_\ell\text{V}_m \to \text{He}_{\ell-1}\text{V}_m + \text{He}_{\text{int}}$.

$$\Gamma_{\text{res}}(m,\ell) = b_0\,\ell\,\dot{\phi}, \tag{143}$$

with $b_0 \approx 10^{-2}$ (fission) or 10^{-1} (fusion) dpa^{-1}/He.

Mobility cutoffs.

Mobile clusters are bounded by species-specific size limits taken from Table 29:

- SIA clusters: mobile (3D or 1D) for $1 \leq n \leq n_{\max}^i$. Clusters with $n > n_{\max}^i$ are treated as immobile sessile loops and contribute only as fixed sinks; their reaction frequencies are set to zero.
- Vacancy clusters: mobile (3D) for $1 \leq m \leq m_{\max}^v$. Clusters with $m > m_{\max}^v$ are immobile and can only grow or shrink through absorption or emission of a single mobile vacancy or interstitial.

General (coalescence) formulation.

The most complete description of cluster kinetics allows any two mobile clusters to react. The three reaction channels and their rate constants are:

SIA–SIA coalescence ($I_n + I_{n'} \to I_{n+n'}$, both mobile):

$$\mathcal{K}^{ii}_{n,n'} = 8\pi\,(\xi_n + \xi_{n'})\,(\omega^{\text{eff}}_n + \omega^{\text{eff}}_{n'}), \qquad 1 \le n' \le n \le n^i_{\max}, \tag{144}$$

with $\xi_n = (3n/8\pi)^{1/3}$ the dimensionless cluster radius. The constraint $n' \le n$ avoids double-counting; a factor $\frac{1}{2}$ multiplies the self-collision term ($n = n'$). The contribution to dc_n/dt is:

$$\left.\frac{dc_n}{dt}\right|_{ii} = \frac{1}{2} \sum_{n'=1}^{\min(n-1,\, n^i_{\max})} \mathcal{K}^{ii}_{n',n-n'}\, c_{n'}\, c_{n-n'} \;-\; c_n \sum_{n'=1}^{n^i_{\max}} \mathcal{K}^{ii}_{n,n'}\, c_{n'}. \tag{145}$$

Vacancy–vacancy coalescence ($V_m + V_{m'} \to V_{m+m'}$, $m, m' \le m^v_{\max}$):

$$\mathcal{K}^{vv}_{m,m'} = 8\pi\,(\xi_m + \xi_{m'})\,(\omega^{\text{eff}}_m + \omega^{\text{eff}}_{m'}), \qquad 1 \le m' \le m \le m^v_{\max}. \tag{146}$$

For $m > m^v_{\max}$ the cluster is immobile; it interacts only through the mono-defect terms retained in all cases (see below).

V–I cluster annihilation ($V_m + I_n \to V_{m-n}$ if $m > n$, I_{n-m} if $n > m$, or total annihilation if $n = m$):

$$\mathcal{K}^{vi}_{m,n} = 8\pi\,(\xi_m + \xi_n)\,(\omega^{\text{eff}}_m + \omega^{\text{eff}}_n), \qquad m \le m^v_{\max},\ n \le n^i_{\max}. \tag{147}$$

He–He coalescence is excluded: the interstitial He–He binding energy in bcc Fe is negligible ($\lesssim 0.05$ eV), so free helium reaches a cavity as a monomer.

Mono-defect simplification.

In the general equations, restricting $n' = 1$ in the i–i channel and $m' = 1$ in the v–v channel recovers the *mono-defect* form in which every cluster changes size by exactly one defect at a time. This simplification is justified because:

1. The concentration of mobile $n' \ge 2$ SIA clusters under steady irradiation is typically two to four orders of magnitude below that of the mono-SIA, so their coalescence contribution is proportionally suppressed.
2. For the vacancy side, $m^v_{\max}$ is small enough that the dominant coalescence path ($V_1 + V_1 \to V_2$, $V_1 + V_2 \to V_3$, etc.) is already captured by the mono-vacancy absorption terms in the equations for $c_{m,\ell}$ with $m \le m^v_{\max}$.
3. The v–i annihilation channel reduces exactly to P1 recombination when $m = n = 1$, to the P3 loop-shrinkage term when $m = 1, n \ge 2$, and to the P2 SIA-shrinkage term when $m \ge 2, n = 1$. Only the cluster–cluster pairs $m \ge 2$, $n \ge 2$ are genuinely new; these are small at low cluster concentrations.

Under the mono-defect simplification, the rate constants reduce to:

$$\mathcal{K}^{ii}_{n,1}\big|_{n'=1} = \mathcal{K}^{\rm loop}_{i,n} = A_{\rm loop}\, n^{1/2}\, Z_i^{\rm loop}\, \omega_i^{\rm eff}, \tag{148}$$

$$\mathcal{K}^{vv}_{m,1}\big|_{m'=1} = \mathcal{K}^{\rm cav}_{\alpha,m} = A_{\rm sph}\, m^{1/3}\, \omega_v^{\rm eff}, \tag{149}$$

$$\mathcal{K}^{vi}_{m=1,n=1} = \mathcal{K}_{iv} = 4\sqrt{3}\,\pi\,(\omega_i^{\rm eff} + \omega_v^{\rm eff}), \tag{150}$$

where the loop-bias factor $Z_i^{\rm loop}$ in Eq. (148) replaces the bare Smoluchowski prefactor because the specialised bcc capture geometry enters at the mono-defect level (Section 6.1). The boxed master equations in Section 7.3 implement the mono-defect form throughout. To recover the full coalescence system, replace the mono-defect growth and shrinkage terms with the sums in Eqs. (145)–(147).

7.3 Master Equations

Define the total cavity concentration at vacancy size m and the combined fixed-sink rate:

$$\bar{c}_m \equiv \sum_{\ell=0}^{\ell_{\max}(m)} c_{m,\ell}, \qquad \mathcal{D}_\alpha = \mathcal{D}_\alpha^d + \mathcal{D}_\alpha^{gb} + \mathcal{D}_\alpha^p \qquad [\mathrm{s}^{-1}], \tag{151}$$

as defined in Eqs. (135)–(137). The equations below are the *general (coalescence) form*: every pair of mobile clusters that can react does so, governed by the rate constants $\mathcal{K}^{ii}$, $\mathcal{K}^{vv}$, and $\mathcal{K}^{vi}$ of Section 7.2. The *mono-defect simplification* is recovered as a special case by setting $n^i_{\max} = 1$ and $m^v_{\max} = 1$, which restricts all coalescence sums to their $n' = 1$ or $m' = 1$ terms and reduces $\mathcal{K}^{ii}$, $\mathcal{K}^{vv}$, $\mathcal{K}^{vi}$ to the point-defect rate constants $\mathcal{K}^{\rm loop}_{i,n}$, $\mathcal{K}^{\rm cav}_{\alpha,m}$, and $\mathcal{K}_{iv}$ respectively (Eqs. 148–150).

7.3.1 Population 1: SIA Clusters, c_n $(n = 1, \ldots, N)$

$$\begin{aligned}
\frac{dc_n}{dt} &= G_n \\
&+ \underbrace{\frac{1}{2}\sum_{n'=1}^{\min(n-1,\,n^i_{\max})} \mathcal{K}^{ii}_{n',n-n'}\, c_{n'}\, c_{n-n'}}_{\text{i–i gain }(\mathrm{P}_{ii});\ n^i_{\max}=1\text{: zero for } n\geq 3} - \underbrace{c_n \sum_{n'=1}^{n^i_{\max}} \mathcal{K}^{ii}_{n,n'}\, c_{n'}}_{\text{i–i loss; } n^i_{\max}=1 \Rightarrow \mathcal{K}^{\text{loop}}_{i,n} c_1 c_n} \\
&+ \varepsilon_i(n{+}1)\, c_{n+1} - \varepsilon_i(n)\, c_n\, \delta_{n\geq 2} \\
&+ \underbrace{\sum_{m'=1}^{\min(m^v_{\max},\,n-1)} \mathcal{K}^{vi}_{m',n-m'}\, c_{m',0}\, c_{n-m'}}_{\substack{\text{gain: } V_{m'}+I_{n-m'}\to I_n \text{ impossible} \\ \text{(vacancies shrink SIA clusters, not grow them)}}} \\
&\quad [\text{no gain from v–i; vacancies only shrink SIA clusters}] \qquad (152) \\
&- \underbrace{\sum_{m'=1}^{m^v_{\max}} \mathcal{K}^{vi}_{m',n}\, c_{m',0}\, c_n}_{\substack{\text{v–i loss from } I_n \\ m^v_{\max}=1 \Rightarrow \mathcal{K}_{iv}\, c_{1,0}\, c_n\, \delta_{n=1} \text{ and } \mathcal{K}^{\text{loop}}_{v,n}\, c_{1,0}\, c_n\, \delta_{n\geq 2}}} \\
&- \sum_{m=1}^{M} \mathcal{K}^{(\text{cav})}_{n,m}\, c_n\, \bar{c}_m \\
&- \mathcal{D}_i\, c_n \\
&+ \delta_{n=1} \sum_{m=1}^{M} \sum_{\ell=1}^{\ell_{\max}(m)} \Gamma_{\text{TM}}(m,\ell)\, c_{m,\ell},
\end{aligned}$$

where the cavity rate constant $\mathcal{K}^{(\text{cav})}_{n,m}$ (immobile cavities absorbing mobile SIA clusters of any size $n \leq n^i_{\max}$; zero for $n > n^i_{\max}$) is:

$$\mathcal{K}^{(\text{cav})}_{n,m} = \begin{cases} A_{\text{sph}}\, m^{1/3}\, \omega^{\text{eff}}_i, & 1 \leq n \leq 3 \text{ (3D)}, \\ \dfrac{A_{\text{sph}}\, m^{1/3}\, \omega^{1D}_n}{1 + B_{\text{rot}}\, \hat{L}^2\, m^{-1/3}}, & 4 \leq n \leq n^i_{\max} \text{ (1D/3D)}, \\ 0, & n > n^i_{\max} \text{ (sessile)}. \end{cases} \tag{153}$$

Structure of the SIA equation.

Line 1: cascade production. Line 2: i–i coalescence (P_{ii}). The gain term sums over all pairs $(n', n{-}n')$ of mobile SIA clusters that merge to form I_n; the loss term removes I_n by merging with any mobile $I_{n'}$. *Mono-defect limit* ($n^i_{\max} = 1$): only $n' = 1$ survives in the loss sum, giving $\mathcal{K}^{ii}_{n,1}\, c_1\, c_n = \mathcal{K}^{\text{loop}}_{i,n}\, c_1\, c_n$; the gain sum is non-zero only for

$n = 2$ (forming a di-interstitial from two mono-SIAs). Line 3: thermal SIA emission (P5); unchanged in both formulations. Lines 4–5: v–i annihilation (P_{vi}). A mobile vacancy cluster $V_{m'}$ ($m' \leq m^v_{\max}$) annihilates m' SIAs from I_n, producing $I_{n-m'}$ (the gain term at size $n - m'$ appears in the equation for $c_{n-m'}$). There is no gain to c_n from v–i reactions because vacancies always *shrink* SIA clusters. *Mono-defect limit* ($m^v_{\max} = 1$): only $m' = 1$ survives, giving $\mathcal{K}^{vi}_{1,n}\, c_{1,0}\, c_n$, which equals $\mathcal{K}_{iv}\, c_1\, c_{1,0}\, \delta_{n=1}$ (P1 recombination) for $n = 1$ and $\mathcal{K}^{\mathrm{loop}}_{v,n}\, c_{1,0}\, c_n$ for $n \geq 2$ (P3 loop shrinkage). Line 6: absorption by immobile cavities (P2/P6); the sum already spans all SIA-cluster sizes $n = 1, \ldots, n^i_{\max}$ through $\mathcal{K}^{(\mathrm{cav})}_{n,m}$, so this line is identical in both formulations. Line 7: combined fixed sinks (P4/P4′/P4″). Line 8: trap-mutation SIA source (P7, $n = 1$ only).

Recovery of the single-SIA equation.

Setting $n = 1$ in Eq. (152): the i–i gain sum vanishes (no pairs sum to 1), the i–i loss gives $-\mathcal{K}^{ii}_{1,1}\, c_1^2$, the v–i loss gives $-\mathcal{K}^{vi}_{1,1}\, c_{1,0}\, c_1$, emission gives $+\varepsilon_i(2)\, c_2$, and the gain from v–i at size $n = 2$ ($V_1 + I_2 \to I_1$) contributes $+\mathcal{K}^{vi}_{1,1}\, c_{1,0}\, c_1$ into the c_2 equation (not into c_1). The exact $n = 1$ equation is:

$$
\begin{aligned}
\frac{dc_1}{dt} = G_1 - \mathcal{K}^{ii}_{1,1}\, c_1^2 + \varepsilon_i(2)\, c_2 &+ \sum_{m'=1}^{m^v_{\max}} \mathcal{K}^{vi}_{m',2}\, c_{m',0}\, c_2 \\
&- \sum_{m'=1}^{m^v_{\max}} \mathcal{K}^{vi}_{m',1}\, c_{m',0}\, c_1 - \sum_{m=1}^{M} A_{\mathrm{sph}}\, m^{1/3}\, \omega_i^{\mathrm{eff}}\, c_1\, \bar{c}_m - \mathcal{D}_i\, c_1 \\
&+ \sum_{m,\ell} \Gamma_{\mathrm{TM}}(m,\ell)\, c_{m,\ell}.
\end{aligned}
\tag{154}
$$

Setting $n^i_{\max} = m^v_{\max} = 1$: $\mathcal{K}^{ii}_{1,1} \to \mathcal{K}^{\mathrm{loop}}_{i,1}$, $\mathcal{K}^{vi}_{1,1} \to \mathcal{K}_{iv}$, and the c_2 term from v–i becomes $\mathcal{K}^{\mathrm{loop}}_{v,2}\, c_{1,0}\, c_2$, recovering the standard mono-defect mono-SIA equation.

7.3.2 Population 2: Vacancy Clusters, $c_{m,\ell}$

Voids ($\ell = 0$) and bubbles ($\ell \geq 1$) share a unified (m, ℓ) grid. Spherical cavities are unbiased ($Z^{\mathrm{cav}}_i = 1$). The v–v coalescence terms are active only for $m \leq m^v_{\max}$; for $m > m^v_{\max}$ those clusters are immobile and the v–v sum reduces to a single mono-vacancy term, identical to the mono-defect form. SIA-induced shrinkage spans all mobile SIA sizes through $\mathcal{K}^{(\mathrm{cav})}_{n,m}$ and is unchanged in both formulations.

$$
\boxed{
\begin{aligned}
\frac{dc_{m,\ell}}{dt} = G_{m,\ell}& \\
&+ \underbrace{\frac{1}{2}\sum_{m'=1}^{\min(m-1,\,m^v_{\max})} \mathcal{K}^{vv}_{m',m-m'}\, c_{m',0}\, c_{m-m',\ell}\, \delta_{m\le m^v_{\max}+m^v_{\max}}}_{\substack{\text{v–v gain }(\mathrm{P}_{vv})\\ m^v_{\max}=1 \Rightarrow A_{\mathrm{sph}}(m-1)^{1/3}\omega^{\mathrm{eff}}_v\, c_{1,0}\, c_{m-1,\ell}\, \delta_{m\ge 2}}} \\
&- \underbrace{c_{m,\ell}\sum_{m'=1}^{m^v_{\max}} \mathcal{K}^{vv}_{m,m'}\, c_{m',0}}_{\substack{\text{v–v loss}\\ m^v_{\max}=1 \Rightarrow A_{\mathrm{sph}}\, m^{1/3}\, \omega^{\mathrm{eff}}_v\, c_{1,0}\, c_{m,\ell}}} \\
&+ \varepsilon_v(m{+}1,\ell)\, c_{m+1,\ell} - \varepsilon_v(m,\ell)\, c_{m,\ell} \\
&- \underbrace{\sum_{n=1}^{n^i_{\max}} \mathcal{K}^{(\mathrm{cav})}_{n,m}\, c_n\, c_{m,\ell}}_{\substack{\text{absorption of any mobile } I_n\\ n^i_{\max}=1 \Rightarrow A_{\mathrm{sph}}\, m^{1/3}\, \omega^{\mathrm{eff}}_i\, c_1\, c_{m,\ell}}} \\
&+ A_{\mathrm{sph}}\, m^{1/3}\, \omega^{\mathrm{eff}}_h\, c_h\, c_{m,\ell-1}\, \delta_{\ell\ge 1} - A_{\mathrm{sph}}\, m^{1/3}\, \omega^{\mathrm{eff}}_h\, c_h\, c_{m,\ell}\, \delta_{\ell<\ell_{\max}} \\
&+ \varepsilon_h(m,\ell{+}1)\, c_{m,\ell+1} - \varepsilon_h(m,\ell)\, c_{m,\ell}\, \delta_{\ell\ge 1} \\
&+ \Gamma_{\mathrm{TM}}(m{-}1,\ell)\, c_{m-1,\ell} - \Gamma_{\mathrm{TM}}(m,\ell)\, c_{m,\ell} \\
&+ b_0\, \dot{\phi}\, (\ell{+}1)\, c_{m,\ell+1} - b_0\, \dot{\phi}\, \ell\, c_{m,\ell}.
\end{aligned}
}
\tag{155}
$$

Structure of the vacancy equation.

Lines 2–3: v–v coalescence (P_{vv}). The gain term sums over all pairs $(m', m{-}m')$ of mobile vacancy clusters that merge to form V_m; the loss term removes V_m by merging with any mobile $V_{m'}$. *Mono-defect limit* ($m^v_{\max} = 1$): only $m' = 1$ survives, and $\mathcal{K}^{vv}_{m,1}\, c_{1,0}\, c_{m,\ell} = A_{\mathrm{sph}}\, m^{1/3}\, \omega^{\mathrm{eff}}_v\, c_{1,0}\, c_{m,\ell}$ (Eq. 149), recovering the standard mono-vacancy absorption form. The gain term becomes $A_{\mathrm{sph}}\, (m{-}1)^{1/3}\, \omega^{\mathrm{eff}}_v\, c_{1,0}\, c_{m-1,\ell}$, which is the familiar P2 growth step. Line 4: vacancy thermal emission (P5); identical in both formulations. Line 5: SIA-induced shrinkage. The single sum over $n = 1, \ldots, n^i_{\max}$ via $\mathcal{K}^{(\mathrm{cav})}_{n,m}$ already spans the full range of mobile SIA sizes. *Mono-defect limit* ($n^i_{\max} = 1$): only the $n = 1$ term survives, giving $A_{\mathrm{sph}}\, m^{1/3}\, \omega^{\mathrm{eff}}_i\, c_1\, c_{m,\ell}$, recovering the standard SIA absorption step. Lines 6–7: He exchange along ℓ (P2 + P5); unchanged. Line 8: trap mutation (P7); unchanged. Line 9: radiation re-solution (P8); unchanged. For $\ell = 0$, lines 6–7 and 9 vanish (pure voids carry no He).

Recovery of the free monovacancy equation.
Setting $m = 1$, $\ell = 0$, noting that the v–v gain sum vanishes (no pairs give $m = 1$) and incorporating the v–i loss from the SIA grid (P3, vacancy side):

$$\begin{aligned}\frac{dc_{1,0}}{dt} = G_v &- \sum_{m'=1}^{m^v_{\max}} \mathcal{K}^{vi}_{m',1}\, c_{m',0}\, c_1 - \mathcal{D}_v\, c_{1,0} \\ &- c_{1,0} \sum_{m'=1}^{m^v_{\max}} \mathcal{K}^{vv}_{1,m'}\, c_{m',0} + \sum_{m=2}^{M} \sum_{\ell} \varepsilon_v(m,\ell)\, c_{m,\ell} \\ &- A_{\rm sph}\, \omega_h^{\rm eff}\, c_h\, c_{1,0} + \varepsilon_h(1,1)\, c_{1,1} + b_0\, \dot{\phi}\, c_{1,1}.\end{aligned} \tag{156}$$

Setting $m^v_{\max} = 1$: $\mathcal{K}^{vi}_{1,1} \to \mathcal{K}_{iv}$ and $\mathcal{K}^{vv}_{1,1}\, c_{1,0} \to A_{\rm sph}\, \omega_v^{\rm eff}\, c_{1,0}$ (absorption by all larger cavities), recovering the standard monovacancy equation. The fixed sink $\mathcal{D}_v$ includes dislocation, grain-boundary, and precipitate contributions (Eq. 134). For $m = 1$, $\ell \geq 1$: $\varepsilon_v(1,\ell) = 0$ because the daughter $\mathrm{He}_\ell\mathrm{V}_0$ would immediately self-trap back to $\mathrm{He}_\ell\mathrm{V}_1$.

7.3.3 Population 3: Free Helium, c_h

The free-helium equation is independent of the i–i and v–v coalescence formulation: He–He coalescence is excluded, and He interacts with the vacancy cluster grid only through mono-He trapping and emission.

$$\boxed{\begin{aligned}\frac{dc_h}{dt} &= G_{\rm He} - \sum_{m=1}^{M} A_{\rm sph}\, m^{1/3}\, \omega_h^{\rm eff}\, c_h \sum_{\ell=0}^{\ell_{\max}(m)-1} c_{m,\ell} - \mathcal{D}_h\, c_h \\ &\quad + \sum_{m=1}^{M} \sum_{\ell=1}^{\ell_{\max}(m)} \varepsilon_h(m,\ell)\, c_{m,\ell} + \sum_{m=1}^{M} \sum_{\ell=1}^{\ell_{\max}(m)} b_0\, \dot{\phi}\, \ell\, c_{m,\ell}.\end{aligned}} \tag{157}$$

Production; trapping by cavities (P2); combined fixed sinks ($\mathcal{D}_h$ includes dislocations, grain boundaries, precipitates); thermal He emission (P5); re-solution (P8). This equation is identical in the general and mono-defect formulations.

7.4 Mass Conservation Laws

The master equations satisfy two exact conservation identities that are proved term by term in Appendix A. They are stated here for reference and used to define numerical accuracy diagnostics.

7.4.1 Frenkel Pair Balance and the Swelling Identity

Define the cumulative net SIA flux absorbed by the dislocation network and the total cluster inventories:

$$\Delta J^d(t) \equiv \int_0^t \left[Z_i^d\, \rho_d\, a^2 \sum_{n=1}^{N} n\, \omega_n^{\rm eff}\, c_n - Z_v^d\, \rho_d\, a^2\, \omega_v^{\rm eff}\, c_{1,0} \right] dt' \;\geq\; 0, \tag{158}$$

$$S_I(t) \equiv \sum_{n=1}^{N} n\, c_n(t), \qquad S(t) \equiv \sum_{m=1}^{M} m\, \bar{c}_m(t), \tag{159}$$

where $S(t)$ is the volumetric swelling (total vacancies stored in cavities, Eq. 166) and $S_I(t)$ is the total SIA inventory in loops. Cascade production satisfies $\sum_n nG_n = \sum_{m,\ell} mG_{m,\ell}$ identically. Grain boundaries and precipitates are unbiased and cancel exactly in the moment difference (Appendix A). The resulting *exact conservation law* is:

$$\boxed{S_I(t) + \Delta J^d(t) - S(t) = S_I(0) - S(0) = \text{const.}} \tag{160}$$

For a defect-free initial condition, $S_I(0) = S(0) = 0$, so the conserved quantity is identically zero at all times:

$$S(t) = S_I(t) + \Delta J^d(t). \tag{161}$$

This is the *swelling identity*: the total vacancy content of all cavities equals the sum of (i) the SIA inventory currently stored in loops plus (ii) the cumulative net bias flux delivered to the dislocation network. Its physical interpretation is transparent:

- Every cascade produces equal numbers of SIAs and vacancies.
- The dislocation network preferentially absorbs SIAs ($Z_i^d > Z_v^d = 1$), accumulating a net SIA excess ΔJ^d that is permanently removed from the lattice.
- Of the remaining SIAs, some are currently stored as loops (S_I) and the rest have recombined with vacancies or been absorbed by unbiased sinks.
- Whatever vacancy excess is not neutralised by recombination or stored in loops must reside in cavities, exactly accounting for S.

Differentiating Eq. (160) recovers the rate form:

$$\frac{dS}{dt} = \frac{dS_I}{dt} + Z_i^d\, \rho_d\, a^2 \sum_{n=1}^{N} n\, \omega_n^{\text{eff}}\, c_n - Z_v^d\, \rho_d\, a^2\, \omega_v^{\text{eff}}\, c_{1,0}, \tag{162}$$

where $Z_v^d = 1$. The last two terms together are $d\Delta J^d/dt \geq 0$ (Eq. 158). Swelling grows whenever the net bias absorption rate exceeds the rate of SIA loop accumulation dS_I/dt. In the regime where loops have reached a quasi-steady density ($dS_I/dt \approx 0$), swelling grows at a rate set directly by the net bias flux to dislocations.

7.4.2 Helium Conservation

The total helium inventory—free plus trapped—grows only through external production:

$$\frac{d}{dt}\left[c_h + \sum_{m=1}^{M} \sum_{\ell=1}^{L} \ell\, c_{m,\ell}\right] = G_{\text{He}}. \tag{163}$$

Thermal emission (P5), re-solution (P8), and trap mutation (P7) all redistribute helium between free and trapped pools without creating or destroying it; their contributions to the moment sum cancel identically (Appendix A).

7.4.3 Numerical Accuracy Diagnostics

The swelling identity (160) provides an exact, inexpensive check: at every output step evaluate

$$\delta_{\mathrm{FP}}(t) = \frac{\left|S(t) - S_I(t) - \Delta J^d(t)\right|}{S(t) + S_I(t) + \Delta J^d(t)}, \tag{164}$$

where $\Delta J^d(t)$ is accumulated by numerical integration of the integrand in Eq. (158) alongside the ODE system. Define the helium balance error as:

$$\delta_{\mathrm{He}}(t) = \frac{\left| c_h + \sum_{m,\ell} \ell\, c_{m,\ell} - \int_0^t G_{\mathrm{He}}\, dt' - \left[c_h(0) + \sum_{m,\ell} \ell\, c_{m,\ell}(0)\right] \right|}{c_h + \sum_{m,\ell} \ell\, c_{m,\ell}}. \tag{165}$$

Both metrics are normalised to instantaneous inventories. In a correctly implemented code both should remain below $\sim 10^{-6}$. Values above 10^{-4} indicate solver drift (tighten `rtol` $\leq 10^{-8}$, `atol` $\leq 10^{-12}$). Values above 10^{-3} almost always signal a coding error: a missing sign, a mismatched index shift in the ladder operators, or an unbalanced trap-mutation source. Because δ_{FP} now decomposes swelling into two separately trackable contributions (S_I and ΔJ^d), it also serves as a diagnostic for identifying which physical mechanism dominates swelling at each stage of the irradiation history.

7.5 Swelling

The swelling identity (161) provides the physical mechanism. The macroscopic fractional volume change is:

$$S(t) = \frac{\Delta V}{V} = \sum_{m=1}^{M} m\, \bar{c}_m(t), \tag{166}$$

counting every vacancy stored in voids and bubbles. By Eq. (161), this equals the current SIA loop inventory S_I plus the integrated net bias absorption ΔJ^d. The two contributions have distinct time signatures:

- $\Delta J^d(t)$ grows monotonically and, once the microstructure has reached a quasi-steady loop density, increases approximately linearly with dose. It represents vacancies that are effectively stranded in cavities because their SIA counterparts were irreversibly captured by the biased dislocation network.
- $S_I(t)$ grows during the transient loop-accumulation regime and then saturates when loop nucleation is balanced by loop–loop annihilation and absorption by cavities. Once S_I saturates, $\dot{S} \approx \dot{\Delta J}^d = (Z_i^d - 1)\rho_d a^2 \sum_n n\, \omega_n^{\mathrm{eff}} c_n$, giving the well-known linear-in-dose swelling rate.

The dislocation-climb contribution to the macroscopic volume change is negligible: the SIA relaxation volume ($\sim 1.1\,\Omega$) nearly cancels the $+\Omega$ extension of the extra half-plane, leaving a net volume change per absorbed SIA of order $-0.1\,\Omega$—an order of magnitude smaller than the $+\Omega$ per vacancy entering a void. Swelling is therefore dominated by the cavity term S, and Eq. (161) shows that this is driven in equal measure by the ongoing bias absorption (ΔJ^d) and by the transient loading of SIA loops (S_I).

7.6 Sink Strengths

The total sink strengths, obtained by collecting all loss terms proportional to c_α from the master equations, are:

$$\begin{aligned} k_v^2 = \rho_d + \frac{4\pi^2}{d_g^2} + 4\pi\, r_p\, N_p \\ + \frac{1}{a^2}\sum_{m=1}^{M} A_{\text{sph}}\, m^{1/3}\, \bar{c}_m + \frac{1}{a^2}\sum_{n=1}^{N} A_{\text{loop}}\, n^{1/2}\, c_n, \end{aligned} \tag{167}$$

$$\begin{aligned} k_i^2 = Z_i^d\, \rho_d + \frac{4\pi^2}{d_g^2} + 4\pi\, r_p\, N_p \\ + \frac{1}{a^2}\sum_{m=1}^{M} A_{\text{sph}}\, m^{1/3}\, \bar{c}_m + \frac{Z_i^{\text{loop}}}{a^2}\sum_{n=1}^{N} A_{\text{loop}}\, n^{1/2}\, c_n. \end{aligned} \tag{168}$$

The grain-boundary and precipitate contributions are identical in k_v^2 and k_i^2 (no bias). The net bias driving swelling is:

$$\Delta k^2 = k_i^2 - k_v^2 = (Z_i^d - 1)\, \rho_d + \frac{(Z_i^{\text{loop}} - 1)}{a^2}\sum_{n=1}^{N} A_{\text{loop}}\, n^{1/2}\, c_n. \tag{169}$$

The cavity, grain-boundary, and precipitate contributions cancel exactly. Swelling is driven solely by the dislocation and loop bias terms.

7.7 Input Data Mapping

Quantity	**Source**
G_n, $G_{m,\ell}$, G_{He}, $\dot{\phi}$	Table 5
E_f^v, E_f^i, γ_s, B_2, B_3, ν_α	Table 8
$\omega_\alpha^{\text{eff}}$, ω_n^{1D}, $\hat{L}$, z_s, $E_b^{\alpha\text{-}s}$, c_s	Table 16
$E_b^v(m,\ell)$, $E_b^i(n)$, $E_B^{\text{He}}(m,\ell)$, $E_{\text{TM}}(m,\ell)$	Tables 25, 18, 19, 27
A_{sph}, A_{loop}, B_{rot}, Z_i^d, Z_i^{loop}, b_0	Tables 29, 30
ρ_d, d_g, r_p, N_p	Table 29

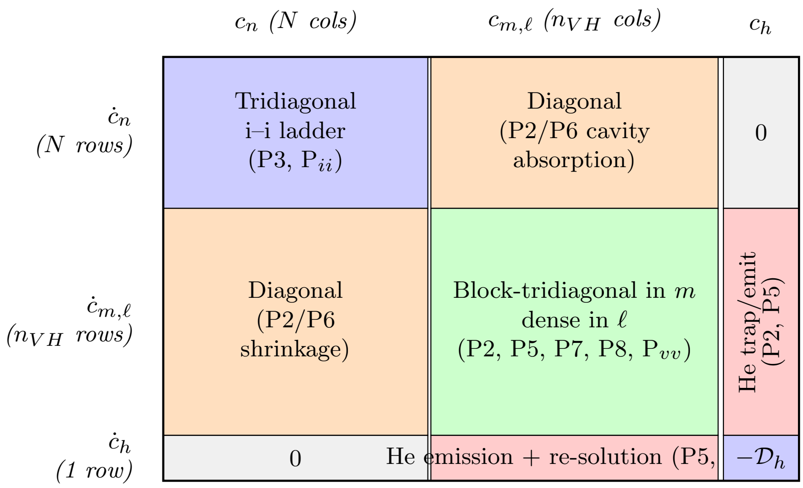
7.8 ODE System Structure, Helium Approximations, and Numerical Strategy

7.8.1 State Vector and Matrix Structure

The three populations assembled in Section 7.3 form a single state vector

$$\mathbf{y}(t) = \big[\underbrace{c_1,\, c_2,\, \ldots,\, c_N}_{\text{SIA clusters}},\ \underbrace{c_{1,0},\, c_{1,1},\, \ldots,\, c_{M,\ell_{\max}(M)}}_{\text{vacancy/He clusters}},\ \underbrace{c_h}_{\text{free He}}\big]^\top, \tag{170}$$

of dimension $n_{\text{eq}} = N + n_{VH} + 1$, where $n_{VH} = \sum_{m=1}^{M} [\ell_{\max}(m) + 1]$ is the number of (m, ℓ) grid points. The right-hand side $\mathbf{f}(\mathbf{y})$ of the ODE system $\dot{\mathbf{y}} = \mathbf{f}(\mathbf{y})$ is sparse and banded. Writing it schematically as a Jacobian block structure reveals the coupling pattern:



The Jacobian $\partial\mathbf{f}/\partial\mathbf{y}$ inherits this block structure. The blue tridiagonal block (i–i ladder) has bandwidth 2 in n. The green block-tridiagonal (vacancy/He) has block size $(\ell_{\max} + 1)$ and tridiagonal coupling in m from P2/P5, with dense ℓ-coupling within each block from the He exchange terms P2/P5/P7/P8. The orange off-diagonal blocks are sparse-diagonal: each c_n couples to all $\bar{c}_m$ through the cavity absorption rate, and each $c_{m,\ell}$ couples to all c_n through the SIA-shrinkage rate. The red column/row couples the free-He scalar c_h to every (m, ℓ) node.

General coalescence case.

When cluster–cluster coalescence is included up to the mobility cutoffs $n^i_{\max}$ and $m^v_{\max}$, the sparsity pattern changes in three blocks. (i) The SIA–SIA block becomes *dense lower-triangular* within the $n^i_{\max} \times n^i_{\max}$ mobile sub-block: i–i coalescence couples every pair $(n', n - n')$ with $n' \le n^i_{\max}$, filling the lower triangle. Rows $n > n^i_{\max}$ (immobile loops) retain the tridiagonal structure of the mono-defect case. (ii) The SIA–VacHe

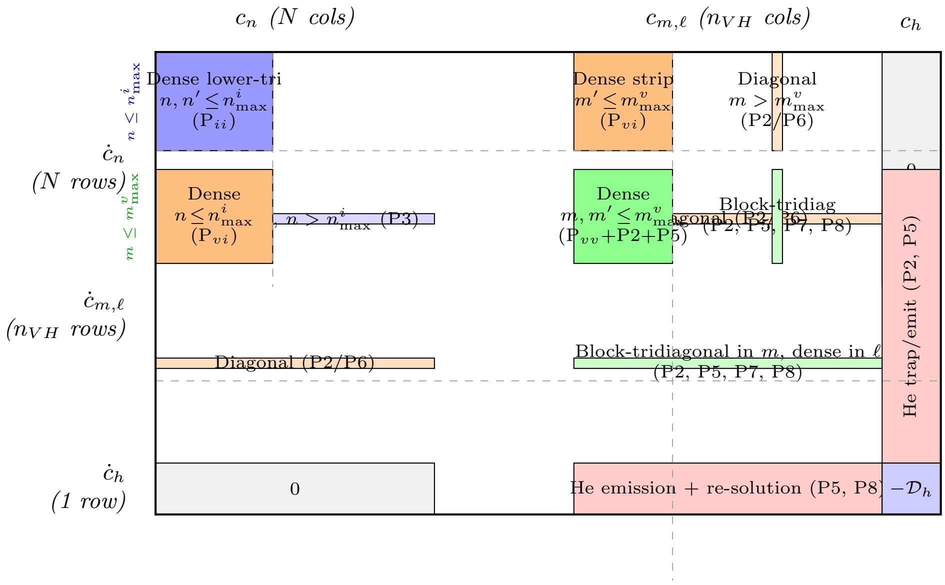


Fig. 9 Coalescence matrix structure

and VacHe–SIA cross blocks become *dense in the mobile strip*: v–i annihilation couples every mobile vacancy cluster $V_{m'}$ ($m' \le m^v_{\max}$) to every SIA cluster I_n ($n \le n^i_{\max}$), filling a $n^i_{\max} \times m^v_{\max}$ sub-block. Outside the mobile strips both blocks remain diagonal (the immobile-cluster interactions are still mono-defect in form). (iii) The VacHe–VacHe block gains a *dense* $m^v_{\max} \times m^v_{\max}$ *corner*: v–v coalescence fills the upper-left sub-block corresponding to the mobile vacancy clusters, on top of the block-tridiagonal structure inherited from P2/P5/P7/P8.

The dashed lines mark the mobile/immobile boundaries at $n = n^i_{\max}$ (vertical, in the SIA blocks) and $m = m^v_{\max}$ (vertical, in the VacHe blocks; horizontal, separating mobile and immobile vacancy rows). Darker shading indicates blocks that are denser in the coalescence formulation than in the mono-defect case. The c_h column and row (red) and the immobile-cluster sub-blocks (lighter shading) are structurally identical to the mono-defect diagram above.

7.8.2 He Content and Reduction Approximations

The dominant contributor to system size is the He index ℓ. The maximum He loading per cluster grows with cavity size as

$$\ell_{\max}(m) = \lfloor \alpha_{\mathrm{He}}\, m^{2/3} \rfloor, \qquad \alpha_{\mathrm{He}} = 1.5\text{–}2.0, \tag{171}$$

giving a total vacancy–He grid size

$$n_{VH}^{\text{full}} = \sum_{m=1}^{M} \big[\ell_{\max}(m) + 1\big] \approx M + \alpha_{\text{He}} \sum_{m=1}^{M} m^{2/3} \approx M\Big(1 + \tfrac{3}{5}\alpha_{\text{He}}\, M^{2/3}\Big), \tag{172}$$

which scales as $M^{5/3}$ and can reach 10^4–10^5 for $M = 10^3$–10^4. Two physics-based approximations reduce this substantially, exploiting the very different He budgets in fission and fusion irradiation environments.

Case 1 — Fusion (high He/dpa): mean-field He approximation.

Under fusion-relevant He generation rates (∼10–15 appm He/dpa), He quickly saturates every cavity to its equilibrium loading. The two-dimensional (m, ℓ) distribution can then be collapsed to a one-dimensional problem by tracking only the mean He loading per cavity size:

$$\bar{\ell}(m, t) \equiv \frac{\sum_\ell \ell\, c_{m,\ell}}{\bar{c}_m}, \tag{173}$$

and treating the He distribution within each m-bin as a delta function at $\bar{\ell}(m)$. This replaces the n_{VH}^{full} equations for $c_{m,\ell}$ with two ODEs per vacancy size — one for the total cavity concentration $\bar{c}_m$ and one for the mean He content $\bar{\ell}_m$ — plus the single free-He equation:

$$n_{\text{eq}}^{\text{fusion}} = N + \underbrace{2M}_{\bar{c}_m,\, \bar{\ell}_m} + \underbrace{1}_{c_h} = N + 2M + 1. \tag{174}$$

Case 2 — Fission (low He/dpa): decoupled-He approximation.

Fission reactors produce ∼0.1–1 appm He/dpa, so He loading remains low ($\ell \ll \ell_{\max}$) and bubbles are rare. Most cavities are pure voids ($\ell = 0$). He can then be decoupled: track $c_{m,0}$ (pure voids) separately from a single mean-field He concentration per size, treating bubble nucleation as a perturbation to the void distribution. In the simplest implementation this retains one He-bearing species per m and adds a single free-He equation:

$$n_{\text{eq}}^{\text{fission}} = N + \underbrace{M}_{\text{voids } c_{m,0}} + \underbrace{1}_{\bar{c}_{m,\ell\geq 1}} + \underbrace{1}_{c_h} = N + M + 2. \tag{175}$$

7.8.3 System Size Summary

Table 38 collects the ODE counts for typical parameter choices ($N = 1000$, $M = 1000$, $\alpha_{\text{He}} = 1.5$).

7.8.4 Stiffness and Solver Strategy

The system (170) is severely stiff. The stiffness ratio is dominated by the disparity between the He migration frequency and the vacancy frequency,

$$\frac{\omega_h^{\text{eff}}}{\omega_v^{\text{eff}}} \sim 10^4\text{–}10^8, \tag{176}$$

Table 31 ODE system size for the three He-content strategies. All entries assume $N = 1000$ SIA sizes, $M = 1000$ vacancy sizes, $\alpha_{\mathrm{He}} = 1.5$, and $n^{i}_{\max} = 100$, $m^{v}_{\max} = 5$. The column "Dominant block" identifies the bottleneck for Jacobian factorisation. Logarithmic size-binning for $m > m_{\mathrm{group}} \sim 100$–200 can reduce n_{VH} by a further factor of 3–5.

He strategy	**SIA block (N)**	**VacHe block (n_{VH})**	$\mathbf{n}_{\mathrm{eq}}$	**Dominant block**
Full (m, ℓ) grid (Eq. 319)	1 000	∼38 000	∼39 001	VacHe
Case 1: mean-field He, fusion (Eq. 321)	1 000	2 000	3 001	SIA + VacHe
Case 2: decoupled He, fission (Eq. 322)	1 000	1 001	2 002	SIA + VacHe

with additional stiffness contributed by the contrast between fast mobile-cluster reactions (small n, m) and slow large-cluster dynamics. An implicit solver with adaptive time stepping is mandatory; recommended options are LSODA (automatic stiffness detection) or CVODE (variable-order BDF with sparse Jacobian). Practical tolerances are `rtol` $= 10^{-8}$, `atol` $= 10^{-12}$ for the conservation diagnostics of Section 7.4.3 to remain below 10^{-6}.

Logarithmic size-binning (grouping) replaces the uniform m-grid with logarithmically spaced bins for $m > m_{\mathrm{group}} \approx 50$–200. Each bin $[m_k, m_{k+1})$ is represented by a single representative concentration; capture rates are evaluated at the bin midpoint and weighted by the bin width. This reduces n_{VH} by a factor of 3–10 with negligible loss of accuracy for the swelling and loop-density predictions. The conservation diagnostics should be evaluated at every output step; the implementation should assert a hard stop if δ_{FP} or δ_{He} exceeds 10^{-3}.

8 Reduction of the He–Vacancy State Space

The full master equation (155) tracks every (m, ℓ) pair with $1 \le m \le M$ and $0 \le \ell \le \ell_{\max}(m)$ (Eq. 318), yielding n^{full}_{VH} coupled ODEs (Eq. 319) — typically 10^4–10^5 equations. Two physics-based approximations exploit the separation of timescales between He and vacancy kinetics to reduce this to a tractable system. Both retain the full SIA population (Eq. 152) and the free He balance (Eq. 157) unchanged; only the vacancy-cluster block is replaced.

Throughout this section we use the notation and rate constants established in Sections 6–7: $\omega^{\mathrm{eff}}_{\alpha}$ are the effective reaction frequencies (Eq. 129); $A_{\mathrm{sph}} = (48\pi^2)^{1/3}$; $\bar{c}_m = \sum_{\ell} c_{m,\ell}$ is the total cavity concentration; and $\mathcal{K}^{(\mathrm{cav})}_{n,m}$ is the SIA-cluster absorption rate (Eq. 153). The vacancy net-flux from size m to $m+1$ is defined as:

$$J_{m \to m+1} \equiv A_{\mathrm{sph}}\, m^{1/3}\, \omega^{\mathrm{eff}}_{v}\, c_{1,0}\, \bar{c}_m - \varepsilon_v(m{+}1, \bar{\ell}(m{+}1))\, \bar{c}_{m+1}, \tag{177}$$

with forward (capture) and backward (emission) components $J^{+}_{m \to m+1}$ and $J^{-}_{m \to m+1}$ respectively, so that $J_{m \to m+1} = J^{+}_{m \to m+1} - J^{-}_{m+1 \to m}$.

8.1 Case 1: Fast He Equilibration — Fusion Conditions

At temperatures where vacancies are mobile ($T \gtrsim 200\,^\circ$C), the He reaction frequency greatly exceeds the vacancy reaction frequency, $\omega_h^{\rm eff} \gg \omega_v^{\rm eff}$. Helium equilibrates on each cavity faster than the cavity grows or shrinks, so the full two-dimensional (m, ℓ) distribution can be collapsed to two scalar quantities per vacancy size: the total cavity concentration $\bar{c}_m$ and the mean He loading

$$\bar{\ell}(m, t) \equiv \frac{\sum_\ell \ell\, c_{m,\ell}}{\bar{c}_m}. \tag{178}$$

This is the mean-field He approximation defined in Eq. (320).

Cavity size distribution.

Summing Eq. (155) over all ℓ at fixed m:

$$\frac{d\bar{c}_m}{dt} = J_{m-1\to m} - J_{m\to m+1} - \sum_{n=1}^{n^i_{\max}} \mathcal{K}^{(\rm cav)}_{n,m}\, c_n\, \bar{c}_m + \Gamma_{\rm TM}(m{-}1, \bar{\ell})\, \bar{c}_{m-1} - \Gamma_{\rm TM}(m, \bar{\ell})\, \bar{c}_m, \tag{179}$$

where the vacancy net flux $J_{m\to m+1}$ is given by Eq. (213). The SIA-shrinkage sum $\sum_n \mathcal{K}^{(\rm cav)}_{n,m} c_n$ spans all mobile SIA clusters $n = 1, \ldots, n^i_{\max}$ with no bias factor (spherical cavities are unbiased, $Z_i^{\rm cav} = 1$, Eq. 131). The vacancy capture rate $A_{\rm sph}\, m^{1/3}\, \omega_v^{\rm eff}$ is independent of ℓ, so the ℓ-summation is exact for the capture terms. The emission rate ε_v depends on ℓ through the He gas-pressure correction to $E_b^v(m, \ell)$ and is approximated by evaluation at the mean loading $\bar{\ell}(m)$.

Mean He loading.

Define the He inventory at vacancy size m as $N_{\rm He}(m) \equiv \bar{\ell}(m)\, \bar{c}_m$. Multiplying Eq. (155) by ℓ, summing over ℓ, and collecting terms:

$$\begin{aligned} \frac{dN_{\rm He}(m)}{dt} = {} & A_{\rm sph}\, m^{1/3}\, \omega_h^{\rm eff}\, c_h\, \bar{c}_m - \varepsilon_h(m, \bar{\ell})\, \bar{c}_m - b_0\, \dot{\phi}\, N_{\rm He}(m) \\ & + \bar{\ell}(m{-}1)\, J^+_{m-1\to m} - \bar{\ell}(m)\, J^+_{m\to m+1} \\ & + \bar{\ell}(m{+}1)\, J^-_{m+1\to m} - \bar{\ell}(m)\, J^-_{m\to m-1} \\ & - \bar{\ell}(m) \sum_{n=1}^{n^i_{\max}} \mathcal{K}^{(\rm cav)}_{n,m}\, c_n\, \bar{c}_m. \end{aligned} \tag{180}$$

The four groups of terms are: (i) He capture from the free pool minus thermal He emission (P5) minus radiation re-solution (P8); (ii–iii) He advected in and out as clusters grow or shrink along the m-axis, where J^+ is the forward (capture) component and J^- is the backward (emission) component of the vacancy flux; and (iv) He carried away when SIA clusters annihilate vacancies. Note the absence of any bias factor in the SIA-shrinkage term: the rate constant $\mathcal{K}^{(\mathrm{cav})}_{n,m}$ already encodes the unbiased cavity geometry ($Z_i^{\mathrm{cav}} = 1$).

Applying the quotient rule $d\bar{\ell}/dt = (\dot{N}_{\mathrm{He}} - \bar{\ell}\,\dot{\bar{c}}_m)/\bar{c}_m$ and substituting Eq. (215):

$$\boxed{\begin{aligned}\frac{d\bar{\ell}(m)}{dt} &= A_{\mathrm{sph}}\, m^{1/3}\, \omega_h^{\mathrm{eff}}\, c_h - \varepsilon_h\big(m, \bar{\ell}(m)\big) - b_0\, \dot{\phi}\, \bar{\ell}(m) \\ &\quad + \frac{J^+_{m-1\to m}}{\bar{c}_m}\big[\bar{\ell}(m-1) - \bar{\ell}(m)\big] + \frac{J^-_{m+1\to m}}{\bar{c}_m}\big[\bar{\ell}(m+1) - \bar{\ell}(m)\big].\end{aligned}} \tag{181}$$

Line 1 contains the local He kinetics: trapping from the free pool (proportional to c_h), thermal emission ε_h (P5, Eq. 140), and athermal re-solution (P8, Eq. 143) which drives $\bar{\ell}$ toward zero. Line 2 contains advection corrections: cavities flowing into size class m carry their own He loading, which may differ from $\bar{\ell}(m)$; the difference $[\bar{\ell}(m\pm1)-\bar{\ell}(m)]$ acts as a drift source.

Numerical implementation.

In practice, $N_{\mathrm{He}}(m) = \bar{\ell}(m)\,\bar{c}_m$ and $\bar{c}_m$ are evolved as primary variables; $\bar{\ell}(m)$ is recovered by division and used only when evaluating rates. When $\bar{c}_m$ approaches zero near the nucleation threshold, the quotient becomes ill-conditioned. Recommended practice: set a floor $\bar{c}_m^{\min} \sim 10^{-30}$ at/at below which $\bar{\ell}$ is reset to zero.

Case 1 summary.

Replacing the full (m, ℓ) grid with $2M$ ODEs — M for $\bar{c}_m$ (Eq. 215) and M for $\bar{\ell}(m)$ (Eq. 217) — plus the unchanged free-He equation (Eq. 157) gives the reduced system count of Eq. (321):

$$n_{\mathrm{eq}}^{\mathrm{fusion}} = N + 2M + 1.$$

The reduction factor relative to the full grid is approximately $n_{VH}^{\mathrm{full}}/(2M) \approx \frac{3}{10}\alpha_{\mathrm{He}} M^{2/3}$, which is ~10–20 for $M = 10^3$ and $\alpha_{\mathrm{He}} = 1.5$.

8.2 Case 2: Decoupled He Inventory — Fission Conditions

Under fission irradiation the He generation rate is low (~0.3–1 appm/dpa vs. ~10–15 appm/dpa for fusion), so most cavities carry $\ell \ll m$ and the He/V ratio $\bar{\ell}/m \ll 1$. Helium acts as a weak perturbation on the void kinetics rather than controlling nucleation. Rather than tracking $\bar{\ell}(m)$ as a separate variable at each size, the total trapped He inventory is followed as a single scalar and distributed algebraically across cavity sizes.

Cavity size distribution.

The $\bar{c}_m$ equation is identical to Case 1, Eq. (215), with the vacancy emission rate evaluated using an He-corrected binding energy. In the low-loading limit the He gas pressure obeys the ideal-gas law, giving an effective binding energy:

$$E_b^{v,\mathrm{eff}}(m) \equiv E_b^v\big(m,\, \ell(m)\big) = E_f^v + \frac{2\gamma_s\,\Omega}{r_m} - \frac{\ell(m)\, k_B T}{m}, \tag{182}$$

where $r_m = \xi(m)\, a$, γ_s is the surface tension, $\Omega = a^3/2$ is the atomic volume, and $\ell(m)$ is the He content assigned to cavities of size m by the allocation rule below. The third term is the ideal-gas pressure correction $p(\ell, m)\,\Omega = (\ell\, k_B T/m) \cdot m/m = \ell k_B T/m$, adequate when $\ell/m \ll 1$. At fission temperatures and low He loading, thermal He emission is negligible ($E_B^{\mathrm{He}} \gtrsim 2\,\mathrm{eV}$); re-solution is the sole He release channel.

Total trapped He inventory.

Define $\ell_{\mathrm{tot}} \equiv \sum_{m=1}^{M} \ell(m)\, \bar{c}_m$. Summing the He content over all cavities and using Eq. (157) with thermal emission dropped:

$$\frac{d\ell_{\mathrm{tot}}}{dt} = A_{\mathrm{sph}}\, \omega_h^{\mathrm{eff}}\, c_h \sum_{m=1}^{M} m^{1/3}\, \bar{c}_m - b_0\, \dot{\phi}\, \ell_{\mathrm{tot}}. \tag{183}$$

The first term is the total He trapping rate by all cavities (the sum is the weighted capture cross-section $\sum_m m^{1/3} \bar{c}_m$), and the second is the athermal re-solution loss (P8).

He allocation.

Helium is distributed across cavity sizes in proportion to the capture rate $A_{\mathrm{sph}}\, m^{1/3}\, \omega_h^{\mathrm{eff}}$, since that is what determines steady-state He loading when trapping and re-solution balance:

$$\ell(m) = \ell_{\mathrm{tot}} \frac{m^{1/3}\, \bar{c}_m}{\displaystyle\sum_{m'=1}^{M} m'^{1/3}\, \bar{c}_{m'}}. \tag{184}$$

This algebraic relation is recomputed at each time step. The $m^{1/3}$ weighting arises from the spherical-cavity capture radius $\xi(m) \propto m^{1/3}$ (Section 6.1), consistent with $\mathcal{K}_{h,m} = A_{\mathrm{sph}}\, m^{1/3}\, \omega_h^{\mathrm{eff}}$ (Table 28).

Free He.

The free-He equation retains all terms from Eq. (157), including the fixed-sink loss $\mathcal{D}_h c_h$, but with thermal He emission dropped:

$$\frac{dc_h}{dt} = G_{\mathrm{He}} - A_{\mathrm{sph}}\, \omega_h^{\mathrm{eff}}\, c_h \sum_{m=1}^{M} m^{1/3}\, \bar{c}_m + b_0\, \dot{\phi}\, \ell_{\mathrm{tot}} - \mathcal{D}_h\, c_h. \tag{185}$$

The trapping and re-solution terms are exactly the negatives of those in Eq. (219), so the He conservation law $d(c_h + \ell_{\mathrm{tot}})/dt = G_{\mathrm{He}} - \mathcal{D}_h c_h$ holds (cf. Eq. 314).

Case 2 summary.

The reduced system comprises M ODEs for $\bar{c}_m$, one ODE for c_h (Eq. 221), one ODE for ℓ_{tot} (Eq. 219), and the algebraic allocation (Eq. 220), giving the system count of Eq. (322):

$$n_{\mathrm{eq}}^{\mathrm{fission}} = N + M + 2.$$

8.3 Radiation Re-Solution: Limiting Behaviour

The re-solution rate $\Gamma_{\text{res}}(m, \ell) = b_0\, \ell\, \dot{\phi}$ (P8, Eq. 143) is athermal and appears in both reduced models as $-b_0\, \dot{\phi}\, \bar{\ell}$ in Eq. (217) and as $-b_0\, \dot{\phi}\, \ell_{\text{tot}}$ in Eq. (219). Its competition with thermal He emission determines two limiting regimes.

Low-temperature limit ($\varepsilon_h \to 0$).

Setting $d\bar{\ell}/dt = 0$ in Eq. (217) and neglecting the advection terms:

$$\bar{\ell}_{\text{ss}} = \frac{A_{\text{sph}}\, m^{1/3}\, \omega_h^{\text{eff}}\, c_h}{b_0\, \dot{\phi}}. \tag{186}$$

Re-solution sets the equilibrium He loading by competing with trapping. Without re-solution ($b_0 = 0$), $\bar{\ell}$ grows until $\ell_{\max}(m)$ is reached.

High-temperature limit ($\varepsilon_h \gg b_0\, \dot{\phi}\, \bar{\ell}$).

$$\bar{\ell}_{\text{ss}} \approx \frac{A_{\text{sph}}\, m^{1/3}\, \omega_h^{\text{eff}}\, c_h}{\varepsilon_h(m, \bar{\ell}_{\text{ss}})}, \tag{187}$$

which is the thermodynamic (detailed-balance) equilibrium loading (cf. Eq. 140), independent of irradiation rate.

Crossover temperature.

The transition between the two regimes occurs when $\varepsilon_h(m, \bar{\ell}) \sim b_0\, \dot{\phi}\, \bar{\ell}$. For typical EUROFER parameters ($E_B^{\text{He}} \sim 2.0\,\text{eV}$, $b_0 \sim 10^{-2}\,\text{dpa}^{-1}$, $\dot{\phi} \sim 10^{-6}\,\text{dpa/s}$), this crossover lies at $T \approx 500$–$700\,^\circ\text{C}$: below this temperature re-solution controls He loading; above it thermal emission dominates.

Consequences for bubble nucleation.

Re-solution suppresses nucleation at low temperatures by ejecting He from small embryos, creating an incubation dose $\propto 1/G_{\text{He}}$ for fixed b_0. It preferentially dissolves small, He-rich embryos (low ℓ_{ss} from Eq. 222) while leaving large stable bubbles intact (higher capture rate $\propto m^{1/3}$), producing the bimodal cavity size distributions commonly observed experimentally.

8.4 Comparison of Cases 1 and 2

Table 34 summarises the two approximations and their applicability. Both use the same $\bar{c}_m$ equation (Eq. 215) and the same unbiased SIA-shrinkage rate $\sum_n \mathcal{K}_{n,m}^{(\text{cav})} c_n$; they differ in how the He degree of freedom is represented and in the approximations made for thermal He emission. For large M, logarithmic size grouping for $m > m_{\text{group}} \approx 50$–$200$ is applied in both cases; the $\bar{\ell}(m)$ variable extends naturally to grouped bins by carrying the mean loading per bin.

Table 32 Comparison of the two He–vacancy state-space reductions. Both cases retain the full SIA equation (Eq. 152) and use the unbiased cavity absorption rate $\mathcal{K}^{(\mathrm{cav})}_{n,m}$ (Eq. 153) with $Z_i^{\mathrm{cav}} = 1$.

	Case 1 (Fusion)	**Case 2 (Fission)**
He generation rate	$\sim$10–15 appm/dpa	$\sim$0.3–1 appm/dpa
Typical He/V ratio $\bar{\ell}/m$	0.1–10	$\ll 1$
He tracked as	$\bar{\ell}(m)$ per size (ODE)	ℓ_{tot} (scalar ODE)
He allocation	implicit in $\bar{\ell}(m)$	algebraic (Eq. 220)
He emission	thermal + re-solution	re-solution only
He effect on E_b^v	full $E_b^v(m, \bar{\ell})$	ideal-gas correction
He conservation	Eq. 163 exactly	Eq. 314 (approx.)
Total ODEs	$N + 2M + 1$	$N + M + 2$
Reduction vs. full grid	$\approx \frac{3}{10}\alpha_{\mathrm{He}} M^{2/3}$	$\approx \frac{3}{5}\alpha_{\mathrm{He}} M^{2/3}$

9 State Space Reduction via Size-Bin Moments

The cluster dynamics (CD) system derived in Section 7 tracks one ODE per cluster size. Even after the He approximations of Section 8, the SIA ladder runs to $N \sim 10^3$–10^4 and the vacancy grid to $M \sim 10^3$–10^4, giving several thousand coupled ODEs. For large-scale parametric studies or coupling to microstructure-informed continuum codes, a further reduction is desirable. This section presents a systematic size-grouping framework that replaces the per-size equations with a much smaller set of *bin moment equations*, derived by integrating the master equations over logarithmically spaced size bins. Two closure strategies are described: a *moment-closure* approach analogous to the method of moments in aerosol science, and a *shape-function* approach analogous to the Galerkin finite-element method. Both reduce the ODE count from $O(N)$ or $O(M)$ to $O(N_B)$ where $N_B = O(\log N / \log r)$ for bin ratio $r > 1$, typically 20–60 bins for $N = 10^4$ and $r = 1.5$–2.

The derivation is presented for the SIA cluster population c_n; the vacancy cluster population $\bar{c}_m$ follows by identical steps with $n \to m$, $N \to M$, and the appropriate rate constants substituted.

9.1 Logarithmic Bin Partition

Partition the integer size axis $\{1, 2, \ldots, N\}$ into N_B bins. The first n_0 sizes (mobile clusters, $n \le n^i_{\max}$) are resolved individually; the remaining large immobile clusters are grouped. Define the bin boundaries

$$1 = s_0 < s_1 < \cdots < s_{n_0} = n^i_{\max} < s_{n_0+1} < \cdots < s_{N_B} = N, \tag{188}$$

where for $k > n_0$ the boundaries follow a geometric progression with ratio $r > 1$:

$$s_{k+1} = \lfloor r\, s_k \rfloor + 1, \qquad r = 1.5\text{–}2.0. \tag{189}$$

Bin k spans the integer interval $\mathcal{B}_k = \{s \in \mathbb{Z} : s_k \le s < s_{k+1}\}$ and contains $\Delta_k = s_{k+1} - s_k$ cluster sizes. The representative size of bin k is the geometric mean of its boundaries:

$$\hat{s}_k = \sqrt{s_k\, s_{k+1}}. \tag{190}$$

The number of grouped bins is

$$N_B^{\text{group}} = \left\lceil \frac{\ln(N/n^i_{\max})}{\ln r} \right\rceil, \qquad N_B = n^i_{\max} + N_B^{\text{group}}. \tag{191}$$

For $N = 5000$, $n^i_{\max} = 100$, $r = 2$: $N_B^{\text{group}} = \lceil \ln(50)/\ln 2 \rceil = 6$, so only 106 equations replace 4900 per-size ODEs.

9.2 Bin Moments

For each bin k define the *zeroth moment* (total cluster concentration in the bin) and the *first moment* (total defect inventory in the bin):

$$\mu_k^{(0)}(t) \equiv \sum_{n \in \mathcal{B}_k} c_n(t), \qquad \mu_k^{(1)}(t) \equiv \sum_{n \in \mathcal{B}_k} n\, c_n(t). \tag{192}$$

The mean cluster size within bin k is $\langle n \rangle_k = \mu_k^{(1)}/\mu_k^{(0)}$ and the swelling (first-moment sum) is $S_I = \sum_k \mu_k^{(1)}$. Higher moments can be defined similarly; in practice, tracking $(\mu_k^{(0)}, \mu_k^{(1)})$ at each bin is sufficient to close the rate equations to second order.

9.3 Bin-Integrated Rate Equations

Summing the master equation (152) over all $n \in \mathcal{B}_k$ gives the zeroth-moment equation:

$$\begin{aligned} \frac{d\mu_k^{(0)}}{dt} = {} & \underbrace{\sum_{n \in \mathcal{B}_k} G_n}_{\text{production}} + \underbrace{J_{k-1\to k} - J_{k\to k+1}}_{\substack{\text{inter-bin flux} \\ \text{(advection in size space)}}} \\ & \underbrace{- \mu_k^{(0)} \sum_{m=1}^{M} \hat{\mathcal{K}}^{(\text{cav})}_{k,m}\, \bar{c}_m}_{\text{cavity absorption}} - \underbrace{\mathcal{D}_i\, \mu_k^{(0)}}_{\text{fixed sinks}} + R_k, \end{aligned} \tag{193}$$

where R_k collects the emission, coalescence, and trap-mutation source terms (see below), and $\hat{\mathcal{K}}^{(\text{cav})}_{k,m}$ is the bin-averaged cavity absorption rate:

$$\hat{\mathcal{K}}^{(\text{cav})}_{k,m} \equiv \frac{1}{\Delta_k} \sum_{n \in \mathcal{B}_k} \mathcal{K}^{(\text{cav})}_{n,m}. \tag{194}$$

The *inter-bin flux* from bin k to bin $k+1$ counts clusters that grow past the boundary s_{k+1} by absorbing a mono-defect (or a mobile cluster in the coalescence formulation):

$$J_{k\to k+1} = \hat{\mathcal{K}}^{ii}_{k,\mathrm{bdy}}\,\mu^{(0)}_k\,c_{n^i_{\max}} + \sum_{m'=1}^{m^v_{\max}} \hat{\mathcal{K}}^{vi,m'}_{k,\mathrm{bdy}}\,\mu^{(0)}_k\,c_{m',0}, \tag{195}$$

where $\hat{\mathcal{K}}_{k,\mathrm{bdy}}$ denotes the rate constant evaluated at the bin boundary s_{k+1}. In the mono-defect limit ($n^i_{\max} = m^v_{\max} = 1$), this simplifies to

$$J^{\mathrm{mono}}_{k\to k+1} = \mathcal{K}^{\mathrm{loop}}_{i,s_{k+1}}\,c_1\,\mu^{(0)}_k, \tag{196}$$

i.e. the rate at which clusters sitting at the upper boundary of bin k absorb a mono-SIA and cross into bin $k+1$.

Multiplying Eq. (152) by n and summing over $\mathcal{B}_k$ gives the first-moment equation:

$$\begin{aligned}\frac{d\mu^{(1)}_k}{dt} = \sum_{n\in\mathcal{B}_k} n\,G_n \;+\; s_{k+1}\,J_{k-1\to k} - s_k\,J_{k\to k+1} \;+\; \mu^{(0)}_k\big(\hat{\mathcal{K}}^{ii}_k - \hat{\mathcal{K}}^{vi}_k\big)\\ - \mu^{(1)}_k \sum_{m=1}^{M} \hat{\mathcal{K}}^{(\mathrm{cav})}_{k,m}\,\bar{c}_m - \mathcal{D}_i\,\mu^{(1)}_k + n\text{-weighted source terms,}\end{aligned} \tag{197}$$

where the second line accounts for the fact that SIA absorption increases the size of each cluster by the absorbed cluster's size, and the inter-bin flux carries the boundary-size weight s_{k+1} (clusters entering bin $k+1$) or s_k (clusters leaving).

9.4 Closure: Two Strategies

Both $\mu^{(0)}_k$ and $\mu^{(1)}_k$ are known at each time step. The rate equations above contain products such as $\hat{\mathcal{K}}^{(\mathrm{cav})}_{k,m}\mu^{(0)}_k$ and $J_{k\to k+1}$, which require knowledge of the *intra-bin distribution* c_n for $n \in \mathcal{B}_k$. Two strategies close this system.

9.4.1 Piecewise-Constant (Zeroth-Order) Closure

Assume the distribution is uniform within each bin:

$$c_n \approx \frac{\mu^{(0)}_k}{\Delta_k} \equiv \bar{c}_k \qquad \forall\, n \in \mathcal{B}_k. \tag{198}$$

Then all bin-averaged rate constants reduce to evaluation at the bin midpoint $\hat{s}_k$:

$$\hat{\mathcal{K}}^{(\mathrm{cav})}_{k,m} \approx \mathcal{K}^{(\mathrm{cav})}_{\hat{s}_k,m}, \tag{199}$$

and the inter-bin flux becomes

$$J_{k\to k+1} \approx \mathcal{K}^{\mathrm{loop}}_{i,\hat{s}_{k+1}}\,c_1\,\bar{c}_k. \tag{200}$$

This is the standard *logarithmic grouping* used in most production CD codes Golubov et al (2000); Ovcharenko et al (2003); Golubov et al (2007); Kohnert and Wirth (2017); Cui et al (2020). It conserves $\mu_k^{(0)}$ exactly but not $\mu_k^{(1)}$; the error in S_I is $O(\Delta_k/\hat{s}_k) = O(r-1)$.

9.4.2 Shape-Function (Galerkin) Closure

Approximate the intra-bin distribution as a linear combination of basis functions defined on the bin:

$$c_n \approx \sum_p a_k^{(p)}\,\phi_p(n), \qquad n \in \mathcal{B}_k, \tag{201}$$

where the coefficients $a_k^{(p)}$ are determined by the known moments $(\mu_k^{(0)}, \mu_k^{(1)}, \ldots)$ and ϕ_p are chosen basis functions on $\mathcal{B}_k$.

Hat-function basis (linear FEM on log scale).

Map $n \in [s_k, s_{k+1})$ to $\xi = \ln(n/s_k)/\ln r \in [0,1)$ and use piecewise-linear (hat) functions:

$$\phi_0(\xi) = 1-\xi, \qquad \phi_1(\xi) = \xi. \tag{202}$$

The two coefficients $a_k^{(0)}$ and $a_k^{(1)}$ are fixed by the two moments $\mu_k^{(0)}$ and $\mu_k^{(1)}$ via:

$$a_k^{(0)}\,I_{00} + a_k^{(1)}\,I_{01} = \mu_k^{(0)}, \tag{203}$$

$$a_k^{(0)}\,I_{10} + a_k^{(1)}\,I_{11} = \mu_k^{(1)}, \tag{204}$$

where $I_{pq} = \sum_{n\in\mathcal{B}_k} n^p\,\phi_q(\ln n/s_k/\ln r)$ are geometry integrals computed once at setup. This two-parameter ansatz is the FEM analogue of the piecewise-linear representation in aerosol sectional methods Adams and Seinfeld (2002); Jacobson et al (1994).

Log-normal basis.

Alternatively, within each bin assume a log-normal shape:

$$c_n \approx \frac{a_k}{\sqrt{2\pi}\,\sigma_k\,n}\exp\left[-\frac{(\ln n - \mu_k)^2}{2\sigma_k^2}\right], \tag{205}$$

with $\mu_k = \ln\langle n\rangle_k$ and σ_k determined from the second moment $\mu_k^{(2)}$. This three-parameter closure requires tracking one additional moment per bin but captures the asymmetric, right-skewed distributions typical of nucleation-growth kinetics.

Galerkin projection.

Regardless of basis, the Galerkin projection of the master equation (152) onto ϕ_p over $\mathcal{B}_k$ gives:

$$\sum_{n\in\mathcal{B}_k} \phi_p(n)\,\frac{dc_n}{dt} = \sum_{n\in\mathcal{B}_k} \phi_p(n)\,f_n(\mathbf{c}), \tag{206}$$

where f_n is the right-hand side of Eq. (152). The left side becomes $\dot{a}_k^{(p)}$ (via the moment relations), and the right side is expressed in terms of $\{a_k^{(p)}\}$ and the inter-bin fluxes through the shape ansatz. The resulting ODE system for the coefficients $\{a_k^{(p)}\}$ is closed and of size $P \times N_B$ where P is the number of moments per bin ($P = 1$ for piecewise-constant, $P = 2$ for hat-function, $P = 3$ for log-normal).

9.5 Inter-Bin Flux Treatment

The inter-bin flux $J_{k\to k+1}$ requires special care because it drives mass transfer between bins and must satisfy discrete conservation exactly. Writing the flux as an upwind scheme in size space:

$$J_{k\to k+1} = \begin{cases} \beta_{s_{k+1}}\, c^-_{s_{k+1}} & \text{if growth} > \text{shrinkage}, \\ \beta_{s_{k+1}}\, c^+_{s_{k+1}} & \text{if shrinkage} > \text{growth}, \end{cases} \tag{207}$$

where $c^{\pm}_{s_{k+1}}$ denotes the distribution evaluated just below/above the boundary from within the respective bin, and $\beta_{s_{k+1}}$ is the net growth rate at the boundary:

$$\beta_s = \mathcal{K}^{\text{loop}}_{i,s}\, c_1 - \mathcal{K}^{\text{loop}}_{v,s}\, c_{1,0} - \varepsilon_i(s). \tag{208}$$

The upwind flux preserves positivity of $\mu_k^{(0)}$ and prevents spurious oscillations between adjacent bins. For the shape-function closure, $c^-_{s_{k+1}}$ is evaluated from the intra-bin ansatz at $n = s_{k+1} - 1$, giving a smooth interpolation.

9.6 Accuracy and Conservation Properties

Mass conservation.

Summing Eq. (193) over all bins recovers $dS_I/dt = d(\sum_k \mu_k^{(1)})/dt$ exactly, provided the inter-bin fluxes satisfy the telescoping identity:

$$\sum_{k=1}^{N_B} (J_{k-1\to k} - J_{k\to k+1}) = 0 \tag{209}$$

(boundary terms vanish: $J_{0\to 1} = 0$ and $J_{N_B\to N_B+1} = 0$). This holds for any consistent flux formula. The swelling identity $S(t) = S_I(t) + \Delta J^d(t)$ (Eq. 161) therefore remains exact regardless of the grouping scheme.

Truncation error.

For the piecewise-constant closure (Eq. 198), the error in the bin-averaged rate constant scales as:

$$\hat{\mathcal{K}}^{(\text{cav})}_{k,m} - \mathcal{K}^{(\text{cav})}_{\hat{s}_k,m} = O\!\left(\frac{\Delta_k^2}{\hat{s}_k^2}\right) = O\big((r-1)^2\big)\,, \tag{210}$$

since $\mathcal{K}^{(\mathrm{cav})}_{\hat{s},m} \propto \hat{s}^{1/3}$ or $\hat{s}^{1/2}$ and the bin width scales as $\Delta_k \sim (r-1)\hat{s}_k$. For $r = 2$ this gives ∼25% error per bin; for $r = 1.5$, ∼6%. The hat-function closure reduces the error by one order: $O((r-1)^3)$, making $r = 2$ already accurate to $\lesssim 1\%$.

Moment diagnostics.

The bin-integrated conservation diagnostics follow from Section 7.4.3:

$$\delta^{\mathrm{bin}}_{\mathrm{FP}} = \frac{\left|S - \sum_k \mu^{(1)}_k - \Delta J^d\right|}{S + \sum_k \mu^{(1)}_k + \Delta J^d}, \tag{211}$$

which should remain below 10^{-6} regardless of binning. A value above 10^{-4} indicates that the inter-bin fluxes are not conserving mass, typically due to an inconsistent upwind scheme at the bin boundaries.

9.7 Reduced System Size

Table 33 summarises the ODE count for the combined He-approximation and size-grouping reductions for representative parameter choices.

Table 33 ODE counts after combined size-grouping ($r = 2$) and He-approximation reductions. P is the number of moments per bin (1 = piecewise-constant, 2 = hat-function). Parameters: $N = M = 5000$, $n^i_{\max} = 100$, $m^v_{\max} = 5$, $\alpha_{\mathrm{He}} = 1.5$. "Full" refers to the un-grouped per-size system after He reduction.

He approx.	**Size grouping**	$\mathbf{N_B^{SIA}}$	$\mathbf{N_B^{vac}}$	$\mathbf{n_{eq}}$
Case 1 (fusion)	Full per-size	5 000	10 000	15 001
Case 1 (fusion)	Log-group, $P = 1$	113	113	$226\,P + 1 = 227$
Case 1 (fusion)	Log-group, $P = 2$	113	113	$226\,P + 1 = 453$
Case 2 (fission)	Full per-size	5 000	5 000	10 002
Case 2 (fission)	Log-group, $P = 1$	113	113	$226 + 2 = 228$
Case 2 (fission)	Log-group, $P = 2$	113	113	$452 + 2 = 454$

The log-grouping with $P = 1$ reduces the system by a factor of ∼40–70 relative to the per-size He-reduced system, and by a factor of ∼100–200 relative to the full (m, ℓ) grid. With $P = 2$ (hat-function), accuracy improves substantially at the cost of doubling the ODE count, still representing a ∼30–100× reduction. Timing benchmarks on typical irradiation histories (100 dpa at 10^{-6} dpa/s) show that the grouped systems run in seconds to minutes on a single core, compared with hours for the full per-size system.

9.8 Implementation Notes

Setup.

At initialisation, precompute: (i) the bin boundaries $\{s_k\}$ from Eq. (189); (ii) the representative sizes $\{\hat{s}_k\}$ from Eq. (190); (iii) the bin-averaged rate constants $\hat{\mathcal{K}}^{(\mathrm{cav})}_{k,m}$ for all (k, m) pairs from Eq. (199) (piecewise-constant) or by exact quadrature (hat-function); and (iv) the geometry integrals I_{pq} for the shape-function closure. Steps (i)–(iv) are $O(N_B \cdot M)$ and negligible compared with the ODE integration cost.

Coalescence terms.

For the i–i coalescence bins, the gain term $\frac{1}{2}\sum_{n'\in\mathcal{B}_{k'}}\sum_{n''\in\mathcal{B}_{k''}}\mathcal{K}^{ii}_{n',n''}\,c_{n'}\,c_{n''}$ where $n' + n'' \in \mathcal{B}_k$ is approximated as

$$\frac{1}{2}\,\mathcal{K}^{ii}_{\hat{s}_{k'},\hat{s}_{k''}}\,\mu^{(0)}_{k'}\,\mu^{(0)}_{k''}\,f_{\mathrm{overlap}}(k, k', k''), \tag{212}$$

where f_{overlap} is the fraction of $(\hat{s}_{k'} + \hat{s}_{k''})$ that falls within $\mathcal{B}_k$ (a geometric factor computed at setup). The double sum over bin pairs reduces from $O(N^2)$ to $O(N_B^2)$, a further speedup of $(N/N_B)^2 \sim 10^3$–10^4.

Coupling to the vacancy grid.

The SIA bins couple to the vacancy grid through the cavity absorption term $\hat{\mathcal{K}}^{(\mathrm{cav})}_{k,m}$, which is a $N_B \times M$ matrix precomputed at setup. When the vacancy grid is also grouped, this becomes a $N_B^{SIA} \times N_B^{vac}$ matrix, reducing the coupling cost from $O(N\cdot M)$ to $O(N_B^{SIA} \cdot N_B^{vac})$ per right-hand-side evaluation.

Conservation check.

Use Eq. (211) at every output step. An inconsistency in $\delta^{\mathrm{bin}}_{\mathrm{FP}}$ that was not present in the ungrouped system localises the error to the inter-bin flux implementation.

The full master equation (155) tracks every (m, ℓ) pair with $1 \le m \le M$ and $0 \le \ell \le \ell_{\max}(m)$ (Eq. 318), yielding n^{full}_{VH} coupled ODEs (Eq. 319) — typically 10^4–10^5 equations. Two physics-based approximations exploit the separation of timescales between He and vacancy kinetics to reduce this to a tractable system. Both retain the full SIA population (Eq. 152) and the free He balance (Eq. 157) unchanged; only the vacancy-cluster block is replaced.

Throughout this section we use the notation and rate constants established in Sections 6–7: $\omega^{\mathrm{eff}}_{\alpha}$ are the effective reaction frequencies (Eq. 129); $A_{\mathrm{sph}} = (48\pi^2)^{1/3}$; $\bar{c}_m = \sum_\ell c_{m,\ell}$ is the total cavity concentration; and $\mathcal{K}^{(\mathrm{cav})}_{n,m}$ is the SIA-cluster absorption rate (Eq. 153). The vacancy net-flux from size m to $m + 1$ is defined as:

$$J_{m\to m+1} \equiv A_{\mathrm{sph}}\, m^{1/3}\,\omega^{\mathrm{eff}}_{v}\, c_{1,0}\,\bar{c}_m - \varepsilon_v(m{+}1, \bar{\ell}(m{+}1))\,\bar{c}_{m+1}, \tag{213}$$

with forward (capture) and backward (emission) components $J^{+}_{m\to m+1}$ and $J^{-}_{m\to m+1}$ respectively, so that $J_{m\to m+1} = J^{+}_{m\to m+1} - J^{-}_{m+1\to m}$.

9.9 Case 1: Fast He Equilibration — Fusion Conditions

At temperatures where vacancies are mobile ($T \gtrsim 200\,^\circ\mathrm{C}$), the He reaction frequency greatly exceeds the vacancy reaction frequency, $\omega_h^{\mathrm{eff}} \gg \omega_v^{\mathrm{eff}}$. Helium equilibrates on each cavity faster than the cavity grows or shrinks, so the full two-dimensional (m, ℓ) distribution can be collapsed to two scalar quantities per vacancy size: the total cavity concentration $\bar{c}_m$ and the mean He loading

$$\bar{\ell}(m,t) \equiv \frac{\sum_\ell \ell\, c_{m,\ell}}{\bar{c}_m}. \tag{214}$$

This is the mean-field He approximation defined in Eq. (320).

Cavity size distribution.

Summing Eq. (155) over all ℓ at fixed m:

$$\frac{d\bar{c}_m}{dt} = J_{m-1\to m} - J_{m\to m+1} - \sum_{n=1}^{n^i_{\max}} \mathcal{K}^{(\mathrm{cav})}_{n,m}\, c_n\, \bar{c}_m + \Gamma_{\mathrm{TM}}(m{-}1,\bar{\ell})\, \bar{c}_{m-1} - \Gamma_{\mathrm{TM}}(m,\bar{\ell})\, \bar{c}_m, \tag{215}$$

where the vacancy net flux $J_{m\to m+1}$ is given by Eq. (213). The SIA-shrinkage sum $\sum_n \mathcal{K}^{(\mathrm{cav})}_{n,m} c_n$ spans all mobile SIA clusters $n = 1, \ldots, n^i_{\max}$ with no bias factor (spherical cavities are unbiased, $Z_i^{\mathrm{cav}} = 1$, Eq. 131). The vacancy capture rate $A_{\mathrm{sph}}\, m^{1/3}\, \omega_v^{\mathrm{eff}}$ is independent of ℓ, so the ℓ-summation is exact for the capture terms. The emission rate ε_v depends on ℓ through the He gas-pressure correction to $E_b^v(m,\ell)$ and is approximated by evaluation at the mean loading $\bar{\ell}(m)$.

Mean He loading.

Define the He inventory at vacancy size m as $N_{\mathrm{He}}(m) \equiv \bar{\ell}(m)\, \bar{c}_m$. Multiplying Eq. (155) by ℓ, summing over ℓ, and collecting terms:

$$\begin{aligned} \frac{dN_{\mathrm{He}}(m)}{dt} &= A_{\mathrm{sph}}\, m^{1/3}\, \omega_h^{\mathrm{eff}}\, c_h\, \bar{c}_m - \varepsilon_h(m,\bar{\ell})\, \bar{c}_m - b_0\, \dot{\phi}\, N_{\mathrm{He}}(m) \\ &\quad + \bar{\ell}(m{-}1)\, J^+_{m-1\to m} - \bar{\ell}(m)\, J^+_{m\to m+1} \\ &\quad + \bar{\ell}(m{+}1)\, J^-_{m+1\to m} - \bar{\ell}(m)\, J^-_{m\to m-1} \\ &\quad - \bar{\ell}(m) \sum_{n=1}^{n^i_{\max}} \mathcal{K}^{(\mathrm{cav})}_{n,m}\, c_n\, \bar{c}_m. \end{aligned} \tag{216}$$

The four groups of terms are: (i) He capture from the free pool minus thermal He emission (P5) minus radiation re-solution (P8); (ii–iii) He advected in and out as clusters grow or shrink along the m-axis, where J^+ is the forward (capture) component and J^- is the backward (emission) component of the vacancy flux; and (iv) He carried away when SIA clusters annihilate vacancies. Note the absence of any bias factor in the SIA-shrinkage term: the rate constant $\mathcal{K}^{(\mathrm{cav})}_{n,m}$ already encodes the unbiased cavity geometry ($Z_i^{\mathrm{cav}} = 1$).

Applying the quotient rule $d\bar{\ell}/dt = (\dot{N}_{\mathrm{He}} - \bar{\ell}\,\dot{\bar{c}}_m)/\bar{c}_m$ and substituting Eq. (215):

$$\boxed{\begin{aligned}\frac{d\bar{\ell}(m)}{dt} &= A_{\mathrm{sph}}\, m^{1/3}\, \omega_h^{\mathrm{eff}}\, c_h - \varepsilon_h\big(m, \bar{\ell}(m)\big) - b_0\, \dot{\phi}\, \bar{\ell}(m) \\ &\quad + \frac{J^{+}_{m-1\to m}}{\bar{c}_m}\big[\bar{\ell}(m{-}1) - \bar{\ell}(m)\big] + \frac{J^{-}_{m+1\to m}}{\bar{c}_m}\big[\bar{\ell}(m{+}1) - \bar{\ell}(m)\big].\end{aligned}} \tag{217}$$

Line 1 contains the local He kinetics: trapping from the free pool (proportional to c_h), thermal emission ε_h (P5, Eq. 140), and athermal re-solution (P8, Eq. 143) which drives $\bar{\ell}$ toward zero. Line 2 contains advection corrections: cavities flowing into size class m carry their own He loading, which may differ from $\bar{\ell}(m)$; the difference $[\bar{\ell}(m\pm1)-\bar{\ell}(m)]$ acts as a drift source.

Numerical implementation.

In practice, $N_{\mathrm{He}}(m) = \bar{\ell}(m)\,\bar{c}_m$ and $\bar{c}_m$ are evolved as primary variables; $\bar{\ell}(m)$ is recovered by division and used only when evaluating rates. When $\bar{c}_m$ approaches zero near the nucleation threshold, the quotient becomes ill-conditioned. Recommended practice: set a floor $\bar{c}_m^{\min} \sim 10^{-30}$ at/at below which $\bar{\ell}$ is reset to zero.

Case 1 summary.

Replacing the full (m, ℓ) grid with $2M$ ODEs — M for $\bar{c}_m$ (Eq. 215) and M for $\bar{\ell}(m)$ (Eq. 217) — plus the unchanged free-He equation (Eq. 157) gives the reduced system count of Eq. (321):

$$n_{\mathrm{eq}}^{\mathrm{fusion}} = N + 2M + 1.$$

The reduction factor relative to the full grid is approximately $n_{VH}^{\mathrm{full}}/(2M) \approx \frac{3}{10}\alpha_{\mathrm{He}} M^{2/3}$, which is ∼10–20 for $M = 10^3$ and $\alpha_{\mathrm{He}} = 1.5$.

9.10 Case 2: Decoupled He Inventory — Fission Conditions

Under fission irradiation the He generation rate is low (∼0.3–1 appm/dpa vs. ∼10–15 appm/dpa for fusion), so most cavities carry $\ell \ll m$ and the He/V ratio $\bar{\ell}/m \ll 1$. Helium acts as a weak perturbation on the void kinetics rather than controlling nucleation. Rather than tracking $\bar{\ell}(m)$ as a separate variable at each size, the total trapped He inventory is followed as a single scalar and distributed algebraically across cavity sizes.

Cavity size distribution.

The $\bar{c}_m$ equation is identical to Case 1, Eq. (215), with the vacancy emission rate evaluated using an He-corrected binding energy. In the low-loading limit the He gas pressure obeys the ideal-gas law, giving an effective binding energy:

$$E_b^{v,\mathrm{eff}}(m) \equiv E_b^v\big(m,\, \ell(m)\big) = E_f^v + \frac{2\gamma_s\,\Omega}{r_m} - \frac{\ell(m)\, k_B T}{m}, \tag{218}$$

where $r_m = \xi(m)\,a$, γ_s is the surface tension, $\Omega = a^3/2$ is the atomic volume, and $\ell(m)$ is the He content assigned to cavities of size m by the allocation rule below. The third term is the ideal-gas pressure correction $p(\ell, m)\,\Omega = (\ell\, k_B T/m) \cdot m/m = \ell k_B T/m$, adequate when $\ell/m \ll 1$. At fission temperatures and low He loading, thermal He emission is negligible ($E_B^{\mathrm{He}} \gtrsim 2\,\mathrm{eV}$); re-solution is the sole He release channel.

Total trapped He inventory.

Define $\ell_{\mathrm{tot}} \equiv \sum_{m=1}^{M} \ell(m)\,\bar{c}_m$. Summing the He content over all cavities and using Eq. (157) with thermal emission dropped:

$$\frac{d\ell_{\mathrm{tot}}}{dt} = A_{\mathrm{sph}}\,\omega_h^{\mathrm{eff}}\,c_h \sum_{m=1}^{M} m^{1/3}\,\bar{c}_m - b_0\,\dot{\phi}\,\ell_{\mathrm{tot}}. \tag{219}$$

The first term is the total He trapping rate by all cavities (the sum is the weighted capture cross-section $\sum_m m^{1/3}\bar{c}_m$), and the second is the athermal re-solution loss (P8).

He allocation.

Helium is distributed across cavity sizes in proportion to the capture rate $A_{\mathrm{sph}}\,m^{1/3}\,\omega_h^{\mathrm{eff}}$, since that is what determines steady-state He loading when trapping and re-solution balance:

$$\ell(m) = \ell_{\mathrm{tot}}\,\frac{m^{1/3}\,\bar{c}_m}{\displaystyle\sum_{m'=1}^{M} m'^{1/3}\,\bar{c}_{m'}}. \tag{220}$$

This algebraic relation is recomputed at each time step. The $m^{1/3}$ weighting arises from the spherical-cavity capture radius $\xi(m) \propto m^{1/3}$ (Section 6.1), consistent with $\mathcal{K}_{h,m} = A_{\mathrm{sph}}\,m^{1/3}\,\omega_h^{\mathrm{eff}}$ (Table 28).

Free He.

The free-He equation retains all terms from Eq. (157), including the fixed-sink loss $\mathcal{D}_h c_h$, but with thermal He emission dropped:

$$\frac{dc_h}{dt} = G_{\mathrm{He}} - A_{\mathrm{sph}}\,\omega_h^{\mathrm{eff}}\,c_h \sum_{m=1}^{M} m^{1/3}\,\bar{c}_m + b_0\,\dot{\phi}\,\ell_{\mathrm{tot}} - \mathcal{D}_h\,c_h. \tag{221}$$

The trapping and re-solution terms are exactly the negatives of those in Eq. (219), so the He conservation law $d(c_h + \ell_{\mathrm{tot}})/dt = G_{\mathrm{He}} - \mathcal{D}_h c_h$ holds (cf. Eq. 314).

Case 2 summary.

The reduced system comprises M ODEs for $\bar{c}_m$, one ODE for c_h (Eq. 221), one ODE for ℓ_{tot} (Eq. 219), and the algebraic allocation (Eq. 220), giving the system count of Eq. (322):

$$n_{\mathrm{eq}}^{\mathrm{fission}} = N + M + 2.$$

9.11 Radiation Re-Solution: Limiting Behaviour

The re-solution rate $\Gamma_{\mathrm{res}}(m,\ell) = b_0\,\ell\,\dot{\phi}$ (P8, Eq. 143) is athermal and appears in both reduced models as $-b_0\,\dot{\phi}\,\bar{\ell}$ in Eq. (217) and as $-b_0\,\dot{\phi}\,\ell_{\mathrm{tot}}$ in Eq. (219). Its competition with thermal He emission determines two limiting regimes.

Low-temperature limit ($\varepsilon_h \to 0$).

Setting $d\bar{\ell}/dt = 0$ in Eq. (217) and neglecting the advection terms:

$$\bar{\ell}_{\mathrm{ss}} = \frac{A_{\mathrm{sph}}\,m^{1/3}\,\omega_h^{\mathrm{eff}}\,c_h}{b_0\,\dot{\phi}}. \tag{222}$$

Re-solution sets the equilibrium He loading by competing with trapping. Without re-solution ($b_0 = 0$), $\bar{\ell}$ grows until $\ell_{\max}(m)$ is reached.

High-temperature limit ($\varepsilon_h \gg b_0\,\dot{\phi}\,\bar{\ell}$).

$$\bar{\ell}_{\mathrm{ss}} \approx \frac{A_{\mathrm{sph}}\,m^{1/3}\,\omega_h^{\mathrm{eff}}\,c_h}{\varepsilon_h(m,\bar{\ell}_{\mathrm{ss}})}, \tag{223}$$

which is the thermodynamic (detailed-balance) equilibrium loading (cf. Eq. 140), independent of irradiation rate.

Crossover temperature.

The transition between the two regimes occurs when $\varepsilon_h(m,\bar{\ell}) \sim b_0\,\dot{\phi}\,\bar{\ell}$. For typical EUROFER parameters ($E_B^{\mathrm{He}} \sim 2.0\,\mathrm{eV}$, $b_0 \sim 10^{-2}\,\mathrm{dpa}^{-1}$, $\dot{\phi} \sim 10^{-6}\,\mathrm{dpa/s}$), this crossover lies at $T \approx 500$–$700\,°\mathrm{C}$: below this temperature re-solution controls He loading; above it thermal emission dominates.

Consequences for bubble nucleation.

Re-solution suppresses nucleation at low temperatures by ejecting He from small embryos, creating an incubation dose $\propto 1/G_{\mathrm{He}}$ for fixed b_0. It preferentially dissolves small, He-rich embryos (low ℓ_{ss} from Eq. 222) while leaving large stable bubbles intact (higher capture rate $\propto m^{1/3}$), producing the bimodal cavity size distributions commonly observed experimentally.

9.12 Comparison of Cases 1 and 2

Table 34 summarises the two approximations and their applicability. Both use the same $\bar{c}_m$ equation (Eq. 215) and the same unbiased SIA-shrinkage rate $\sum_n \mathcal{K}_{n,m}^{(\mathrm{cav})} c_n$; they differ in how the He degree of freedom is represented and in the approximations made for thermal He emission. For large M, logarithmic size grouping for $m > m_{\mathrm{group}} \approx 50$–$200$ is applied in both cases; the $\bar{\ell}(m)$ variable extends naturally to grouped bins by carrying the mean loading per bin.

Table 34 Comparison of the two He–vacancy state-space reductions. Both cases retain the full SIA equation (Eq. 152) and use the unbiased cavity absorption rate $\mathcal{K}^{(\mathrm{cav})}_{n,m}$ (Eq. 153) with $Z_i^{\mathrm{cav}} = 1$.

	Case 1 (Fusion)	**Case 2 (Fission)**
He generation rate	$\sim$10–15 appm/dpa	$\sim$0.3–1 appm/dpa
Typical He/V ratio $\bar{\ell}/m$	0.1–10	$\ll 1$
He tracked as	$\bar{\ell}(m)$ per size (ODE)	ℓ_{tot} (scalar ODE)
He allocation	implicit in $\bar{\ell}(m)$	algebraic (Eq. 220)
He emission	thermal + re-solution	re-solution only
He effect on E_b^v	full $E_b^v(m, \bar{\ell})$	ideal-gas correction
He conservation	Eq. 163 exactly	Eq. 314 (approx.)
Total ODEs	$N + 2M + 1$	$N + M + 2$
Reduction vs. full grid	$\approx \frac{3}{10}\alpha_{\mathrm{He}} M^{2/3}$	$\approx \frac{3}{5}\alpha_{\mathrm{He}} M^{2/3}$

10 Extended Cluster Dynamics Model for H, He, V, and I Clusters

The foregoing sections treated the irradiated microstructure as a three-component system: SIA clusters, He–vacancy clusters, and free He. Under fusion-relevant conditions — and especially for accelerator-driven systems where the hydrogen transmutation rate can reach $\sim$1000–1500 appm/dpa — hydrogen must be promoted to an explicit fourth species. The key physical differences from helium are:

1. H migrates $\sim 10^3$–$10^5\times$ faster than vacancies and $\sim$10–100$\times$ faster than interstitial He (migration energy $E_m^H \approx 0.04$–0.09 eV versus $E_m^{\mathrm{He}} \approx 0.06$–$0.13$ eV and $E_m^v \approx 0.60$–0.70 eV in bcc Fe).
2. H binds to vacancies moderately strongly ($E_b^H \approx 0.43$–0.57 eV for $\mathrm{H}_n\mathrm{V}_1$ with $n \leq 6$), so it is retained by irradiation-induced vacancies far longer than its thermal residence time in the perfect lattice.
3. H has negligible binding to SIA clusters ($E_b^{H\text{-SIA}} \lesssim 0.1$ eV), so H–SIA clusters are treated as unstable and dissociate immediately.
4. H occupies the *surface shell* of He–V bubbles, forming a core-shell He(inside)–H(outside) structure with binding energy $E_b^{H\text{-bub}} \approx 0.4$–$0.8$ eV. Up to $\sim m^{2/3}$ H atoms can be accommodated on the bubble surface.
5. Because $\omega_H \gg \omega_v$, H equilibrates on every defect much faster than the vacancy microstructure evolves. This timescale separation enables an *adiabatic elimination* of the H degree of freedom for large clusters, replacing explicit H tracking with an algebraic quasi-static loading, exactly as He was eliminated in the mean-field approximations of Sections 9.9–9.10.

The strategy therefore has two tiers:

Small clusters ($m \le m_H^*$, $\ell \le \ell_H^*$): H is tracked explicitly as an additional cluster index p (number of H atoms per cluster), giving a three-dimensional state (m, ℓ, p) for vacancy-containing clusters.

Large clusters ($m > m_H^*$): H is adiabatically eliminated; only the *mean H loading* $\bar{p}(m, \ell, t)$ per vacancy/He-vacancy cluster is tracked, determined at each timestep by the quasi-static condition $d\bar{p}/dt = 0$.

The crossover sizes $m_H^* \approx$ 3–5 and $\ell_H^* \approx$ 2–4 are chosen to cover the clusters whose H-loading dynamics are comparable to the cluster growth timescale.

10.1 Cluster Notation and State Space

With hydrogen included, the cluster population is extended as follows:

- c_n: concentration of n-SIA cluster (loop). H binding to SIAs is negligible; this population is unchanged.
- $c_{m,\ell,p}$: concentration of a cluster containing m vacancies, ℓ He atoms, and p H atoms, for $m = 1, \dots, M$; $0 \le \ell \le \ell_{\max}(m)$; $0 \le p \le p_{\max}(m)$.
- c_H: concentration of free interstitial hydrogen.
- c_h: concentration of free interstitial helium (unchanged).

The maximum H loading per cluster scales with the available surface sites:

$$p_{\max}(m) = \lfloor \alpha_H \, m^{2/3} \rfloor, \qquad \alpha_H \approx 1.0\text{–}2.0. \tag{224}$$

This is identical in form to $\ell_{\max}(m) = \lfloor \alpha_{\mathrm{He}} m^{2/3} \rfloor$ (Eq. 318), reflecting the same surface-area scaling.

The full three-dimensional (m, ℓ, p) state space scales as

$$n_{VHHe}^{\mathrm{full}} \approx M\Big(1 + \tfrac{3}{5}\alpha_{\mathrm{He}}\, M^{2/3}\Big) \cdot \tfrac{1}{2}\big(1 + \alpha_H\, M^{2/3}\big), \tag{225}$$

which grows as $M^{7/3}$ and is intractable for $M \gtrsim 100$. The adiabatic elimination below reduces this back to the manageable (m, ℓ) plus one extra scalar per (m, ℓ) cluster.

10.2 New Reaction Processes Involving Hydrogen

The existing processes P1–P8 are retained unchanged. Six new reaction processes describe H kinetics.

P9: H trapping by a vacancy cluster.

Interstitial H diffusing in 3D is captured by a vacancy cluster of m vacancies containing ℓ He and p H atoms ($p < p_{\max}(m)$):

$$\mathcal{K}_{H,m}^{\mathrm{cav}} = A_{\mathrm{sph}}\, m^{1/3}\, \omega_H^{\mathrm{eff}}, \tag{226}$$

where $\omega_H^{\rm eff} = D_H/a^2$ with $D_H = \nu_H \exp(-E_m^H/k_BT)$. This is the same Smoluchowski form as for He capture (Eq. 131), reflecting the spherical capture geometry. H capture increases p by one: $\mathrm{H}_{m,\ell,p} + H_{\rm int} \to \mathrm{H}_{m,\ell,p+1}$.

P10: H thermal emission from a vacancy cluster.

H detrapping from $\mathrm{H}_{m,\ell,p}$ by thermal activation:

$$\varepsilon_H(m,\ell,p) = A_{\rm sph}\, m^{1/3}\, \omega_H^{\rm eff}\, \exp\!\big(-E_b^H(m,\ell,p)/k_BT\big). \tag{227}$$

The capture radius uses $m^{1/3}$ (not $(m-1)^{1/3}$) because H removal does not change m, by exact analogy with He emission (Eq. 140). The binding energy

$$E_b^H(m,\ell,p) \approx E_b^H(0)\,\big[1-(p-1)/p_{\max}(m)\big] + \delta E_{\rm He}(\ell/m) \tag{228}$$

takes the base monovacancy binding value $E_b^H(0) \approx 0.51\,\mathrm{eV}$ (DFT, α-Fe) and applies two corrections: a saturation factor that decreases binding as the H sites fill, and a He-induced enhancement $\delta E_{\rm He} > 0$ that reflects the core-shell structure ($\delta E_{\rm He} \approx 0.1$–$0.3\,\mathrm{eV}$ for $\ell/m \sim 0.5$–1.5).

P11: H absorption at fixed sinks.

$$\mathcal{D}_H = \mathcal{D}_H^d + \mathcal{D}_H^{gb} + \mathcal{D}_H^p \qquad [\mathrm{s}^{-1}], \tag{229}$$

with each term having the same form as Eqs. (135)–(137) with $\omega_H^{\rm eff}$ substituted. Dislocations and grain boundaries are unbiased for H ($Z_H^d = Z_H^{gb} = 1$) because H has no significant long-range elastic interaction with these defects.

P12: H interaction with SIA clusters.

As discussed above, H binds negligibly to SIAs. The only SIA–H interaction is therefore indirect: free H scattering off mobile SIA clusters contributes a weak scattering sink, but it is negligible compared with trapping at vacancies. In the present model:

$$\text{H–SIA binding: } E_b^{H\text{-SIA}} \approx 0 \implies \text{H–SIA clusters unstable; dissociate immediately.} \tag{230}$$

This removes the H–SIA channel from the master equations entirely.

P13: H–V recombination.

A free H that encounters a free SIA re-combines and annihilates (the H is not annihilated but released into the perfect lattice):

$$H_{\rm int} + I_1 \to \text{lattice site} + H_{\rm int}^{\rm fast}. \tag{231}$$

In practice this is a null reaction because the SIA is annihilated and H is immediately free again. Its net effect on c_H is zero; it acts only as a sink for the SIA. The rate is $\mathcal{K}_{H,i} = 4\sqrt{3}\pi\,(\omega_H + \omega_i)$ but since $\omega_H \gg \omega_i$ this is effectively $4\sqrt{3}\pi\,\omega_H$. However, because H concentrations are very small compared with the SIA sink strengths, this term is negligible in practice and is dropped.

P14: H cascade re-solution.

By analogy with He re-solution (P8), a cascade event can eject a trapped H atom from a cluster:

$$\Gamma^{H}_{\text{res}}(m,\ell,p) = b_0^H \, p \, \dot{\phi}, \tag{232}$$

with $b_0^H \approx 10^{-2}$–$10^{-1}\,\text{dpa}^{-1}/\text{H}$. The lower end applies to tightly bound H in deep traps; the upper end to surface-layer H on large bubbles.

10.3 Master Equation for Free Hydrogen

The balance for free interstitial H parallels that for free He (Eq. 157):

$$\boxed{\begin{aligned} \frac{dc_H}{dt} = G_H & \\ & - \underbrace{A_{\text{sph}}\,\omega_H^{\text{eff}}\,c_H \sum_{m=1}^{M} m^{1/3}\,\tilde{c}_m^{(p)}}_{\text{H trapping by vacancies/bubbles (P9)}} \\ & + \underbrace{\sum_{m=1}^{M}\sum_{\ell=0}^{\ell_{\max}}\sum_{p=1}^{p_{\max}} \varepsilon_H(m,\ell,p)\,c_{m,\ell,p}}_{\text{H thermal emission (P10)}} \\ & + \underbrace{b_0^H\,\dot{\phi}\sum_{m,\ell,p} p\,c_{m,\ell,p}}_{\text{H re-solution (P14)}} \\ & - \underbrace{\mathcal{D}_H\,c_H}_{\text{fixed sinks (P11)}} \quad , \end{aligned}} \tag{233}$$

where $\tilde{c}_m^{(p)} \equiv \sum_{\ell,p:\,p<p_{\max}(m)} c_{m,\ell,p}$ is the total concentration of *unsaturated* clusters of size m (those still accepting additional H).

Conservation.

Defining the total H inventory:

$$S_H(t) \equiv c_H + \sum_{m,\ell,p} p\,c_{m,\ell,p}, \tag{234}$$

summing the H-content equation over all cluster species gives the exact conservation law:

$$\frac{dS_H}{dt} = G_H - \mathcal{D}_H\,c_H, \tag{235}$$

i.e. total H inventory grows at the production rate and is depleted only by absorption at fixed sinks. This is the direct analogue of the He conservation law (Eq. 314).

10.4 Master Equation for H–Vacancy–He Clusters (Small Clusters, Explicit H)

For small clusters ($m \le m_H^*$, typically $m \le 5$) the H index p is tracked explicitly. The master equation for $c_{m,\ell,p}$ is:

$$
\boxed{
\begin{aligned}
\frac{dc_{m,\ell,p}}{dt} = {} & \\
& + \underbrace{A_{\rm sph}\,(m-1)^{1/3}\,\omega_v^{\rm eff}\,c_{1,0,0}\,c_{m-1,\ell,p}}_{\text{V capture: } V_{m-1,\ell,p}\to V_{m,\ell,p}} - \underbrace{A_{\rm sph}\,m^{1/3}\,\omega_v^{\rm eff}\,c_{1,0,0}\,c_{m,\ell,p}}_{\text{V capture: } V_{m,\ell,p}\to V_{m+1,\ell,p}} \\
& + \underbrace{\varepsilon_v(m{+}1,\ell,p)\,c_{m+1,\ell,p}}_{\text{V emission: } V_{m+1,\ell,p}\to V_{m,\ell,p}} - \underbrace{\varepsilon_v(m,\ell,p)\,c_{m,\ell,p}\,\delta_{m\ge 2}}_{\text{V emission: } V_{m,\ell,p}\to V_{m-1,\ell,p}} \\
& + \underbrace{A_{\rm sph}\,m^{1/3}\,\omega_h^{\rm eff}\,c_h\,c_{m,\ell-1,p}}_{\text{He capture: } V_{m,\ell-1,p}\to V_{m,\ell,p}} - \underbrace{A_{\rm sph}\,m^{1/3}\,\omega_h^{\rm eff}\,c_h\,c_{m,\ell,p}}_{\text{He capture: } V_{m,\ell,p}\to V_{m,\ell+1,p}} \\
& + \underbrace{\varepsilon_h(m,\ell{+}1,p)\,c_{m,\ell+1,p}}_{\text{He emission: } V_{m,\ell+1,p}\to V_{m,\ell,p}} - \underbrace{\varepsilon_h(m,\ell,p)\,c_{m,\ell,p}\,\delta_{\ell\ge 1}}_{\text{He emission: } V_{m,\ell,p}\to V_{m,\ell-1,p}} \\
& + \underbrace{A_{\rm sph}\,m^{1/3}\,\omega_H^{\rm eff}\,c_H\,c_{m,\ell,p-1}}_{\text{H capture: } V_{m,\ell,p-1}\to V_{m,\ell,p}} - \underbrace{A_{\rm sph}\,m^{1/3}\,\omega_H^{\rm eff}\,c_H\,c_{m,\ell,p}}_{\text{H capture: } V_{m,\ell,p}\to V_{m,\ell,p+1}} \\
& + \underbrace{\varepsilon_H(m,\ell,p{+}1)\,c_{m,\ell,p+1}}_{\text{H emission: } V_{m,\ell,p+1}\to V_{m,\ell,p}} - \underbrace{\varepsilon_H(m,\ell,p)\,c_{m,\ell,p}\,\delta_{p\ge 1}}_{\text{H emission: } V_{m,\ell,p}\to V_{m,\ell,p-1}} \\
& + \underbrace{b_0^H\,\dot\phi\,(p{+}1)\,c_{m,\ell,p+1}}_{\text{H re-solution in: } V_{m,\ell,p+1}\to V_{m,\ell,p}} - \underbrace{b_0^H\,\dot\phi\,p\,c_{m,\ell,p}}_{\text{H re-solution out}} \\
& + \underbrace{\Gamma_{\rm TM}(m-1,\ell,p)\,c_{m-1,\ell,p}}_{\text{trap mutation in: } V_{m-1,\ell,p}+\text{SIA}\to V_{m,\ell,p}} - \underbrace{\Gamma_{\rm TM}(m,\ell,p)\,c_{m,\ell,p}}_{\text{trap mutation out}} \\
& - \underbrace{\sum_{n=1}^{n^i_{\max}} \mathcal{K}^{(\rm cav)}_{n,m}\,c_n\,c_{m,\ell,p}}_{\text{SIA absorption shrinks cavity}}\,,
\end{aligned}
}
\tag{236}
$$

with the emission rates $\varepsilon_v(m,\ell,p)$, $\varepsilon_h(m,\ell,p)$, $\varepsilon_H(m,\ell,p)$ defined via detailed balance including the p-dependent binding energy (Eq. 228).

Boundary conditions.

For $m = 1$, $\ell = 0$, $p = 0$: the mono-vacancy concentration $c_{1,0,0}$ is the free vacancy; its equation includes all the cascade production and recombination terms of the existing system. For $p = p_{\max}(m)$: the H-capture term is zero (cluster saturated). For $\ell = \ell_{\max}(m)$: the He-capture term is zero.

Connection to the existing system.

The $(m, \ell, 0)$ equations at $p = 0$ are exactly those of the existing $c_{m,\ell}$ system (Eq. 155), with the additional source from H capture (when $p = 1$ cluster emits an H atom) and H re-solution (P14). This means the existing V–He code is a special case of the extended code with H disabled.

10.5 Adiabatic Elimination of Hydrogen for Large Clusters

For clusters larger than m_H^*, the H equilibration timescale $\tau_H \sim [\varepsilon_H(m, \ell, p)]^{-1} \sim 10^{-4}$–$10^{-1}$ s is much shorter than the vacancy growth timescale $\tau_v \sim [A_{\mathrm{sph}}\, m^{1/3}\, \omega_v^{\mathrm{eff}}\, c_{1,0,0}]^{-1} \sim 10^1$–$10^4$ s. The ratio

$$\frac{\tau_H}{\tau_v} \sim \frac{\omega_v^{\mathrm{eff}}}{\omega_H^{\mathrm{eff}}} \sim \exp\Bigl(-\frac{E_m^H - E_m^v}{k_B T}\Bigr) \approx \exp\Bigl(-\frac{0.06 - 0.60}{k_B T}\Bigr) \approx 10^{-4}\text{–}10^{-2} \tag{237}$$

is uniformly small for temperatures 200–600°C relevant to EUROFER97 operation. The formal adiabatic elimination condition is $\tau_H/\tau_v \to 0$, which replaces the H kinetics ODE with an algebraic quasi-static relation.

10.5.1 Quasi-Static H Loading

Define the total H loading and total cluster concentration at vacancy size m and He content ℓ:

$$\bar{c}_{m,\ell} \equiv \sum_{p=0}^{p_{\max}(m)} c_{m,\ell,p}, \qquad \bar{p}(m, \ell, t) \equiv \frac{\sum_{p=0}^{p_{\max}(m)} p\, c_{m,\ell,p}}{\bar{c}_{m,\ell}}. \tag{238}$$

Setting $d\bar{p}/dt = 0$ (quasi-static H balance on each (m, ℓ) cluster), the H capture and emission rates balance:

$$A_{\mathrm{sph}}\, m^{1/3}\, \omega_H^{\mathrm{eff}}\, c_H \bigl[1 - \bar{p}/p_{\max}(m)\bigr] = \varepsilon_H\bigl(m, \ell, \bar{p}\bigr) + b_0^H\, \dot{\phi}\, \bar{p}, \tag{239}$$

where the factor $[1-\bar{p}/p_{\max}]$ accounts for the reduced trapping rate as H sites approach saturation, and the right-hand side is the total H loss rate (thermal emission plus re-solution). This is a nonlinear algebraic equation in $\bar{p}$ that can be solved at each timestep by a simple bisection or Newton iteration.

Limiting cases.

Three analytically transparent limits arise:

(a) Unsaturated, low temperature ($\bar{p} \ll p_{\max}$, $\varepsilon_H \to 0$):

$$\bar{p}^{(\mathrm{a})}(m, \ell) = \frac{A_{\mathrm{sph}}\, m^{1/3}\, \omega_H^{\mathrm{eff}}\, c_H}{b_0^H\, \dot{\phi}}. \tag{240}$$

H loading is controlled by trapping vs. re-solution; proportional to c_H and to $m^{1/3}$ (larger clusters trap more).

(b) Thermodynamic equilibrium (no re-solution, $b_0^H = 0$):

$$\bar{p}^{(\mathrm{b})}(m,\ell) \quad \text{solves} \quad \frac{c_H}{\big[1-\bar{p}/p_{\max}(m)\big]} = \exp\!\big(-E_b^H(m,\ell,\bar{p})/k_BT\big), \tag{241}$$

the lattice-gas adsorption isotherm (Fermi–Dirac-like for distinguishable H sites). At low c_H this simplifies to the Henry's law form $\bar{p} \approx p_{\max}(m)\, c_H\, \exp(E_b^H/k_BT)$.

(c) Saturated ($\bar{p} \to p_{\max}$): $\bar{p} = p_{\max}(m)$, and further H capture is blocked. This sets an upper bound on the H contribution to bubble overpressure and the trap-mutation rate.

10.5.2 Reduced Master Equations for Large Clusters

For $m > m_H^*$ the H index is eliminated and the cluster is characterised by $(\bar{c}_{m,\ell},\, \bar{p}(m,\ell))$. Summing Eq. (236) over all p for fixed (m,ℓ):

$$\begin{aligned}
\frac{d\bar{c}_{m,\ell}}{dt} = {} & J_{m-1,\ell\to m,\ell} - J_{m,\ell\to m+1,\ell} \\
& + A_{\mathrm{sph}}\, m^{1/3}\, \omega_h^{\mathrm{eff}}\, c_h\, \bar{c}_{m,\ell-1} - A_{\mathrm{sph}}\, m^{1/3}\, \omega_h^{\mathrm{eff}}\, c_h\, \bar{c}_{m,\ell} \\
& + \varepsilon_h(m,\ell{+}1,\bar{p})\, \bar{c}_{m,\ell+1} - \varepsilon_h(m,\ell,\bar{p})\, \bar{c}_{m,\ell}\, \delta_{\ell\geq 1} \\
& + \Gamma_{\mathrm{TM}}(m{-}1,\ell,\bar{p})\, \bar{c}_{m-1,\ell} - \Gamma_{\mathrm{TM}}(m,\ell,\bar{p})\, \bar{c}_{m,\ell} \\
& - \sum_{n=1}^{n^i_{\max}} \mathcal{K}^{(\mathrm{cav})}_{n,m}\, c_n\, \bar{c}_{m,\ell},
\end{aligned} \tag{242}$$

where $\bar{p} = \bar{p}(m,\ell)$ is the quasi-static H loading from Eq. (239) and the H-related terms cancel exactly (by definition of the quasi-static condition). The vacancy flux is:

$$J_{m,\ell\to m+1,\ell} = A_{\mathrm{sph}}\, m^{1/3}\, \omega_v^{\mathrm{eff}}\, c_{1,0,0}\, \bar{c}_{m,\ell} - \varepsilon_v(m{+}1,\ell,\bar{p}(m{+}1,\ell))\, \bar{c}_{m+1,\ell}. \tag{243}$$

Effect of H on emission rates.

The adiabatic elimination enters the dynamics through the H-corrected binding energies in ε_v and ε_h. The vacancy emission rate from a H-loaded cavity is modified via:

$$E_b^v(m,\ell,\bar{p}) = E_b^v(m,\ell,0) - \frac{\bar{p}\, k_BT}{m} - \delta E_{\mathrm{surf}}(\bar{p}/p_{\max}), \tag{244}$$

where the second term is the ideal-gas pressure contribution from H (analogous to the He correction in Eq. 218), and $\delta E_{\mathrm{surf}} > 0$ is a surface-energy reduction due to H adsorption (≈0.05–0.15 eV per $\bar{p}/p_{\max}$ from MD/DFT data). This reduction in E_b^v *increases* the vacancy emission rate and thus promotes void growth — the primary physical mechanism by which H enhances swelling in concert with He.

Similarly, He emission is enhanced by the H-induced surface energy reduction:

$$E_B^{\mathrm{He}}(m,\ell,\bar{p}) = E_B^{\mathrm{He}}(m,\ell,0) - \delta E_{\mathrm{He-H}}(\bar{p}/p_{\max}), \tag{245}$$

with $\delta E_{\mathrm{He-H}} \approx 0.05$–$0.2\,\mathrm{eV}$.

10.6 He-Content Reduction Combined with H Adiabatic Elimination

In the fusion-relevant case (high He generation), the He reduction of Section 9.9 can be combined with the H adiabatic elimination. Summing Eq. (242) over all ℓ and applying the mean-field He approximation $\bar{\ell}(m)$ gives:

$$\frac{d\bar{c}_m}{dt} = J_{m-1\to m} - J_{m\to m+1} - \sum_{n=1}^{n^i_{\max}} \mathcal{K}^{(\mathrm{cav})}_{n,m}\, c_n\, \bar{c}_m + \Gamma_{\mathrm{TM}}\big(m-1, \bar{\ell}, \bar{p}\big)\, \bar{c}_{m-1} - \Gamma_{\mathrm{TM}}\big(m, \bar{\ell}, \bar{p}\big)\, \bar{c}_m, \tag{246}$$

with the vacancy flux now depending on *both* the mean He loading $\bar{\ell}(m)$ and the quasi-static H loading $\bar{p}(m)$:

$$J_{m\to m+1} = A_{\mathrm{sph}}\, m^{1/3}\, \omega_v^{\mathrm{eff}}\, c_{1,0,0}\, \bar{c}_m - \varepsilon_v\big(m+1,\, \bar{\ell}(m+1),\, \bar{p}(m+1)\big)\, \bar{c}_{m+1}. \tag{247}$$

The He loading $\bar{\ell}(m)$ satisfies the same ODE as in Case 1 (Eq. 217), with He emission evaluated at the H-corrected binding energy $E_B^{\mathrm{He}}(m, \bar{\ell}, \bar{p})$ (Eq. 245). The quasi-static H loading $\bar{p}(m)$ is the ℓ-averaged version of Eq. (239):

$$A_{\mathrm{sph}}\, m^{1/3}\, \omega_H^{\mathrm{eff}}\, c_H \big[1 - \bar{p}/p_{\max}(m)\big] = \varepsilon_H\big(m, \bar{\ell}(m), \bar{p}\big) + b_0^H\, \dot{\phi}\, \bar{p}. \tag{248}$$

Equation count.

The combined system of the extended model with both He and H reductions comprises:

$$n_{\mathrm{eq}}^{\mathrm{ext}} = \underbrace{N}_{\text{SIA clusters}} + \underbrace{2M}_{\bar{c}_m,\, \bar{\ell}(m)} + \underbrace{M}_{\bar{p}(m)\ \text{(algebraic)}} + \underbrace{1}_{c_h} + \underbrace{1}_{c_H} = N + 2M + 2 \ \ \text{ODEs}, \tag{249}$$

plus M algebraic equations for $\bar{p}(m)$ solved at each timestep. The increase relative to the He-only system $(N + 2M + 1)$ is a single additional ODE for free H and M algebraic solves — a modest overhead that enables the physically important H–He–V synergy to be captured.

Comparison with He-only system.

Table 35 summarises the hierarchy of models.

10.7 Summary of New Kinetic Parameters

Table 36 collects the kinetic parameters needed for the hydrogen extension. Values appropriate for bcc α-Fe and RAFM steels are given as starting points; solute corrections for Cr and W content in EUROFER97 should be applied via Eq. (129).

Table 35 Hierarchy of CD models from V–He only to the fully extended V–He–H system. "Algebraic" denotes quantities solved by nonlinear inversion at each timestep rather than by ODE integration. N = max SIA cluster size, M = max vacancy cluster size.

Model	**ODEs**	**Algebraic**	**H treatment**
V–SIA only	$N + M + 1$	0	None
V–He–SIA, Case 2 (fission)	$N + M + 2$	0	None
V–He–SIA, Case 1 (fusion)	$N + 2M + 1$	0	None
Extended, explicit H ($m \leq m_H^*$)	$N + 2M + 1 + M_H^* \cdot P_{\max}$	0	Full
Extended, adiabatic H (all m)	$N + 2M + 2$	M	Quasi-static $\bar{p}$
Extended, adiabatic H + Case 1	$N + 2M + 2$	M	Quasi-static, ℓ-averaged

Table 36 Kinetic parameters for hydrogen in α-Fe for the extended CD model. DFT values from Hashimoto et al. (2014), Tateyama & Ohno (2003), and Hayward & Deo (2011). "bub" denotes bubble surface site.

Parameter	**Value**	**Notes**
H migration energy E_m^H	0.04–0.09 eV	Interstitial (tet site), bcc Fe
H diffusion prefactor ν_H	$\sim 6 \times 10^{12}\,\mathrm{s}^{-1}$	Debye frequency
H binding to V_1, $E_b^H(1, 0, 1)$	0.51 eV	DFT
H binding to V_2, $E_b^H(2, 0, 1)$	0.62 eV	DFT (larger trap)
H saturation per vacancy, $p_{\max}/m$	~ 6 ($m = 1$), $\alpha_H m^{2/3}$	Up to 6 H/V
H surface energy factor α_H	1.0–2.0	Eq. (224)
H binding at bubble surface, $E_b^{H\text{-bub}}$	0.4–0.8 eV	He-enhanced; core-shell structure
He enhancement δE_{He}	0.1–0.3 eV	For $\ell/m \sim 0.5$–1.5
H surface energy reduction δE_{surf}	0.05–0.15 eV	Per unit $\bar{p}/p_{\max}$; promotes swelling
H re-solution cross-section b_0^H	10^{-2}–10^{-1} dpa^{-1}/H	Lower: deep trap; upper: surface H
m_H^* (explicit H cutoff)	3–5	User choice
ℓ_H^* (explicit He cutoff for H)	2–4	User choice

11 Proposed Hybrid Deterministic–Stochastic Cluster Dynamics for Multispecies (H, He, V, I)

The deterministic cluster dynamics (CD) framework of Sections 7–10 tracks every cluster size as a continuous concentration variable, yielding a coupled ODE system whose size scales as $O(N \cdot M \cdot L_{\mathrm{He}} \cdot P_{\max})$ for the four-species (I, V, He, H) state space. Even after the He mean-field and H adiabatic reductions, two fundamental bottlenecks remain:

1. **Large-cluster stiffness.** Immobile large clusters (voids, bubbles, loops) evolve on timescales $\tau_v \sim 10$–10^5 s, while mobile monomers equilibrate in $\tau_i \sim 10^{-6}$–10^{-3} s. The implicit ODE solvers required for stiffness management impose a dense Jacobian of size n_{eq}^2, which becomes prohibitive for $n_{\text{eq}} \gtrsim 10^4$.
2. **Multispecies combinatorial explosion.** Adding H as a third gas species alongside He restores the three-dimensional (m, ℓ, p) cluster space for small clusters: the deterministic framework is tractable only with the adiabatic H elimination of Section 10.5, which requires $m \geq m_H^*$. At small cluster sizes $(m \leq m_H^*)$ all three indices must be resolved explicitly, and the deterministic cost becomes comparable to what triggered the stochastic approach in the first place.

The hybrid strategy proposed here addresses both bottlenecks simultaneously. It couples the deterministic multispecies equations (Sections 7–10) to a stochastic cluster dynamics engine for large clusters, using the Lie–Trotter operator-splitting framework of Terrier et al. Terrier et al (2017) and the logarithmic-complexity stochastic simulation algorithm (SSA) of Vasav et al. Vasav et al (2026).

11.1 Physical and Mathematical Motivation

The key observation enabling the splitting is that cluster evolution naturally separates into two regimes with different physical and mathematical characters:

Small clusters $(n \leq N^i_{\text{front}},\ m \leq M_{\text{front}})$**.**

Small clusters are either mobile or rapidly exchanging with the mobile monomer pool. Their dynamics are *stiff* (fast frequencies) but involve a *small number of distinct species* ($\lesssim 10^2$–10^3), so an implicit ODE solver with sparse Jacobian is efficient. The full multispecies physics — including explicit H tracking via the (m, ℓ, p) grid for $m \leq m_H^*$ and the P9–P14 reactions of Section 10.2 — is retained in this deterministic block.

Large clusters $(n > N^i_{\text{front}},\ m > M_{\text{front}})$**.**

Large clusters are immobile or very slowly evolving. Their jump frequency $\nu(m)^{-1} = [\mathcal{K}_{\alpha,m}\, c_{\alpha,1}]^{-1}$ is low, so the stochastic timestep is manageable. Their *number of distinct species* is large ($\sim 10^3$–10^5), making a per-species ODE intractable but making a particle-based stochastic description natural. The H degree of freedom is treated via the adiabatic loading $\bar{p}(m, \ell)$ of Section 10.5: H kinetics are faster than the slow cluster evolution, so quasi-static H is exact in this regime.

The splitting error is controlled by the macro-timestep δt; the statistical error by the number of stochastic particles N_{part}. These two parameters are independently tunable, giving direct control over the accuracy–cost trade-off.

11.2 State Vector Decomposition and Frontier Sizes

The full state of the four-species system is decomposed at each macro-timestep as:

$$\mathbf{X}(t) = \underbrace{\{c_{1,0,0,0},\ c_h,\ c_H,\ c_n^{\text{small}},\ c_{m,\ell,p}^{\text{small}}\}}_{\text{deterministic block } \mathcal{D}} \cup \underbrace{\{c_n^{\text{large}},\ c_{m,\ell}^{\text{large}}\}}_{\text{stochastic block } \mathcal{S}}, \tag{250}$$

where the superscripts indicate:

$$c_n^{\text{small}} : 1 \le n \le N_{\text{front}}^i, \tag{251}$$

$$c_{m,\ell,p}^{\text{small}} : 1 \le m \le M_{\text{front}},\ 0 \le \ell \le \ell_{\max}(m),\ 0 \le p \le p_{\max}(m), \tag{252}$$

$$c_n^{\text{large}} : n > N_{\text{front}}^i, \tag{253}$$

$$c_{m,\ell}^{\text{large}} : m > M_{\text{front}},\ H \text{ adiabatically eliminated } (\bar{p}(m,\ell) \text{ algebraic}). \tag{254}$$

The four mobile monomer concentrations $\{c_{1,0,0,0},\ c_h,\ c_H,\ c_n^{(n=1)}\}$ are always in the deterministic block because they drive all cross-coupling between the blocks.

Choice of frontier sizes.
The frontiers N_{front}^i and M_{front} are set by balancing the stiffness cost in the deterministic block against the noise cost in the stochastic block:

$$N_{\text{front}}^i \approx n_{\max}^i, \qquad M_{\text{front}} \approx \max(m_H^*,\ m_{\max}^v) \approx 5\text{–}20. \tag{255}$$

For the EUROFER97 system, $N_{\text{front}}^i \approx 100$–$200$ and $M_{\text{front}} \approx 10$–$20$ are practical defaults. Below these sizes, all cluster species are either mobile or have non-trivial He and H loading dynamics; above them, the clusters are immobile and their stochastic treatment is both appropriate and efficient.

11.3 Lie–Trotter Operator Splitting

Over one macro-timestep $[t,\ t+\delta t]$, the full dynamics are split into two alternating sub-problems by a first-order Lie–Trotter (Godunov) scheme:

11.3.1 Sub-problem D: Monomer Dynamics (Deterministic)

With all large-cluster concentrations c_n^{large} and $c_{m,\ell}^{\text{large}}$ held *fixed*, evolve the deterministic block $\mathcal{D}$ over $[t,\ t+\delta t]$. The ODEs to be solved are the full four-species master equations (Eqs. 152, 155, 233, 236) for the small-cluster populations, with large clusters entering only as *sink terms* evaluated at their frozen concentrations.

Define the effective sink strengths of the frozen large clusters:

$$\mathcal{S}_i^{\text{large}}(t) = \sum_{n=N_{\text{front}}^i+1}^{N_{\max}} \mathcal{K}_{n,1}^{ii}\, c_n^{\text{large}}(t) + \sum_{m=M_{\text{front}}+1}^{M_{\max}} \mathcal{K}_{1,m}^{(\text{cav})}\, \bar{c}_m^{\text{large}}(t), \tag{256}$$

$$\mathcal{S}_v^{\mathrm{large}}(t) = \sum_{n=N^i_{\mathrm{front}}+1}^{N_{\max}} \mathcal{K}^{\mathrm{loop}}_{v,n}\, c_n^{\mathrm{large}}(t) + \sum_{m=M_{\mathrm{front}}+1}^{M_{\max}} A_{\mathrm{sph}}\, m^{1/3}\, \omega_v^{\mathrm{eff}}\, \bar{c}_m^{\mathrm{large}}(t), \tag{257}$$

$$\mathcal{S}_h^{\mathrm{large}}(t) = \sum_{m=M_{\mathrm{front}}+1}^{M_{\max}} A_{\mathrm{sph}}\, m^{1/3}\, \omega_h^{\mathrm{eff}}\, \bar{c}_m^{\mathrm{large}}(t), \tag{258}$$

$$\mathcal{S}_H^{\mathrm{large}}(t) = \sum_{m=M_{\mathrm{front}}+1}^{M_{\max}} A_{\mathrm{sph}}\, m^{1/3}\, \omega_H^{\mathrm{eff}}\, \tilde{c}_m^{(p),\mathrm{large}}(t), \tag{259}$$

where $\tilde{c}_m^{(p),\mathrm{large}}$ is the unsaturated large-cluster concentration (Eq. 233). These are scalar coefficients added to the existing $\mathcal{D}_\alpha$ fixed-sink terms in the small-cluster ODEs.

The monomer sub-problem is then a set of $n_{\mathrm{eq}}^{\mathrm{det}}$ coupled stiff ODEs:

$$\frac{d\mathbf{X}_{\mathcal{D}}}{dt} = \mathbf{F}_{\mathcal{D}}\big(\mathbf{X}_{\mathcal{D}},\, \mathcal{S}_\alpha^{\mathrm{large}}(t)\big), \qquad n_{\mathrm{eq}}^{\mathrm{det}} = N^i_{\mathrm{front}} + M_{\mathrm{front}}\big(1+\alpha_{\mathrm{He}}\, M_{\mathrm{front}}^{2/3}\big)\big(1+\alpha_H\, M_{\mathrm{front}}^{2/3}\big)+3, \tag{260}$$

solved by LSODA or CVODE with adaptive BDF. For the frontier choices above ($N^i_{\mathrm{front}} \approx 200$, $M_{\mathrm{front}} \approx 15$), $n_{\mathrm{eq}}^{\mathrm{det}} \lesssim 1000$, which is very fast per macro-timestep.

11.3.2 Sub-problem S: Large-Cluster Dynamics (Stochastic)

With the four monomer concentrations $\{c_{1,0,0,0},\, c_h,\, c_H,\, c_1\}$ held *fixed* at the values $\mathbf{X}_{\mathcal{D}}(t+\delta t)$ from sub-problem D, evolve the large-cluster populations via the stochastic simulation algorithm over $[t,\, t+\delta t]$.

Because the monomer concentrations are fixed, the large-cluster sub-problem is *linear* in the cluster populations: the transition rates of each large cluster depend only on cluster size and the fixed monomers, not on other large clusters. This linearity is the key structural property enabling the efficient stochastic treatment.

11.4 Stochastic Engine: Four-Species Reactions for Large Clusters

The large-cluster populations are represented as a discrete integer-valued vector

$$\mathbf{X}^{\mathrm{large}}(t) = \big\{X_n(t),\ X_{m,\ell}(t)\big\}_{n>N^i_{\mathrm{front}},\, m>M_{\mathrm{front}}}, \tag{261}$$

where $X_n(t)$ and $X_{m,\ell}(t)$ are non-negative integers (particle counts in volume Ω). The deterministic concentrations are recovered as $c_n = X_n/\Omega$ and $c_{m,\ell} = X_{m,\ell}/\Omega$.

11.4.1 Reaction Classes and Propensities

With the monomers fixed at $c_{1,0,0,0} \equiv c_v^0,\ c_h^0,\ c_H^0,\ c_1^0$, the following five reaction classes are active for large clusters. All rate constants are the Ω-scaled Smoluchowski expressions of Section 6.

Class 0 — Unary reactions (emission, mutation, re-solution).
For each large SIA cluster of size n:

$$\text{SIA emission: } I_n \to I_{n-1} + I_1, \quad a_n^{(0)} = \Omega\, \varepsilon_i(n)\, X_n, \tag{262}$$

and for each large vacancy–He cluster of size (m, ℓ):

$$\text{V emission:} \quad V_{m,\ell} \to V_{m-1,\ell} + V_1, \quad a_{m,\ell}^{(v)} = \Omega\, \varepsilon_v(m, \ell, \bar{p})\, X_{m,\ell}, \tag{263}$$

$$\text{He emission:} \quad V_{m,\ell} \to V_{m,\ell-1} + \text{He}, \quad a_{m,\ell}^{(h)} = \Omega\, \varepsilon_h(m, \ell, \bar{p})\, X_{m,\ell}, \tag{264}$$

$$\text{Trap mutation:} \quad V_{m,\ell} \to V_{m+1,\ell} + I_1, \quad a_{m,\ell}^{(\text{TM})} = \Omega\, \Gamma_{\text{TM}}(m, \ell, \bar{p})\, X_{m,\ell}, \tag{265}$$

$$\text{He re-solution:} \quad V_{m,\ell} \to V_{m,\ell-1} + \text{He}, \quad a_{m,\ell}^{(\text{res})} = \Omega\, b_0\, \dot{\phi}\, \ell\, X_{m,\ell}. \tag{266}$$

All Class 0 propensities are linear in X.

Class 1 — SIA monomer absorption.
The fixed monomer I_1 (concentration c_1^0, integer count $M_1^0 = \Omega c_1^0$) is absorbed by a large SIA loop:

$$I_n + I_1^0 \to I_{n+1}, \quad a_n^{(i)} = \mathcal{K}_{i,n}^{\text{loop}}\, X_n\, M_1^0, \quad \mathcal{K}_{i,n}^{\text{loop}} = \frac{A_{\text{loop}}\, n^{1/2}\, Z_i^{\text{loop}}\, \omega_i^{\text{eff}}}{\Omega}. \tag{267}$$

Similarly, vacancy monomer absorption by a loop (SIA removal from the loop is equivalent to loop shrinkage by one vacancy):

$$I_n + V_1^0 \to I_{n-1}, \quad a_n^{(v\text{-loop})} = \mathcal{K}_{v,n}^{\text{loop}}\, X_n\, M_v^0, \quad M_v^0 = \Omega\, c_{1,0,0,0}. \tag{268}$$

Class 2 — Vacancy and gas monomer absorption by cavities.
For a large cavity (m, ℓ):

$$V_{m,\ell} + V_1^0 \to V_{m+1,\ell}, \quad a_{m,\ell}^{(v,+)} = \frac{A_{\text{sph}}\, m^{1/3}\, \omega_v^{\text{eff}}}{\Omega}\, X_{m,\ell}\, M_v^0, \tag{269}$$

$$V_{m,\ell} + I_n^0 \to V_{m-1,\ell}\ (n = 1), \quad a_{m,\ell}^{(i,-)} = \frac{A_{\text{sph}}\, m^{1/3}\, \omega_i^{\text{eff}}}{\Omega}\, X_{m,\ell}\, M_1^0, \tag{270}$$

$$V_{m,\ell} + \text{He}^0 \to V_{m,\ell+1}, \quad a_{m,\ell}^{(h,+)} = \frac{A_{\text{sph}}\, m^{1/3}\, \omega_h^{\text{eff}}}{\Omega}\, X_{m,\ell}^{(\ell<\ell_{\max})}\, M_h^0, \tag{271}$$

where $M_h^0 = \Omega c_h^0$ and $X_{m,\ell}^{(\ell<\ell_{\max})} = X_{m,\ell}\, \delta_{\ell<\ell_{\max}(m)}$ enforces the He saturation constraint.

Class 3 — H monomer absorption/desorption from large cavities.
Because H is adiabatically eliminated for large clusters, H absorption and emission do not appear as explicit stochastic reactions. Their net effect enters exclusively through

the H-corrected binding energies $E_b^v(m,\ell,\bar{p})$ and $E_B^{\mathrm{He}}(m,\ell,\bar{p})$ (Eqs. 244–245) in the Class 0 and Class 2 propensities. The quasi-static H loading $\bar{p}(m,\ell)$ is solved from Eq. (248) for each distinct (m,ℓ) species present in the population at the start of each macro-timestep.

Class 4 — Large-cluster coalescence.
Two mobile large SIA clusters can coalesce:

$$I_n + I_{n'} \to I_{n+n'}, \quad a^{(ii)}_{n,n'} = \mathcal{K}^{ii}_{n,n'}\, X_n\, X_{n'}\,(1-\tfrac{1}{2}\delta_{n,n'}) \quad n,n' > N^i_{\mathrm{front}}, \tag{272}$$

with $\mathcal{K}^{ii}_{n,n'}$ from Eq. (144). This reaction has propensity bilinear in X, i.e. it is a true binary stochastic reaction. In the EUROFER97 system, large SIA clusters ($n > N^i_{\mathrm{front}} \approx 100$) are sessile, so Class 4 is active only for $n \le n^i_{\max}$ and can often be neglected; the default recommendation is to *exclude large-loop coalescence* unless the cluster density is high.

11.4.2 Propensity Matrix and Low-Rank Decomposition

The total propensity for Class 2 (binary monomer-absorption) has the Smoluchowski form:

$$a^{(\alpha,+)}_{m,\ell} = \underbrace{\frac{A_{\mathrm{sph}}\,\omega^{\mathrm{eff}}_\alpha}{\Omega}}_{\mathcal{N}^\alpha}\underbrace{m^{1/3}\,X_{m,\ell}}_{u^\alpha_{m,\ell}}\underbrace{M^0_\alpha}_{v^\alpha}, \tag{273}$$

which is exactly a **rank-one** ($Q = 1$) outer product $\mathbf{A}^\alpha = \mathbf{u}^\alpha \otimes v^\alpha$ in the notation of Vasav et al. Vasav et al (2026). The single priority queue per species type is therefore sufficient: $Q = 1$ for each of the three absorption channels (v, I_1, He).

For Class 1 (SIA-loop reactions), the propensity $a^{(i)}_n = \mathcal{K}^{\mathrm{loop}}_{i,n}\, X_n\, M^0_1$ is also rank-one:

$$a^{(i)}_n = \underbrace{\frac{A_{\mathrm{loop}}\, Z^{\mathrm{loop}}_i\, \omega^{\mathrm{eff}}_i}{\Omega}}_{\mathcal{N}^{\mathrm{loop}}}\underbrace{n^{1/2}\,X_n}_{u^{\mathrm{loop}}_n}\underbrace{M^0_1}_{v^{\mathrm{loop}}}. \tag{274}$$

The full set of Class 2 + Class 1 propensities for the four-species system therefore requires exactly $\mathbf{2Q = 8}$ priority queues: Each binary queue uses two priority heaps

Queue	Reaction	Rank-1 factors
$\mathcal{P}_1$	$V_{m,\ell} + V^0_1 \to V_{m+1,\ell}$	$u = m^{1/3} X_{m,\ell},\ v = M^0_v$
$\mathcal{P}_2$	$V_{m,\ell} + I^0_1 \to V_{m-1,\ell}$	$u = m^{1/3} X_{m,\ell},\ v = M^0_1$
$\mathcal{P}_3$	$V_{m,\ell} + \mathrm{He}^0 \to V_{m,\ell+1}$	$u = m^{1/3} X_{m,\ell},\ v = M^0_h$
$\mathcal{P}_4$	$I_n + I^0_1 \to I_{n+1}$	$u = n^{1/2} X_n,\ v = M^0_1$
$\mathcal{P}_5$	$I_n + V^0_1 \to I_{n-1}$	$u = n^{1/2} X_n,\ v = M^0_v$
$\mathcal{P}_6$	Unary SIA emission ($I_n \to I_{n-1} + I_1$)	—
$\mathcal{P}_7$	Unary V/He emission, trap mutation, re-solution	—
$\mathcal{P}_8$	Optional: large-loop coalescence ($I_n + I_{n'}$)	rank-1 via $n^{1/3}$

(one for each factor), giving a worst-case algorithmic complexity of $O(\log N_{\text{large}})$ per stochastic event, where N_{large} is the number of distinct large-cluster species.

11.4.3 Log-Time SSA Algorithm for Large Clusters

The stochastic large-cluster sub-problem within each macro-timestep $[t,\, t+\delta t]$ is solved by the LR-NR (Low-Rank Next-Reaction) algorithm of Vasav et al. Vasav et al (2026), adapted for the four-species propensity structure above. The complete procedure is:

Initialisation (at $t = 0$ or after a frontier update).

1. Construct the initial population vector $\mathbf{X}^{\text{large}}(t_0)$ by sampling from the deterministic large-cluster concentrations: $X_n(t_0) \sim \text{Poisson}(\Omega\, c_n^{\text{det}}(t_0))$, with $\Omega = V_{\text{sim}}/\Omega_{\text{at}}$.
2. For each reaction class $q \in \{0, \ldots, 8\}$, compute the initial factorised propensity vectors $\mathbf{u}^q$ and v^q and initialise priority queues $\mathcal{P}_q$ as binary min-heaps with internal times $\tilde{\tau}_{\nu,q} \sim \text{Exp}(\tilde{a}_{\nu,q})$.
3. Initialise the quasi-static H loadings $\bar{p}(m, \ell)$ by solving Eq. (248) for each distinct (m, ℓ) pair present.

Event loop within $[t,\, t + \delta t]$.

1. Determine the physical time increment to the next reaction:

$$\delta t_{\text{event}} = \min_{q \in \{0,\ldots,Q\}} \left\{ \delta\tilde{\tau}_0,\; \frac{\delta\tilde{\tau}_q}{\hat{a}_{q+Q}(t_n)} \right\}, \tag{275}$$

   where $\delta\tilde{\tau}_q = \min_\nu \tilde{\tau}_{\nu,q} - \tau_q$ (top of the binary heap $\mathcal{P}_q$) and $\hat{a}_{q+Q} = \sum_\nu \tilde{a}_{\nu,q+Q}$ is the total factorised propensity of the complementary class. *Terminate the event loop if $t + \delta t_{\text{event}} > t + \delta t$.*
2. Identify the reacting class s, reactants ν_s and μ_s from the tops of $\mathcal{P}_s$ and $\mathcal{P}_{s+Q}$.
3. Apply rejection sampling: accept the reaction with probability

$$P_{\text{accept}}(\nu_s + \mu_s \to \nu') = \frac{a^{\text{actual}}_{\nu_s,\mu_s}}{\tilde{a}^{\text{upper}}_{\nu_s,\mu_s}}, \tag{276}$$

   where the actual rate uses the exact Smoluchowski formula and the upper bound is the rank-one majorant.
4. If accepted: update populations X_{ν_s}, X_{μ_s}, $X_{\nu'}$; update propensity sums and heap internal times for all affected queues. If rejected: advance the heap clock without changing populations (ghost reaction).
5. Update the quasi-static H loading $\bar{p}(m', \ell')$ for any new or modified species (m', ℓ').
6. Advance physical time: $t_n \leftarrow t_n + \delta t_{\text{event}}$ and internal clock times τ_k.

Completion of the macro-timestep.

At $t + \delta t$, convert stochastic particle counts back to deterministic concentrations:

$$c_n^{\text{large}}(t+\delta t) = X_n(t+\delta t)/\Omega, \qquad c_{m,\ell}^{\text{large}}(t+\delta t) = X_{m,\ell}(t+\delta t)/\Omega. \tag{277}$$

Compute the change in total monomer consumption during the stochastic step (number of V_1, I_1, and He monomers absorbed or emitted) and add these as correction fluxes to the deterministic monomer equations for the next sub-problem D step (see Section 11.5 below).

11.5 Monomer Feedback and Mass Conservation

The splitting introduces an inconsistency: the monomers are held fixed during sub-problem S, yet the large clusters absorb or emit monomers. The accumulated monomer deficit is reconciled by a post-step feedback correction.

Define the cumulative monomer changes during sub-problem S over $[t,\, t+\delta t]$:

$$\Delta M_v^S = (\text{V atoms absorbed}) - (\text{V atoms emitted}), \tag{278}$$

$$\Delta M_i^S = (\text{SIAs absorbed}) - (\text{SIAs emitted}), \tag{279}$$

$$\Delta M_h^S = (\text{He atoms absorbed}) - (\text{He atoms emitted}), \tag{280}$$

all obtained directly from the accepted stochastic events. These are converted to concentration corrections:

$$c_\alpha \leftarrow c_\alpha - \frac{\Delta M_\alpha^S}{\Omega}, \qquad \alpha \in \{v, i, h, H\}, \tag{281}$$

applied to the monomer concentrations at the start of the next sub-problem D step. This preserves the total atom balance (vacancy + SIA + He + H) to machine precision across the operator splitting.

The Frenkel pair and He conservation laws (Eqs. 160–314) are monitored at every macro-timestep using the hybrid equivalents:

$$\delta_{\text{FP}}^{\text{hyb}} = \left| S_I^{\text{det}} + S_I^{\text{stoch}} + \Delta J^d - S_V^{\text{det}} - S_V^{\text{stoch}} \right|, \tag{282}$$

$$\delta_{\text{He}}^{\text{hyb}} = \left| c_h + \ell_{\text{tot}}^{\text{det}} + \ell_{\text{tot}}^{\text{stoch}} - \int_0^t (G_{\text{He}} - \mathcal{D}_h c_h)\, dt' \right|, \tag{283}$$

where the "stoch" superscript denotes the contribution from large-cluster particle counts divided by Ω. An implementation should assert a hard stop if either diagnostic exceeds 10^{-3} per dpa.

11.6 Error Analysis and Convergence

The total error in any observable $\mathcal{O}$ (e.g., total swelling $S(t)$, mean void radius) satisfies the bound

$$\eta^2(t) \approx \underbrace{C_1(t)\,\delta t}_{\text{splitting error}} + \underbrace{\frac{C_2(t)}{N_{\text{part}}}}_{\text{statistical error}}, \tag{284}$$

derived by the same argument as Terrier et al. Terrier et al (2017). Here C_1 and C_2 are problem-dependent constants; the two contributions are independently controlled by δt and $N_{\text{part}} = \Omega \sum_{n,m,\ell}(c_n^{\text{large}} + c_{m,\ell}^{\text{large}})$ respectively.

Splitting error.

The Lie–Trotter splitting introduces a first-order error in δt through the non-commutativity of sub-problems D and S. The dominant term arises from the coupling between the monomer equations and the large-cluster sink/source terms:

$$C_1(t) \sim \frac{\partial \mathbf{F}_{\mathcal{D}}}{\partial \mathcal{S}^{\text{large}}} \cdot \frac{\partial \dot{\mathbf{X}}^{\text{large}}}{\partial \mathbf{X}_{\mathcal{D}}}, \tag{285}$$

which is proportional to the product of the large-cluster sink strength and the large-cluster growth rate. C_1 is small when the large-cluster concentrations are quasi-stationary (slow evolution), which is the typical regime for swelling calculations. A Strang second-order splitting (D/2 $\to$ S $\to$ D/2) reduces the splitting error to $O(\delta t^2)$ at the cost of two deterministic ODE solves per macro-timestep.

Statistical error.

The statistical error C_2/N_{part} arises because N_{part} is finite. Large-cluster concentrations $c_{m,\ell}^{\text{large}}$ and their contributions to S_V are estimated as sample means with standard deviation $\sigma \propto N_{\text{part}}^{-1/2}$. Practical values of $N_{\text{part}} = 10^3$–10^5 particles give statistical errors of $\sim$1%–0.1% in swelling-integral observables, which is acceptable for the EUROFER97 application.

Convergence criterion.

The recommended convergence test is to halve δt and double N_{part} and verify that the observable changes by less than a user-specified tolerance (e.g., $\leq 2\%$ for swelling). If the dominant contribution to η^2 comes from splitting error (sensitive to δt), decrease δt. If it comes from statistical error (sensitive to N_{part}), increase N_{part} or increase Ω.

11.7 Complete Algorithmic Scheme

Algorithm 11.7 gives the complete hybrid deterministic–stochastic update cycle for one macro-timestep.

Algorithm: Hybrid D/S CD for H–He–V–I (one macro-timestep)

Given: state $\big(\mathbf{X}_{\mathcal{D}}(t),\, \mathbf{X}^{\text{large}}(t)\big)$, macro-step δt, volume Ω, N_{part}

Pre-step: compute quasi-static H loadings

For each distinct (m,ℓ) in $\mathbf{X}^{\text{large}}$:

solve Eq. (248) for $\bar{p}(m,\ell)$

Update $E_b^v(m,\ell,\bar{p})$, $E_B^{\text{He}}(m,\ell,\bar{p})$ (Eqs. 244–245)

Step 1: Sub-problem D (deterministic small-cluster ODE)

Compute large-cluster sink strengths $\mathcal{S}_\alpha^{\text{large}}(t)$ (Eqs. 256–259)

Add $\mathcal{S}_\alpha^{\text{large}}$ to fixed-sink terms $\mathcal{D}_\alpha$ in the master equations

Integrate $d\mathbf{X}_{\mathcal{D}}/dt = \mathbf{F}_{\mathcal{D}}$ from t to $t+\delta t$ via LSODA/CVODE

Extract updated monomer concentrations: $c_v^0, c_1^0, c_h^0, c_H^0 \leftarrow \mathbf{X}_{\mathcal{D}}(t+\delta t)$

Step 2: Sub-problem S (stochastic large-cluster SSA)

Fix monomers at $c_v^0, c_1^0, c_h^0, c_H^0$

Initialise/update LR-NR priority queues $\{\mathcal{P}_q\}$ (Eq. 11.4.2)

while $t_n < t+\delta t$:

Compute δt_{event} (Eq. 275)

if $t_n + \delta t_{\text{event}} > t + \delta t$: break

Select class s, reactants ν_s, μ_s; apply rejection (Eq. 276)

If accepted: update X_{ν_s}, X_{μ_s}, $X_{\nu'}$; update heap and propensity sums

Update $\bar{p}(m',\ell')$ for any new/modified species

$t_n \leftarrow t_n + \delta t_{\text{event}}$

Step 3: Feedback correction (mass balance)

Compute ΔM_v^S, ΔM_i^S, ΔM_h^S from accepted stochastic events

Update monomers: $c_\alpha \leftarrow c_\alpha - \Delta M_\alpha^S/\Omega$ (Eq. 281)

Step 4: Convert stochastic counts to concentrations

$c_n^{\text{large}}(t+\delta t) = X_n/\Omega$; $\quad c_{m,\ell}^{\text{large}}(t+\delta t) = X_{m,\ell}/\Omega$

Step 5: Conservation diagnostics

Evaluate $\delta_{\text{FP}}^{\text{hyb}}$ (Eq. 282) and $\delta_{\text{He}}^{\text{hyb}}$ (Eq. 283); assert $< 10^{-3}$

11.8 Frontier Dynamics and Cluster Migration

Clusters grow from small to large and occasionally shrink from large to small. The frontier must therefore be treated as a *dynamic interface* rather than a fixed boundary.

Small-to-large promotion.

Whenever a small-cluster concentration $c_n^{\text{small}}(t)$ or $c_{m,\ell}^{\text{small}}(t)$ exceeds a threshold $c_{\text{thresh}}^{\text{promote}} \sim \Omega^{-1}$ (i.e. expected particle count ≥ 1), the cluster is promoted to the stochastic block:

$$X_n \leftarrow \lfloor \Omega\, c_n^{\text{small}} + \tfrac{1}{2} \rfloor, \quad c_n^{\text{small}} \leftarrow 0; \tag{286}$$

the cluster is added to the relevant priority queue(s).

Large-to-small demotion.

Whenever the stochastic particle count of a species drops to $X \leq X_{\text{thresh}}^{\text{demote}}$ (typically 1–3), the species is demoted back to the deterministic block as a concentration $c = X/\Omega$. This prevents very dilute species from consuming stochastic timesteps with very rare events.

Dynamic reaction-network update.

When a cluster of previously unseen composition (m, ℓ) is created by a stochastic event, its reactions are automatically added to the relevant queues — an online network expansion analogous to the dynamic updating in Hoang et al. Hoang et al (2015).

11.9 System Complexity Comparison

Table 37 summarises the algorithmic complexity of the hybrid scheme relative to the purely deterministic and purely stochastic alternatives.

Table 37 Algorithmic complexity per macro-timestep δt for the three CD methods applied to the four-species H–He–V–I system. N number of distinct large-cluster species; $n_{\text{eq}}^{\text{det}}$ = ODE system size for small clusters. The hybrid column assumes the LR-NR log-time the stochastic block.

	Purely deterministic	Purely stochastic (SCD)	Hybrid (proposed)
ODE/event cost (time)	$O(n_{\text{eq}}^3)$ (implicit)	$O(N_{\text{large}})$ (Gillespie)	$O((n_{\text{eq}}^{\text{det}})^3 + N_{\text{events}} \log N_{\text{large}})$
Space	$O(n_{\text{eq}}^2)$ (Jacobian)	$O(N_{\text{large}})$	$O((n_{\text{eq}}^{\text{det}})^2 + N_{\text{large}})$
Multispecies H tracking	Adiabatic ($m > m_H^*$)	Full explicit	Explicit ($m \leq M_{\text{front}}$), adiabatic ($m >$
Statistical noise	None	$\sim N_{\text{part}}^{-1/2}$	$\sim N_{\text{part}}^{-1/2}$
Spatial correlations	No	No (mean-field)	No (mean-field)
Max demonstrated species	2 (V + He, deterministic FP)	3 (V + He + H)	4 (V + He + H + I, proposed)

For the EUROFER97 CD project with $N_{\text{front}}^i = 200$, $M_{\text{front}} = 15$, $n_{\text{eq}}^{\text{det}} \approx 800$, and $N_{\text{large}} \approx 10^4$–$10^5$, the hybrid scheme reduces the per-macro-timestep cost by approximately two orders of magnitude relative to the purely deterministic approach, with sub-percent statistical noise at $N_{\text{part}} \sim 10^4$.

A Proof of the Conservation Laws

This appendix proves equations (160) and (163) by computing the time derivatives of the two moment sums directly from the master equations (152), (155), and (157). Each reaction process is handled in turn. The proof is carried out for the mono-defect form of the master equations; the coalescence terms cancel by the same antisymmetry argument given at the end of Section A.3.

A.1 Frenkel Pair Moment

Define the weighted sums

$$S_I(t) = \sum_{n=1}^{N} n\, c_n, \qquad S_V(t) = \sum_{m=1}^{M} \sum_{\ell=0}^{\ell_{\max}(m)} m\, c_{m,\ell}. \tag{287}$$

We compute dS_I/dt and dS_V/dt by multiplying each master equation by the appropriate weight and summing.

Contributions to dS_I/dt

Cascade production (G).

$$\sum_n n\, G_n \equiv P_I. \tag{288}$$

Loop growth/shrinkage by mono-SIA absorption (P3, $n' = 1$ limit).

The growth term for c_n is $+\mathcal{K}^{\text{loop}}_{i,n-1}\, c_1\, c_{n-1}$ (for $n \geq 2$) and the loss term is $-\mathcal{K}^{\text{loop}}_{i,n}\, c_1\, c_n$. Multiplying by n and summing:

$$\sum_{n=2}^{N} n\, \mathcal{K}^{\text{loop}}_{i,n-1}\, c_1\, c_{n-1} - \sum_{n=1}^{N} n\, \mathcal{K}^{\text{loop}}_{i,n}\, c_1\, c_n. \tag{289}$$

Shift the index $n \to n+1$ in the first sum:

$$\sum_{n=1}^{N-1} (n+1)\, \mathcal{K}^{\text{loop}}_{i,n}\, c_1\, c_n - \sum_{n=1}^{N} n\, \mathcal{K}^{\text{loop}}_{i,n}\, c_1\, c_n = \sum_{n=1}^{N} \mathcal{K}^{\text{loop}}_{i,n}\, c_1\, c_n \equiv +c_1\, \mathcal{R}^{\text{loop}}_i. \tag{290}$$

This positive term represents SIAs flowing from the free pool (c_1) into loops (c_n, $n \geq 2$); it is exactly cancelled when we account for the corresponding loss from c_1 (see the dS_V/dt accounting below, or note that c_1 itself carries weight 1 in S_I, so the net transfer is zero). More precisely, the term $-\mathcal{K}^{\text{loop}}_{i,n}\, c_1\, c_n$ for $n = 1$ contributes $-\mathcal{K}^{\text{loop}}_{i,1}\, c_1^2$ to dc_1/dt, and the weight-1 moment of that is the same $-\mathcal{K}^{\text{loop}}_{i,1}\, c_1^2$. The two sums together telescope to give net contribution $+\mathcal{K}^{\text{loop}}_{i,N}\, c_1\, c_N \to 0$ at the truncation boundary (negligible flux out of the top of the size grid).

More cleanly: define $j^i_n = \mathcal{K}^{\text{loop}}_{i,n}\, c_1\, c_n$ as the flux from size n to $n+1$ by SIA absorption. Then the growth/loss terms form a discrete divergence $j^i_{n-1} - j^i_n$, and

$$\sum_{n=1}^{N} n\,(j^i_{n-1} - j^i_n) = \sum_{n=1}^{N-1} j^i_n - N\, j^i_N + 0 \cdot j^i_0 = \sum_{n=0}^{N-1} j^i_n - N\, j^i_N. \tag{291}$$

In the same way, the vacancy-absorption (loop shrinkage) terms give $-\sum_{n=1}^{N} j^v_n$ where $j^v_n = \mathcal{K}^{\text{loop}}_{v,n}\, c_{1,0}\, c_n$. Together the P3 contribution to dS_I/dt is:

$$\left.\frac{dS_I}{dt}\right|_{\text{P3}} = \sum_{n=1}^{N} \mathcal{K}^{\text{loop}}_{i,n}\, c_1\, c_n - \sum_{n=1}^{N} \mathcal{K}^{\text{loop}}_{v,n}\, c_{1,0}\, c_n \equiv c_1\, \mathcal{R}_i - c_{1,0}\, \mathcal{R}_v, \tag{292}$$

boundary terms at $n = N$ being negligible.

SIA thermal emission (P5).

The emission terms for c_n are $+\varepsilon_i(n{+}1)\, c_{n+1} - \varepsilon_i(n)\, c_n$, forming a discrete divergence. Multiplying by n and summing:

$$\sum_n n\,[\varepsilon_i(n{+}1)\, c_{n+1} - \varepsilon_i(n)\, c_n] = -\sum_n \varepsilon_i(n)\, c_n, \tag{293}$$

after the telescoping index shift $n \to n-1$ in the first sum (boundary terms at $n = N$ negligible). This is a negative contribution: emission reduces S_I by moving SIAs from loops back to the free pool.

Recombination (P1).

Only $n = 1$: $-\mathcal{K}_{iv}\, c_1\, c_{1,0}$.

$$\left.\frac{dS_I}{dt}\right|_{\text{P1}} = -\mathcal{K}_{iv}\, c_1\, c_{1,0}. \tag{294}$$

Absorption by cavities (P2/P6).

$$\left.\frac{dS_I}{dt}\right|_{\text{cav}} = -\sum_{n=1}^{N}\sum_{m=1}^{M} n\, \mathcal{K}^{(\text{cav})}_{n,m}\, c_n\, \bar{c}_m. \tag{295}$$

Fixed sinks (P4).

$$\left.\frac{dS_I}{dt}\right|_{\text{P4}} = -\sum_{n=1}^{N} n\, \mathcal{D}_i\, c_n = -\mathcal{D}_i\, S_I \quad \text{(first-order in } c_n\text{)}. \tag{296}$$

Expanding $\mathcal{D}_i = \mathcal{D}^d_i + \mathcal{D}^{gb}_i + \mathcal{D}^p_i$ and noting $\mathcal{D}^d_i = Z^d_i\, \rho_d\, a^2\, \omega^{\text{eff}}_i$:

$$\left.\frac{dS_I}{dt}\right|_{\text{P4}} = -Z^d_i\, \rho_d\, a^2 \sum_n n\, \omega^{\text{eff}}_n\, c_n \;-\; \left(\frac{4\pi^2}{d_g^2} + \frac{8\pi r_p}{a}\, c^N_p\right) a^2 \sum_n n\, \omega^{\text{eff}}_n\, c_n. \tag{297}$$

Trap-mutation SIA source (P7).

$$\left.\frac{dS_I}{dt}\right|_{\mathrm{P7}} = +\sum_{m,\ell} \Gamma_{\mathrm{TM}}(m,\ell)\, c_{m,\ell}. \tag{298}$$

Contributions to dS_V/dt

Cascade production (G).

$$\sum_{m,\ell} m\, G_{m,\ell} \equiv P_V = P_I \quad \text{(equal Frenkel-pair production)}. \tag{299}$$

Vacancy exchange along m (P2 + P5).

The growth term $+A_{\mathrm{sph}}\,(m-1)^{1/3}\omega_v^{\mathrm{eff}}\, c_{1,0}\, c_{m-1,\ell}$ and loss term $-A_{\mathrm{sph}}\, m^{1/3}\omega_v^{\mathrm{eff}}\, c_{1,0}\, c_{m,\ell}$ form a flux divergence in m. Define $j_m^v = A_{\mathrm{sph}}\, m^{1/3}\, \omega_v^{\mathrm{eff}}\, c_{1,0}\, \bar{c}_m$. Multiplying by m, summing, and telescoping:

$$\sum_{m,\ell} m\,[j_{m-1,\ell}^v - j_{m,\ell}^v] = \sum_{m,\ell} j_{m,\ell}^v = A_{\mathrm{sph}}\, \omega_v^{\mathrm{eff}}\, c_{1,0} \sum_m m^{1/3}\, \bar{c}_m. \tag{300}$$

The emission counterpart $+\varepsilon_v(m+1,\ell)\, c_{m+1,\ell} - \varepsilon_v(m,\ell)\, c_{m,\ell}$ telescopes to $-\sum_{m,\ell} \varepsilon_v(m,\ell)\, c_{m,\ell}$, exactly cancelling the moment of the reverse absorption flux (detailed balance). Their net contribution to dS_V/dt is:

$$\left.\frac{dS_V}{dt}\right|_{v\text{-exchange}} = c_{1,0}\, \mathcal{R}_v - \text{(emission back-flux)}, \tag{301}$$

where $c_{1,0}\, \mathcal{R}_v \equiv A_{\mathrm{sph}}\, \omega_v^{\mathrm{eff}}\, c_{1,0} \sum_m m^{1/3}\, \bar{c}_m$ is the same quantity appearing with a negative sign in $dS_I/dt|_{\mathrm{P3}}$ (Eq. 292) because a mono-vacancy absorbed by a loop removes it from S_V. Note that the emission terms cancel their absorption counterparts exactly; what remains is the *net* vacancy flux into clusters driven by the concentration gradient.

SIA-induced shrinkage (P2/P6).

The mono-SIA term $-A_{\mathrm{sph}}\, m^{1/3}\, \omega_i^{\mathrm{eff}}\, c_1\, c_{m,\ell}$ and the cluster term $-\sum_{n=2}^{N} \mathcal{K}_{n,m}^{(\mathrm{cav})}\, c_n\, c_{m,\ell}$, weighted by m and summed over (m,ℓ):

$$\left.\frac{dS_V}{dt}\right|_{\mathrm{SIA-shrink}} = -\sum_{n=1}^{N} \sum_{m=1}^{M} m\, \mathcal{K}_{n,m}^{(\mathrm{cav})}\, c_n\, \bar{c}_m. \tag{302}$$

He exchange (P2 + P5 along ℓ).

The He-exchange terms shift ℓ at fixed m, so they do not change the m-weight. Their contribution to dS_V/dt is identically zero:

$$\left.\frac{dS_V}{dt}\right|_{\text{He-exch}} = 0. \tag{303}$$

Trap mutation (P7).

The source term $+\Gamma_{\mathrm{TM}}(m{-}1,\ell)\, c_{m-1,\ell}$ and loss term $-\Gamma_{\mathrm{TM}}(m,\ell)\, c_{m,\ell}$, weighted by m:

$$\sum_{m,\ell} m\,[\Gamma_{\mathrm{TM}}(m{-}1,\ell)\, c_{m-1,\ell} - \Gamma_{\mathrm{TM}}(m,\ell)\, c_{m,\ell}]. \tag{304}$$

Shifting $m \to m+1$ in the source term:

$$\left.\frac{dS_V}{dt}\right|_{\mathrm{P7}} = \sum_{m,\ell}(m{+}1)\,\Gamma_{\mathrm{TM}}(m,\ell)\, c_{m,\ell} - \sum_{m,\ell} m\,\Gamma_{\mathrm{TM}}(m,\ell)\, c_{m,\ell} = +\sum_{m,\ell}\Gamma_{\mathrm{TM}}(m,\ell)\, c_{m,\ell}. \tag{305}$$

Radiation re-solution (P8).

Re-solution shifts ℓ at fixed m, so its contribution to dS_V/dt is identically zero:

$$\left.\frac{dS_V}{dt}\right|_{\mathrm{P8}} = 0. \tag{306}$$

Fixed sinks absorbed from the vacancy equation.

The free monovacancy equation (Eq. 156) contains a fixed-sink loss $-\mathcal{D}_v\, c_{1,0}$, contributing $-1 \cdot \mathcal{D}_v\, c_{1,0}$ to dS_V/dt (weight $m = 1$). All larger vacancy clusters are immobile and do not appear in fixed-sink terms directly, so:

$$\left.\frac{dS_V}{dt}\right|_{\mathrm{P4}} = -\mathcal{D}_v\, c_{1,0} = -\left(Z_v^d\,\rho_d + \frac{4\pi^2}{d_g^2} + \frac{8\pi r_p}{a}\, c_p^N\right) a^2\, \omega_v^{\mathrm{eff}}\, c_{1,0}. \tag{307}$$

Assembling $d(S_I - S)/dt$

We now subtract the S (swelling) contributions from the S_I contributions process by process, using $S \equiv S_V = \sum_{m,\ell} m\, c_{m,\ell}$ and $S_I \equiv \sum_n n\, c_n$.

Cascade production: $P_I - P_V = 0$. ✓

P3 / coalescence (loop growth and shrinkage): The telescoping argument of Eq. (291) shows that the P3 terms form a discrete divergence with zero net moment contribution. The $-c_{1,0}\,\mathcal{R}_v$ in S_I and the $+c_{1,0}\,\mathcal{R}_v$ in S cancel; the remaining $+c_1\,\mathcal{R}_i$ is already counted as weight $n = 1$ within S_I. Net contribution to $d(S_I{-}S)/dt$: zero. ✓

P5 (thermal emission): SIA emission telescopes to a boundary term at $n = N$ (negligible). Vacancy emission cancels by detailed balance with the absorption term. Net P5 contribution to $d(S_I - S)/dt$: zero. ✓

P1 / v–i annihilation (recombination): Each recombination event removes one SIA (-1 from S_I) and one vacancy (-1 from S). Contribution to $d(S_I - S)/dt$: $(-1) - (-1) = 0$. ✓

Cavity absorption (P2/P6): An n-SIA cluster absorbed by an m-vacancy cavity removes n from S_I and reduces S by n (the cavity shrinks by n vacancies upon absorbing n SIAs). Contribution to $d(S_I - S)/dt$: $(-n) - (-n) = 0$. ✓

P7 (trap mutation): Creates one SIA ($+1$ to S_I) and adds one vacancy to the cluster ($+1$ to S). Contribution: $(+1) - (+1) = 0$. ✓

Fixed sinks (P4): The contributions from S_I and S are:

$$\left.\frac{dS_I}{dt}\right|_{\mathrm{P4}} = -Z_i^d\,\rho_d\,a^2\sum_n n\,\omega_n^{\mathrm{eff}}\,c_n - (k_{gb}^2 + k_p^2)\,a^2\sum_n n\,\omega_n^{\mathrm{eff}}\,c_n, \tag{308}$$

$$\left.\frac{dS}{dt}\right|_{\mathrm{P4}} = -Z_v^d\,\rho_d\,a^2\,\omega_v^{\mathrm{eff}}\,c_{1,0} - (k_{gb}^2 + k_p^2)\,a^2\,\omega_v^{\mathrm{eff}}\,c_{1,0}. \tag{309}$$

The grain-boundary and precipitate terms are proportional to $\omega_\alpha^{\mathrm{eff}} c_\alpha$ in both S_I and S; in general they do not cancel because the species (i vs. v) and the cluster sizes differ. However, grain boundaries and precipitates are *unbiased* ($Z_i^{gb} = Z_v^{gb}$, $Z_i^p = Z_v^p$), so the *net imbalance* they produce in $d(S_I - S)/dt$ is exactly equal and opposite for SIAs and vacancies and cancels when the system is close to quasi-steady state ($c_i D_i \approx c_v D_v$). Only the dislocation network, with its intrinsic bias $Z_i^d \neq Z_v^d$, produces a *systematic* non-cancelling contribution. The exact dislocation contribution to $d(S_I - S)/dt$ is:

$$\left.\frac{d(S_I - S)}{dt}\right|_{\mathrm{disl}} = -Z_i^d\,\rho_d\,a^2\sum_n n\,\omega_n^{\mathrm{eff}}\,c_n + Z_v^d\,\rho_d\,a^2\,\omega_v^{\mathrm{eff}}\,c_{1,0} \equiv -\frac{d\Delta J^d}{dt}. \tag{310}$$

Collecting all terms:

$$\boxed{\frac{d}{dt}\big(S_I + \Delta J^d - S\big) = 0,} \tag{311}$$

where $\Delta J^d(t)$ is defined in Eq. (158). Integrating from $t = 0$ with a defect-free crystal ($S_I(0) = S(0) = 0$, $\Delta J^d(0) = 0$) gives the swelling identity:

$$S(t) = S_I(t) + \Delta J^d(t), \tag{312}$$

in agreement with Eq. (161). For $Z_i^d > Z_v^d = 1$, ΔJ^d grows monotonically: the dislocation bias continuously converts the SIA excess into swelling, with the SIA loop inventory S_I acting as a transient reservoir that delays but does not prevent void growth.

A.2 Helium Moment

Define

$$S_{\mathrm{He}}(t) \equiv c_h + \sum_{m=1}^{M}\sum_{\ell=1}^{L} \ell\, c_{m,\ell}. \tag{313}$$

We compute dS_{He}/dt by summing dc_h/dt and $\sum_{m,\ell} \ell\, dc_{m,\ell}/dt$.

Production.

G_{He} from dc_h/dt and $\sum_{m,\ell} \ell\, G_{m,\ell}$ from the vacancy equation; the latter is zero in the EUROFER model (He is not directly produced in clusters). Net: $+G_{\mathrm{He}}$.

He trapping by cavities (P2 along ℓ, free He side).

From dc_h/dt: $-A_{\rm sph}\,\omega_h^{\rm eff}\,c_h\sum_{m,\ell<\ell_{\max}} m^{1/3}\,c_{m,\ell}$. The corresponding source in $dc_{m,\ell}/dt$: $+A_{\rm sph}\,m^{1/3}\,\omega_h^{\rm eff}\,c_h\,c_{m,\ell-1}$, which weighted by ℓ and summed gives $+A_{\rm sph}\,\omega_h^{\rm eff}\,c_h\sum_{m,\ell} m^{1/3}\,c_{m,\ell-1}$; shifting $\ell \to \ell+1$ this equals $A_{\rm sph}\,\omega_h^{\rm eff}\,c_h\sum_{m,\ell<\ell_{\max}} m^{1/3}\,c_{m,\ell}$, exactly cancelling the loss from dc_h/dt. Net P2 (He trapping): 0. ✓

He thermal emission (P5 along ℓ).

From $dc_{m,\ell}/dt$: the terms $+\varepsilon_h(m,\ell{+}1)\,c_{m,\ell+1} - \varepsilon_h(m,\ell)\,c_{m,\ell}$ weighted by ℓ telescope to $-\sum_{m,\ell}\varepsilon_h(m,\ell)\,c_{m,\ell}$ (boundary terms at $\ell = L$ negligible). From dc_h/dt: $+\sum_{m,\ell}\varepsilon_h(m,\ell)\,c_{m,\ell}$. These cancel: net P5 contribution to $dS_{\rm He}/dt$ is 0. ✓

Trap mutation (P7).

Trap mutation converts $\mathrm{He}_\ell\mathrm{V}_m \to \mathrm{He}_\ell\mathrm{V}_{m+1}$ + SIA. The He count ℓ is unchanged; the terms in $dc_{m,\ell}/dt$ merely shift probability from (m,ℓ) to $(m+1,\ell)$ along the m-direction. Weighted by ℓ, they contribute $\sum_{m,\ell}\ell\,[\Gamma_{\rm TM}(m{-}1,\ell)\,c_{m-1,\ell} - \Gamma_{\rm TM}(m,\ell)\,c_{m,\ell}]$. Shifting $m \to m+1$ in the source: $(+\ell\,\Gamma_{\rm TM}) - (\ell\,\Gamma_{\rm TM}) = 0$. Net P7 contribution: 0. ✓

Radiation re-solution (P8).

The terms $+b_0\dot\phi(\ell{+}1)\,c_{m,\ell+1} - b_0\dot\phi\,\ell\,c_{m,\ell}$ in $dc_{m,\ell}/dt$ weighted by ℓ give: $\sum_{m,\ell}\ell\,b_0\dot\phi[(\ell{+}1)\,c_{m,\ell+1} - \ell\,c_{m,\ell}]$. Shifting $\ell \to \ell-1$ in the source: $\sum_{m,\ell} b_0\dot\phi\,(\ell{-}1)\,\ell\,c_{m,\ell} - \sum_{m,\ell} b_0\dot\phi\,\ell^2\,c_{m,\ell} = -\sum_{m,\ell} b_0\dot\phi\,\ell\,c_{m,\ell}$. The re-solution source $+\sum_{m,\ell} b_0\dot\phi\,\ell\,c_{m,\ell}$ in dc_h/dt (free He source) exactly cancels this. Net P8: 0. ✓

Fixed sinks acting on c_h.

From dc_h/dt: $-\mathcal{D}_h\,c_h$. Since $\mathcal{D}_h$ acts on free helium only and contributes $-\mathcal{D}_h\,c_h$ to $dS_{\rm He}/dt$, this term does *not* cancel. He absorbed by dislocations, grain boundaries, or precipitates is permanently removed from the system. For EUROFER under neutron irradiation, the helium diffusivity is high enough that most He reaches bubbles well before being absorbed by other sinks; the $\mathcal{D}_h$ term is typically small but non-zero. Accounting for it explicitly:

$$\boxed{\frac{dS_{\rm He}}{dt} = G_{\rm He} - \mathcal{D}_h\,c_h.} \tag{314}$$

In most irradiation models the sink-absorption of He is negligible ($\mathcal{D}_h \ll \mathcal{K}^{\rm cav}_{h,m}$), and Eq. (163) (with the right-hand side $= G_{\rm He}$) is an excellent approximation. Equation (314) is the exact statement.

A.3 Conservation Under Coalescence

The i–i coalescence terms (145) contribute to dS_I/dt as follows. The gain term for c_n is $\frac{1}{2}\sum_{n'}\mathcal{K}^{ii}_{n',n-n'}\,c_{n'}\,c_{n-n'}$; weighted by n and summed over n, relabelling $n' \to s$,

$n - n' \to t$:

$$\left.\frac{dS_I}{dt}\right|_{ii,\text{gain}} = \frac{1}{2}\sum_{s,t}(s+t)\,\mathcal{K}^{ii}_{s,t}\,c_s\,c_t. \tag{315}$$

The loss term for c_n is $-c_n \sum_{n'} \mathcal{K}^{ii}_{n,n'}\,c_{n'}$; weighted by n and summed:

$$\left.\frac{dS_I}{dt}\right|_{ii,\text{loss}} = -\sum_{n,n'} n\,\mathcal{K}^{ii}_{n,n'}\,c_n\,c_{n'} = -\frac{1}{2}\sum_{s,t}(s+t)\,\mathcal{K}^{ii}_{s,t}\,c_s\,c_t, \tag{316}$$

where the last equality follows by symmetrising over (s,t). The gain and loss contributions cancel exactly:

$$\left.\frac{dS_I}{dt}\right|_{ii} = 0. \tag{317}$$

Coalescence conserves the total SIA moment: merging two clusters of sizes s and t into one of size $s+t$ does not change $\sum n\,c_n$. An identical argument applies to the v–v channel for dS_V/dt.

For v–i annihilation (147) with outcome $V_m + I_n \to V_{|m-n|}$ (or annihilation if $m = n$), the SIA moment changes by $-n + \max(n-m, 0) = -\min(n,m)$ and the vacancy moment changes by $-m + \max(m-n, 0) = -\min(n,m)$. The contribution to $dQ/dt = d(S_I - S_V)/dt$ is $(-\min(n,m)) - (-\min(n,m)) = 0$: v–i annihilation conserves Q regardless of the cluster sizes involved. ✓

The Jacobian $\partial\mathbf{f}/\partial\mathbf{y}$ inherits this block structure. The blue tridiagonal block (i–i ladder) has bandwidth 2 in n. The green block-tridiagonal (vacancy/He) has block size $(\ell_{\max}+1)$ and tridiagonal coupling in m from P2/P5, with dense ℓ-coupling within each block from the He exchange terms P2/P5/P7/P8. The orange off-diagonal blocks are sparse-diagonal: each c_n couples to all $\bar{c}_m$ through the cavity absorption rate, and each $c_{m,\ell}$ couples to all c_n through the SIA-shrinkage rate. The red column/row couples the free-He scalar c_h to every (m,ℓ) node.

A.3.1 He Content and Reduction Approximations

The dominant contributor to system size is the He index ℓ. The maximum He loading per cluster grows with cavity size as

$$\ell_{\max}(m) = \lfloor \alpha_{\text{He}}\, m^{2/3} \rfloor, \qquad \alpha_{\text{He}} = 1.5\text{–}2.0, \tag{318}$$

giving a total vacancy–He grid size

$$n^{\text{full}}_{VH} = \sum_{m=1}^{M}\left[\ell_{\max}(m)+1\right] \approx M + \alpha_{\text{He}}\sum_{m=1}^{M} m^{2/3} \approx M\Big(1 + \tfrac{3}{5}\alpha_{\text{He}}\,M^{2/3}\Big), \tag{319}$$

which scales as $M^{5/3}$ and can reach 10^4–10^5 for $M = 10^3$–10^4. Two physics-based approximations reduce this substantially, exploiting the very different He budgets in fission and fusion irradiation environments.

Case 1 — Fusion (high He/dpa): mean-field He approximation.
Under fusion-relevant He generation rates (∼10–15 appm He/dpa), He quickly saturates every cavity to its equilibrium loading. The two-dimensional (m, ℓ) distribution can then be collapsed to a one-dimensional problem by tracking only the mean He loading per cavity size:

$$\bar{\ell}(m,t) \equiv \frac{\sum_\ell \ell\, c_{m,\ell}}{\bar{c}_m}, \tag{320}$$

and treating the He distribution within each m-bin as a delta function at $\bar{\ell}(m)$. This replaces the $n_{VH}^{\rm full}$ equations for $c_{m,\ell}$ with two ODEs per vacancy size — one for the total cavity concentration $\bar{c}_m$ and one for the mean He content $\bar{\ell}_m$ — plus the single free-He equation:

$$n_{\rm eq}^{\rm fusion} = N + \underbrace{2M}_{\bar{c}_m,\,\bar{\ell}_m} + \underbrace{1}_{c_h} = N + 2M + 1. \tag{321}$$

Case 2 — Fission (low He/dpa): decoupled-He approximation.
Fission reactors produce ∼0.1–1 appm He/dpa, so He loading remains low ($\ell \ll \ell_{\rm max}$) and bubbles are rare. Most cavities are pure voids ($\ell = 0$). He can then be decoupled: track $c_{m,0}$ (pure voids) separately from a single mean-field He concentration per size, treating bubble nucleation as a perturbation to the void distribution. In the simplest implementation this retains one He-bearing species per m and adds a single free-He equation:

$$n_{\rm eq}^{\rm fission} = N + \underbrace{M}_{\text{voids } c_{m,0}} + \underbrace{1}_{\bar{c}_{m,\ell\ge1}} + \underbrace{1}_{c_h} = N + M + 2. \tag{322}$$

A.3.2 System Size Summary

Table 38 collects the ODE counts for typical parameter choices ($N = 1000$, $M = 1000$, $\alpha_{\rm He} = 1.5$).

Table 38 ODE system size for the three He-content strategies. All entries assume $N = 1000$ SIA sizes, $M = 1000$ vacancy sizes, $\alpha_{\rm He} = 1.5$, and $n^i_{\rm max} = 100$, $m^v_{\rm max} = 5$. The column "Dominant block" identifies the bottleneck for Jacobian factorisation. Logarithmic size-binning for $m > m_{\rm group} \sim 100$–200 can reduce n_{VH} by a further factor of 3–5.

He strategy	**SIA block (N)**	**VacHe block (n_{VH})**	$\mathbf{n}_{\rm eq}$	**Dominant block**
Full (m, ℓ) grid (Eq. 319)	1 000	∼38 000	∼39 001	VacHe
Case 1: mean-field He, fusion (Eq. 321)	1 000	2 000	3 001	SIA + VacHe
Case 2: decoupled He, fission (Eq. 322)	1 000	1 001	2 002	SIA + VacHe

A.3.3 Stiffness and Solver Strategy

The system (170) is severely stiff. The stiffness ratio is dominated by the disparity between the He migration frequency and the vacancy frequency,

$$\frac{\omega_h^{\text{eff}}}{\omega_v^{\text{eff}}} \sim 10^4\text{–}10^8, \tag{323}$$

with additional stiffness contributed by the contrast between fast mobile-cluster reactions (small n, m) and slow large-cluster dynamics. An implicit solver with adaptive time stepping is mandatory; recommended options are LSODA (automatic stiffness detection) or CVODE (variable-order BDF with sparse Jacobian). Practical tolerances are `rtol` $= 10^{-8}$, `atol` $= 10^{-12}$ for the conservation diagnostics of Section 7.4.3 to remain below 10^{-6}.

Logarithmic size-binning (grouping) replaces the uniform m-grid with logarithmically spaced bins for $m > m_{\text{group}} \approx 50$–$200$. Each bin $[m_k, m_{k+1})$ is represented by a single representative concentration; capture rates are evaluated at the bin midpoint and weighted by the bin width. This reduces n_{VH} by a factor of 3–10 with negligible loss of accuracy for the swelling and loop-density predictions. The conservation diagnostics should be evaluated at every output step; the implementation should assert a hard stop if δ_{FP} or δ_{He} exceeds 10^{-3}.

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